

Analytical Geometry

by sciqb

**Complete Lecture Notes, Rigorous Derivations, TikZ Diagrams,
and Solved Examination Problems**

Transcribed and Enhanced Page-by-Page from Manuscript (Pages 1–69)

sciqb

<https://sciqb.com>

Academic Session 2026

Abstract

This compendium presents a definitive, rigorous, and visually enriched set of lecture notes in **Analytical Geometry**, faithfully capturing the entire 69-page handwritten curriculum. Topics span two-dimensional polar conics (focus as pole, chords, tangents, normals, asymptotes, chord of contact, director circle), three-dimensional rectangular Cartesian coordinate systems (direction cosines, direction ratios, projection of lines, angular formulas), the analytical and vector geometry of the plane (normal, intercept, three-point forms, dihedral angle bisectors, systems of planes), straight lines in space and skew lines (symmetrical reduction, line-plane intersections, shortest distance derivations), and the three-dimensional geometry of the sphere (general forms, diametric forms, plane sections, tangency conditions, normals, and tangent planes). All derivations are presented with complete algebraic steps, accompanied by custom high-precision TikZ vector graphics and mathematical remarks highlighting slips in handwritten records.

Contents

1	Polar Equations of Conic Sections (Pages 1–14)	6
1.1	Page 1: Focus-Directrix Definition and Standard Polar Equations	6
1.1.1	Geometric Definition and Polar Setup	6
1.1.2	Step-by-Step Derivation of the Standard Equation	7
1.1.3	Standard Orientations and Rotations	7
1.2	Page 2: Focal Chords and the Semi-Latus Rectum	9
1.2.1	Geometric Formulation	9
1.3	Page 3: Equation of a Chord Joining Points $(\alpha \pm \beta)$	11
1.3.1	Algebraic Derivation	11
1.4	Page 4: Alternate Chord Form and the Tangent at Point α	13
1.4.1	Chord Joining Vectorial Angles α and β	13
1.4.2	Derivation of the Tangent at Point α	13
1.4.3	Condition of Tangency for a Straight Line	13
1.5	Page 5: Condition of Tangency and the Normal Line	14
1.5.1	Tangency Condition in Standard Form $r \cos(\theta - \phi) = p$	14
1.5.2	Derivation of the Equation of the Normal Line	14
1.6	Page 6: Chord of Contact and the Polar Line	16
1.6.1	Derivation of the Chord of Contact	16
1.7	Page 7: Classification of Conics and Analysis of Eccentricity	17
1.7.1	Canonical Reduction Method	17
1.7.2	Detailed Analysis of the Manuscript Example	17
1.8	Page 8: Asymptotes of a Hyperbola in Polar Coordinates	18
1.8.1	Mathematical Derivation	18
1.9	Page 9: Problem 1 — Tangent Segment Intercepted by Directrix	19
1.10	Page 10: Problem 2 — Sum of Reciprocal Products of Focal Chords	20
1.11	Page 11: Problem 3 — Tangents at Extremities of a Focal Chord	21
1.12	Page 12: Problem 4 — Perpendicular Focal Tangents of a Parabola	22
1.13	Page 13: Problem 5 — Director Circle in Polar Coordinates	23
1.14	Page 14: Problem 6 — Locus of the Pole of a Subtended Chord	24
2	3D Cartesian Coordinates and Direction Cosines (Pages 15–24)	25
2.1	Page 15: Introduction to 3D Cartesian Coordinates and Section Formulae	25
2.1.1	The Spatial Cartesian Coordinate System	25
2.1.2	The 3D Distance Formula	26
2.1.3	Section Formula (Internal and External Division)	26
2.2	Page 16: Transformation of Coordinates — Translation of Axes	27
2.2.1	Transformation Equations for Translation	27
2.2.2	Invariance of Spatial Distances	28
2.3	Page 17: Projection of a Directed Line Segment on Coordinate Axes	29
2.3.1	Geometric Definition of Orthogonal Projection	29

2.3.2	Projections of the Radius Vector OP	29
2.4	Page 18: Projection of a Broken Line — The Contour Theorem	30
2.4.1	The Fundamental Projection Theorem	30
2.4.2	Application: Projections of Arbitrary Spatial Displacements	30
2.5	Page 19: Direction Cosines and the Fundamental Identity $l^2 + m^2 + n^2 = 1$	31
2.5.1	Definition of Direction Cosines	31
2.5.2	The Fundamental Quadratic Identity	31
2.6	Page 20: Projection of Segment PQ on a Line with Direction Cosines (l, m, n) . . .	32
2.6.1	Step-by-Step Derivation of the Projection Formula	32
2.6.2	Derivation of the Distance Formula via Self-Projection	33
2.7	Page 21: Direction Ratios and Normalization to Direction Cosines	33
2.7.1	Definition of Direction Ratios	33
2.7.2	Normalization Formula	34
2.7.3	Direction Ratios of the Line Joining Two Points	34
2.7.4	Condition for Perpendicularity of Two Lines	34
2.8	Page 22: Angle Between Two Directed Lines in Space	34
2.8.1	Derivation of the Cosine Formula	35
2.9	Page 23: Sine of the Angle, Lagrange's Identity, and Parallelism	35
2.9.1	Lagrange's Identity and the Sine Formula	36
2.9.2	Condition for Parallelism of Two Lines	36
2.10	Page 24: Solved Problem — The Four Diagonals of a Cube	36
3	The Analytical Geometry of the Plane (Pages 25–36)	39
3.1	Page 25: Definition of a Plane and the General Equation of First Degree	39
3.1.1	Geometric Definition of a Plane	39
3.1.2	Every First-Degree Equation Represents a Plane	39
3.1.3	Number of Independent Arbitrary Constants	40
3.2	Page 26: Equation of a Plane Passing Through a Given Point	40
3.2.1	One-Point Form of the Plane Equation	40
3.2.2	Geometrical Meaning of the Coefficients (A, B, C)	41
3.3	Page 27: The Normal Form of the Equation of a Plane	41
3.3.1	Derivation of the Normal Form	42
3.4	Page 28: Reduction of the General Equation to the Normal Form	42
3.4.1	The Reduction Algorithm	43
3.4.2	Sign Convention for the Perpendicular Distance $p \geq 0$	43
3.5	Page 29: Intercept Form and the Three-Point Determinant Form	44
3.5.1	The Intercept Form of a Plane	44
3.5.2	Plane Through Three Non-Collinear Points	45
3.6	Page 30: Solved Examination Problems on Plane Equations	45
3.7	Page 31: Perpendicular Distance of a Point from a Plane	46
3.7.1	Derivation of the Distance Formula	46
3.7.2	Distance Between Two Parallel Planes	47
3.8	Page 32: Position of Points Relative to a Plane and Systems of Planes	47
3.8.1	Position of Two Points Relative to a Plane	47
3.8.2	Plane Passing Through the Intersection of Two Given Planes	48
3.9	Page 33: Planes Bisecting the Dihedral Angles Between Two Planes	49
3.9.1	Definition and Geometric Derivation	49
3.9.2	Orthogonality of the Two Bisector Planes	50
3.10	Page 34: Discrimination Between Acute and Obtuse Angle Bisectors	50
3.10.1	The Standard University Algorithm	50
3.11	Page 35: Method 2 (Point Method) and Comprehensive Worked Problem	51

3.11.1	Method 2: The Tangent Half-Angle (Point) Method	51
3.11.2	Worked Examination Problem Setup	52
3.12	Page 36: Continued Solution of Angle Bisector Problem and Summary	52
3.12.1	Algebraic Simplification of the Bisectors	52
3.12.2	Determination of Acute and Obtuse Character	53
3.12.3	Verification of Mutual Perpendicularity	53
4	Straight Lines in Space and Skew Lines (Pages 37–52)	54
4.1	Page 37: The Straight Line in Space — Symmetrical and Unsymmetrical Forms . .	54
4.1.1	The Line as the Intersection of Two Planes	54
4.1.2	Derivation of the Symmetrical Form	54
4.1.3	Two-Point Form of a Straight Line	55
4.2	Page 38: Parametric Coordinates and the General Point on a Line	55
4.2.1	Parametric Coordinates	55
4.2.2	Geometrical Significance	56
4.3	Page 39: Transformation from Unsymmetrical Form to Symmetrical Form	56
4.3.1	The Reduction Algorithm	56
4.4	Page 40: Angle Between a Line and a Plane	57
4.4.1	Derivation of the Sine Formula	57
4.4.2	Conditions of Parallelism and Perpendicularity	58
4.5	Page 41: Intersection of a Line and a Plane, Foot of Perpendicular, and Image . .	58
4.5.1	Point of Intersection of a Line and a Plane	58
4.5.2	Foot of Perpendicular and Image of a Point	58
4.6	Page 42: Condition of Coplanarity of Two Straight Lines	59
4.6.1	Derivation of the Coplanarity Condition	59
4.6.2	Equation of the Plane Containing Two Coplanar Lines	60
4.7	Page 43: Condition for a Line to Lie Wholly in a Plane	61
4.8	Page 44: Plane Containing a Given Line and Parallel to Another Line	61
4.8.1	Construction and Derivation	61
4.9	Page 45: Line Intersecting Two Given Lines and Satisfying a Geometric Condition	62
4.9.1	Transversal Passing Through a Given Point	62
4.9.2	Transversal Parallel to a Given Line	62
4.10	Page 46: Skew Lines and the Concept of Shortest Distance	63
4.10.1	Definition of Skew Lines	63
4.10.2	The Common Perpendicular	63
4.11	Page 47: Geometric Configuration and Direction of the Common Perpendicular . .	63
4.11.1	Direction Cosines of the Common Perpendicular	64
4.12	Page 48: Analytical Derivation of the Magnitude of Shortest Distance	64
4.13	Page 49: Equations of the Line of Shortest Distance	65
4.14	Page 50: Comprehensive Solved Examination Problem on Skew Lines	66
4.15	Page 51: Equations of the Line of S.D. and Feet of Common Perpendicular	67
4.15.1	Equations of the Line of Shortest Distance	67
4.15.2	Coordinates of the Feet G and H	68
4.16	Page 52: Distance Between Parallel Lines and Chapter Summary	68
4.16.1	Shortest Distance Between Parallel Lines	69
4.16.2	Summary of Spatial Line Geometry	69
5	Vector Treatment of Planes and Lines (Pages 53–62)	70
5.1	Page 53: Vector Equation of a Plane — Normal Form and One-Point Form	70
5.1.1	The Vector Normal Form of a Plane	70
5.1.2	Connection with Cartesian Coordinates	71
5.1.3	Plane Passing Through a Point with Given Normal	71

5.2	Page 54: General Vector Plane Equations and Perpendicular Distance	71
5.2.1	General Vector Equation and Normalization	71
5.2.2	Perpendicular Distance of a Point from a Vector Plane	71
5.2.3	Distance Between Two Parallel Planes in Vector Form	72
5.3	Page 55: Angle Between Two Planes and Plane Through Three Points	72
5.3.1	Angle Between Two Planes	72
5.3.2	Plane Passing Through Three Points in Vector Form	73
5.4	Page 56: Vector Family of Planes Through an Intersection Line	73
5.5	Page 57: Vector Equations of Dihedral Angle Bisectors	74
5.5.1	Derivation of the Vector Bisector Equations	74
5.5.2	Orthogonality of the Vector Bisectors	75
5.6	Page 58: Vector Equation of a Straight Line	75
5.6.1	Point-Direction Vector Form	75
5.6.2	Non-Parametric Cross-Product Form	75
5.6.3	Two-Point Vector Form	76
5.7	Page 59: Perpendicular Distance of a Point from a Vector Line	76
5.7.1	Foot of the Perpendicular in Vector Form	76
5.8	Page 60: Coplanarity of Vector Lines and the Plane Containing Them	77
5.8.1	Plane Containing the Two Coplanar Lines	77
5.9	Page 61: Shortest Distance Between Skew Lines in Vector Form	77
5.9.1	Shortest Distance Between Parallel Lines in Vector Form	78
5.10	Page 62: Solved Vector Examination Problems	78
6	The Geometry of the Sphere in 3D (Pages 63–69)	80
6.1	Page 63: Definition of the Sphere and the General Equation	80
6.1.1	Definition of a Sphere	80
6.1.2	The Center-Radius Form	80
6.1.3	The General Second-Degree Equation of a Sphere	81
6.1.4	Classification of Spheres	81
6.2	Page 64: Diametric Form of the Sphere and Plane Sections	81
6.2.1	The Diametric Form of a Sphere	81
6.2.2	Theorem: Plane Section of a Sphere is a Circle	82
6.3	Page 65: Great Circles, Small Circles, and the Radical Plane	83
6.3.1	Great Circles and Small Circles	83
6.3.2	The Radical Plane of Two Spheres	83
6.4	Page 66: Condition of Tangency Between a Plane and a Sphere	84
6.4.1	Standard Sphere Centered at the Origin	84
6.5	Page 67: Tangent Plane to a Sphere at a Given Point	84
6.6	Page 68: Normal to a Sphere, Length of Tangent, and Plane of Contact	85
6.6.1	Equations of the Normal Line to a Sphere	86
6.6.2	Length of Tangent from an External Point	86
6.6.3	Plane of Contact	86
6.7	Page 69: Comprehensive Solved Examination Problem and Course Summary . . .	87
6.7.1	Curriculum Summary of Analytical Geometry (Pages 1–69)	88
A	Master Formula Cheat Sheet and Exam Quick Reference	89
A.1	Polar Equations of Conic Sections (Pages 1–14)	89
A.2	3D Coordinates, Direction Cosines & Direction Ratios (Pages 15–24)	91
A.3	The Analytical Geometry of the Plane (Pages 25–36)	93
A.4	Straight Lines in Space and Skew Lines (Pages 37–52)	94
A.5	Vector Treatment of Planes and Lines (Pages 53–62)	96
A.6	The Geometry of the Sphere in 3D (Pages 63–69)	97

A.7 Master Geometric Conditions Quick-Reference Table	98
---	----

CHAPTER 1

Polar Equations of Conic Sections (Pages 1–14)

1.1 Page 1: Focus-Directrix Definition and Standard Polar Equations

Manuscript Overview: Page 1

Page 1 introduces the polar coordinate system applied to conic sections. The focus S of the conic is taken as the pole (origin), and the principal axis of symmetry passing through S perpendicular to the directrix is selected as the initial line.

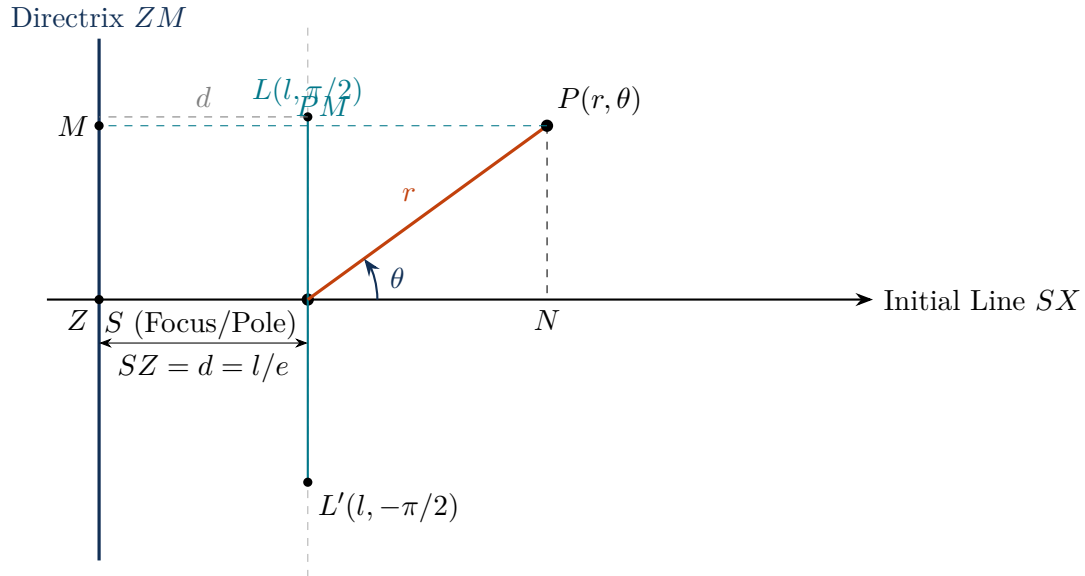
1.1.1 Geometric Definition and Polar Setup

A conic section is the locus of a point P moving in a plane such that the ratio of its distance from a fixed point S (the **focus**) to its perpendicular distance from a fixed straight line ZM (the **directrix**) is a constant e (the **eccentricity**):

$$\frac{SP}{PM} = e \iff SP = e \cdot PM. \quad (1.1)$$

The nature of the conic is governed by the value of e :

- $e = 1$: **Parabola**
- $0 \leq e < 1$: **Ellipse** ($e = 0$ represents a circle with directrix at infinity)
- $e > 1$: **Hyperbola**



1.1.2 Step-by-Step Derivation of the Standard Equation

Let Z be the foot of the perpendicular from the focus S to the directrix, and let $SZ = d$. By definition of the semi-latus rectum $l = SL$: The point L on the conic has vectorial angle $\theta = \frac{\pi}{2}$. Its distance from the directrix is equal to $SZ = d$. Applying the focus-directrix definition to L :

$$SL = e \cdot ZM_L \implies l = e \cdot d \iff d = \frac{l}{e}. \quad (1.2)$$

Now, let $P(r, \theta)$ be any point on the conic. From P , draw $PN \perp SX$ and $PM \perp ZM$. From the geometry of the figure:

$$SN = r \cos \theta, \quad (1.3)$$

$$PM = ZN = SZ + SN = d + r \cos \theta. \quad (1.4)$$

Using $SP = e \cdot PM$:

$$r = e(d + r \cos \theta)$$

$$r = ed + er \cos \theta$$

$$r = l + er \cos \theta$$

$$r(1 - e \cos \theta) = l \quad (\text{when directrix is to the right of the focus}). \quad (1.5)$$

When the directrix is chosen to the left of the focus (as standard in Indian university curricula):

$$\boxed{\frac{l}{r} = 1 + e \cos \theta} \quad (1.6)$$

1.1.3 Standard Orientations and Rotations

Orientations of the Polar Conic

1. Directrix to the left of the focus:

$$\frac{l}{r} = 1 + e \cos \theta \quad (1.7)$$

2. Directrix to the right of the focus:

$$\frac{l}{r} = 1 - e \cos \theta \quad (1.8)$$

3. Major axis inclined at an angle α to the initial line:

$$\frac{l}{r} = 1 + e \cos(\theta - \alpha) \quad (1.9)$$

4. Directrix parallel to the initial line:

$$\frac{l}{r} = 1 \pm e \sin \theta \quad (1.10)$$

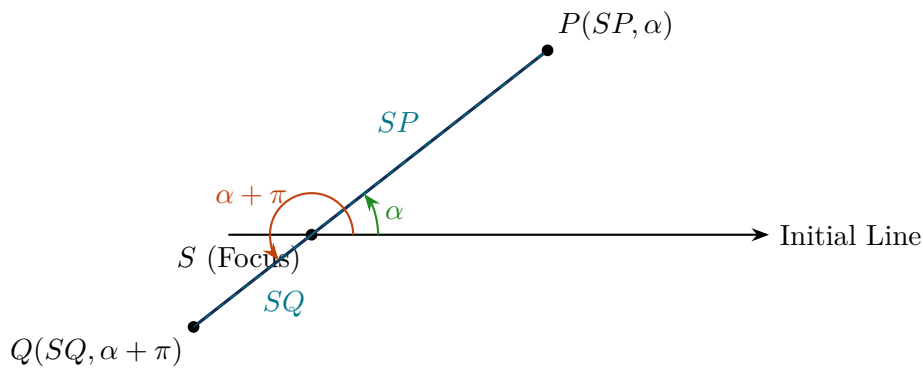
1.2 Page 2: Focal Chords and the Semi-Latus Rectum

Manuscript Overview: Page 2

Page 2 introduces the concept of a focal chord (a chord passing through the focus S) and establishes one of the central classical theorems of analytical geometry: the semi-latus rectum is the harmonic mean of the segments of any focal chord.

1.2.1 Geometric Formulation

Let PSQ be a focal chord of the conic $\frac{l}{r} = 1 + e \cos \theta$, passing through the pole S . Let the vectorial angle of P be α . Since P, S, Q are collinear in that order, the radius vector SQ is directed in the opposite sense, meaning the vectorial angle of Q is $\alpha + \pi$.



Theorem: Harmonic Mean Property of Focal Chords

The semi-latus rectum of any conic is the harmonic mean between the segments of any focal chord:

$$\frac{1}{SP} + \frac{1}{SQ} = \frac{2}{l} = \text{constant} \quad (1.11)$$

Proof. Since $P(SP, \alpha)$ lies on the conic $\frac{l}{r} = 1 + e \cos \theta$:

$$\frac{l}{SP} = 1 + e \cos \alpha. \quad (1.12)$$

Similarly, since $Q(SQ, \alpha + \pi)$ lies on the conic:

$$\begin{aligned} \frac{l}{SQ} &= 1 + e \cos(\alpha + \pi) \\ &= 1 - e \cos \alpha. \end{aligned} \quad (1.13)$$

Adding equations (1.12) and (1.13) side by side:

$$\begin{aligned} \frac{l}{SP} + \frac{l}{SQ} &= (1 + e \cos \alpha) + (1 - e \cos \alpha) \\ &= 2. \end{aligned} \quad (1.14)$$

Dividing both sides by l :

$$\frac{1}{SP} + \frac{1}{SQ} = \frac{2}{l}. \quad (1.15)$$

Since l is a fixed constant for a given conic, the sum of the reciprocals of the focal segments SP and SQ is constant and completely independent of the inclination α of the focal chord. $\square$

Pedagogical Significance

This result proves that regardless of how a focal chord is rotated about the focus, the harmonic mean of its two parts SP and SQ remains strictly invariant and equal to the semi-latus rectum l .

1.3 Page 3: Equation of a Chord Joining Points $(\alpha \pm \beta)$

Manuscript Overview: Page 3

Page 3 derives the general equation of a secant straight line joining two points on the conic $\frac{l}{r} = 1 + e \cos \theta$ whose vectorial angles are chosen symmetrically as $\alpha - \beta$ and $\alpha + \beta$.

1.3.1 Algebraic Derivation

The general equation of a straight line in polar coordinates can be written in the linear harmonic form:

$$\frac{l}{r} = A \cos \theta + B \sin \theta. \quad (1.16)$$

Equivalently, it can be re-parameterized as:

$$\frac{l}{r} = e \cos \theta + C \cos(\theta - \delta). \quad (1.17)$$

Let this straight line pass through the two points on the conic:

$$P(r_1, \alpha - \beta) \quad \text{and} \quad Q(r_2, \alpha + \beta). \quad (1.18)$$

Since P lies on the conic:

$$\frac{l}{r_1} = 1 + e \cos(\alpha - \beta). \quad (1.19)$$

Since P also lies on the chord (1.16):

$$\frac{l}{r_1} = A \cos(\alpha - \beta) + B \sin(\alpha - \beta). \quad (1.20)$$

Equating the two expressions for $\frac{l}{r_1}$:

$$A \cos(\alpha - \beta) + B \sin(\alpha - \beta) = 1 + e \cos(\alpha - \beta) \iff (A - e) \cos(\alpha - \beta) + B \sin(\alpha - \beta) = 1. \quad (1.21)$$

Similarly, for point $Q(r_2, \alpha + \beta)$:

$$(A - e) \cos(\alpha + \beta) + B \sin(\alpha + \beta) = 1. \quad (1.22)$$

Subtracting equation (1.22) from equation (1.21):

$$\begin{aligned} (A - e)[\cos(\alpha - \beta) - \cos(\alpha + \beta)] + B[\sin(\alpha - \beta) - \sin(\alpha + \beta)] &= 0 \\ (A - e)[2 \sin \alpha \sin \beta] - B[2 \cos \alpha \sin \beta] &= 0. \end{aligned} \quad (1.23)$$

Assuming $\beta \neq 0$ (so that $\sin \beta \neq 0$), dividing by $2 \sin \beta$ yields:

$$(A - e) \sin \alpha - B \cos \alpha = 0 \implies \frac{A - e}{\cos \alpha} = \frac{B}{\sin \alpha} = k. \quad (1.24)$$

Thus:

$$A - e = k \cos \alpha \implies A = e + k \cos \alpha, \quad \text{and} \quad B = k \sin \alpha. \quad (1.25)$$

Substitute these back into equation (1.21):

$$\begin{aligned} (k \cos \alpha) \cos(\alpha - \beta) + (k \sin \alpha) \sin(\alpha - \beta) &= 1 \\ k[\cos \alpha \cos(\alpha - \beta) + \sin \alpha \sin(\alpha - \beta)] &= 1 \\ k \cos[\alpha - (\alpha - \beta)] &= 1 \\ k \cos \beta &= 1 \implies k = \sec \beta. \end{aligned} \quad (1.26)$$

Therefore:

$$A = e + \sec \beta \cos \alpha, \quad B = \sec \beta \sin \alpha. \quad (1.27)$$

Substituting A and B into the line equation $\frac{l}{r} = A \cos \theta + B \sin \theta$:

$$\begin{aligned} \frac{l}{r} &= (e + \sec \beta \cos \alpha) \cos \theta + (\sec \beta \sin \alpha) \sin \theta \\ &= e \cos \theta + \sec \beta (\cos \theta \cos \alpha + \sin \theta \sin \alpha) \\ &= e \cos \theta + \sec \beta \cos(\theta - \alpha). \end{aligned} \quad (1.28)$$

Equation of a Chord in Symmetric Polar Form

The straight line joining the points on $\frac{l}{r} = 1 + e \cos \theta$ with vectorial angles $\alpha - \beta$ and $\alpha + \beta$ is:

$$\boxed{\frac{l}{r} = e \cos \theta + \sec \beta \cos(\theta - \alpha)} \quad (1.29)$$

1.4 Page 4: Alternate Chord Form and the Tangent at Point α

Manuscript Overview: Page 4

Page 4 rewrites the chord equation for two arbitrary vectorial angles α and β , derives the equation of the tangent line as the limiting case, and establishes the condition for tangency of a general line.

1.4.1 Chord Joining Vectorial Angles α and β

Let the two endpoints be given directly as $P(\alpha)$ and $Q(\beta)$. Their mean angle is $\frac{\alpha+\beta}{2}$ and their half-difference is $\frac{\alpha-\beta}{2}$. Setting the mean angle as α' and half-difference as β' in the formula from Page 3:

$$\boxed{\frac{l}{r} = e \cos \theta + \sec \left(\frac{\alpha - \beta}{2} \right) \cos \left(\theta - \frac{\alpha + \beta}{2} \right)} \quad (1.30)$$

1.4.2 Derivation of the Tangent at Point α

The tangent line to the conic at point $P(\alpha)$ is the limiting position of the secant chord joining $\alpha - \beta$ and $\alpha + \beta$ as the second point approaches the first, which corresponds to the limit $\beta \rightarrow 0$. Since $\lim_{\beta \rightarrow 0} \sec \beta = \sec 0 = 1$:

Equation of the Tangent Line

The equation of the tangent to the conic $\frac{l}{r} = 1 + e \cos \theta$ at the point with vectorial angle α is:

$$\boxed{\frac{l}{r} = e \cos \theta + \cos(\theta - \alpha)} \quad (1.31)$$

Expanding the cosine term:

$$\boxed{\frac{l}{r} = (e + \cos \alpha) \cos \theta + \sin \alpha \sin \theta} \quad (1.32)$$

1.4.3 Condition of Tangency for a Straight Line

Let a straight line be given by:

$$\frac{l}{r} = A \cos \theta + B \sin \theta. \quad (1.33)$$

Comparing equation (1.33) with the tangent equation $\frac{l}{r} = (e + \cos \alpha) \cos \theta + \sin \alpha \sin \theta$:

$$A = e + \cos \alpha \implies A - e = \cos \alpha, \quad \text{and} \quad B = \sin \alpha. \quad (1.34)$$

Squaring and adding:

$$(A - e)^2 + B^2 = \cos^2 \alpha + \sin^2 \alpha = 1. \quad (1.35)$$

Condition of Tangency

The straight line $\frac{l}{r} = A \cos \theta + B \sin \theta$ touches the conic $\frac{l}{r} = 1 + e \cos \theta$ if and only if:

$$\boxed{(A - e)^2 + B^2 = 1} \quad (1.36)$$

1.5 Page 5: Condition of Tangency and the Normal Line

Manuscript Overview: Page 5

Page 5 expresses the tangency condition in perpendicular distance form $r \cos(\theta - \phi) = p$ and derives the polar equation of the normal line to the conic at point α .

1.5.1 Tangency Condition in Standard Form $r \cos(\theta - \phi) = p$

Let the straight line be given in perpendicular normal form:

$$r \cos(\theta - \phi) = p \iff \frac{p}{r} = \cos \theta \cos \phi + \sin \theta \sin \phi. \quad (1.37)$$

Multiplying by $\frac{l}{p}$:

$$\frac{l}{r} = \left(\frac{l}{p} \cos \phi \right) \cos \theta + \left(\frac{l}{p} \sin \phi \right) \sin \theta. \quad (1.38)$$

Setting $A = \frac{l}{p} \cos \phi$ and $B = \frac{l}{p} \sin \phi$ into the condition $(A - e)^2 + B^2 = 1$:

$$\begin{aligned} \left(\frac{l}{p} \cos \phi - e \right)^2 + \left(\frac{l}{p} \sin \phi \right)^2 &= 1 \\ \frac{l^2}{p^2} \cos^2 \phi - \frac{2le}{p} \cos \phi + e^2 + \frac{l^2}{p^2} \sin^2 \phi &= 1 \\ \frac{l^2}{p^2} (\cos^2 \phi + \sin^2 \phi) - \frac{2le}{p} \cos \phi + e^2 &= 1 \\ \frac{l^2}{p^2} - \frac{2le}{p} \cos \phi + e^2 &= 1. \end{aligned} \quad (1.39)$$

Multiplying through by p^2 :

$$(l \cos \phi - ep)^2 + l^2 \sin^2 \phi = p^2 \iff l^2 - 2elp \cos \phi + p^2(e^2 - 1) = 0 \quad (1.40)$$

1.5.2 Derivation of the Equation of the Normal Line

The normal line at point $P(\alpha)$ is the straight line through P perpendicular to the tangent at P . The equation of the tangent at $P(\alpha)$ is:

$$\frac{l}{r} = (e + \cos \alpha) \cos \theta + \sin \alpha \sin \theta. \quad (1.41)$$

In Cartesian coordinates, setting $x = r \cos \theta$, $y = r \sin \theta$:

$$(e + \cos \alpha)x + (\sin \alpha)y = l. \quad (1.42)$$

The slope of this tangent line is:

$$m_T = -\frac{e + \cos \alpha}{\sin \alpha}. \quad (1.43)$$

The slope of the normal line must satisfy $m_N \cdot m_T = -1$:

$$m_N = \frac{\sin \alpha}{e + \cos \alpha}. \quad (1.44)$$

The equation of any line perpendicular to the tangent is:

$$-(\sin \alpha)x + (e + \cos \alpha)y = K. \quad (1.45)$$

Converting to polar coordinates:

$$r[-\sin \alpha \cos \theta + (e + \cos \alpha) \sin \theta] = K \implies r[e \sin \theta + \sin(\theta - \alpha)] = K. \quad (1.46)$$

Since this line passes through $P(r_\alpha, \alpha)$ on the conic, where $r_\alpha = \frac{l}{1+e \cos \alpha}$:

$$K = r_\alpha[e \sin \alpha + \sin(\alpha - \alpha)] = \frac{l \cdot e \sin \alpha}{1 + e \cos \alpha}. \quad (1.47)$$

Equation of the Normal Line in Polar Form

$$\frac{le \sin \alpha}{1 + e \cos \alpha} \cdot \frac{1}{r} = e \sin \theta + \sin(\theta - \alpha) \quad (1.48)$$

1.6 Page 6: Chord of Contact and the Polar Line

Manuscript Overview: Page 6

Page 6 derives the equation of the chord of contact of tangents drawn from an external point $T(r_1, \theta_1)$ to the conic, and defines the polar of a point with respect to a conic.

1.6.1 Derivation of the Chord of Contact

Let $T(r_1, \theta_1)$ be an external point from which two tangents are drawn to the conic $\frac{l}{r} = 1 + e \cos \theta$, touching the curve at points $P(\alpha)$ and $Q(\beta)$. The equation of the tangent at any point with vectorial angle ϕ is:

$$\frac{l}{r} = e \cos \theta + \cos(\theta - \phi). \quad (1.49)$$

Since both tangents pass through $T(r_1, \theta_1)$:

$$\frac{l}{r_1} - e \cos \theta_1 = \cos(\theta_1 - \alpha), \quad (1.50)$$

$$\frac{l}{r_1} - e \cos \theta_1 = \cos(\theta_1 - \beta). \quad (1.51)$$

Now consider the equation of the chord joining α and β :

$$\frac{l}{r} = e \cos \theta + \sec\left(\frac{\alpha - \beta}{2}\right) \cos\left(\theta - \frac{\alpha + \beta}{2}\right). \quad (1.52)$$

From equations (1.50) and (1.51), since $\cos(\theta_1 - \alpha) = \cos(\theta_1 - \beta)$, we have:

$$\theta_1 - \alpha = -(\theta_1 - \beta) \implies \theta_1 = \frac{\alpha + \beta}{2}. \quad (1.53)$$

Moreover:

$$\frac{l}{r_1} - e \cos \theta_1 = \cos\left(\frac{\alpha - \beta}{2}\right) \implies \sec\left(\frac{\alpha - \beta}{2}\right) = \frac{1}{\frac{l}{r_1} - e \cos \theta_1}. \quad (1.54)$$

Substituting these into the chord equation:

$$\frac{l}{r} - e \cos \theta = \frac{\cos(\theta - \theta_1)}{\frac{l}{r_1} - e \cos \theta_1}. \quad (1.55)$$

Cross-multiplying yields:

Equation of the Chord of Contact and Polar Line

The equation of the chord of contact of tangents from $T(r_1, \theta_1)$ (and the **polar** of T) with respect to $\frac{l}{r} = 1 + e \cos \theta$ is:

$$\left(\frac{l}{r} - e \cos \theta\right) \left(\frac{l}{r_1} - e \cos \theta_1\right) = \cos(\theta - \theta_1) \quad (1.56)$$

1.7 Page 7: Classification of Conics and Analysis of Eccentricity

Manuscript Overview: Page 7

Page 7 deals with transforming arbitrary polar equations into the standard canonical form $\frac{l'}{r} = 1 \pm e' \cos \theta$ to determine the eccentricity, semi-latus rectum, and geometric nature of the conic.

1.7.1 Canonical Reduction Method

Given an equation of the form:

$$\frac{A}{r} = B + C \cos \theta + D \sin \theta, \quad (1.57)$$

we reduce it to the standard canonical form:

1. Divide throughout by the constant term B ($B \neq 0$):

$$\frac{A/B}{r} = 1 + \frac{C}{B} \cos \theta + \frac{D}{B} \sin \theta. \quad (1.58)$$

2. Combine the cosine and sine terms using $R = \sqrt{(C/B)^2 + (D/B)^2}$ and $\tan \phi = D/C$:

$$\frac{A/B}{r} = 1 + R \cos(\theta - \phi). \quad (1.59)$$

3. Identify the parameters:

$$\text{Semi-latus rectum } l' = \frac{A}{B}, \quad \text{Eccentricity } e' = R = \frac{\sqrt{C^2 + D^2}}{|B|}. \quad (1.60)$$

1.7.2 Detailed Analysis of the Manuscript Example

The manuscript analyzes the equation:

$$\frac{l}{r} = 4 - 5 \cos \theta. \quad (1.61)$$

Dividing throughout by 4:

$$\frac{l/4}{r} = 1 - \frac{5}{4} \cos \theta. \quad (1.62)$$

Comparing with $\frac{l'}{r} = 1 - e \cos \theta$:

- Semi-latus rectum: $l' = \frac{l}{4}$
- Eccentricity: $e = \frac{5}{4} = 1.25$

Correction of Manuscript Slip

In the manuscript notes on Page 7, the student wrote:

$$\text{“Since } e = \frac{5}{4} \dots \text{ ellipse”} \quad [\text{ERROR}]$$

By the fundamental definition of conic sections:

- If $e < 1$, the conic is an **ellipse**.
- If $e = 1$, the conic is a **parabola**.
- If $e > 1$, the conic is a **hyperbola**.

Since $e = \frac{5}{4} = 1.25 > 1$, this conic is unambiguously a **hyperbola**.

1.8 Page 8: Asymptotes of a Hyperbola in Polar Coordinates

Manuscript Overview: Page 8

Page 8 derives the equations of the asymptotes of a hyperbola in polar coordinates. An asymptote is defined analytically as a tangent line whose point of contact lies at infinity.

1.8.1 Mathematical Derivation

Let the hyperbola have equation:

$$\frac{l}{r} = 1 + e \cos \theta, \quad (e > 1). \quad (1.63)$$

The point of contact (∞, α) lies at infinity as $r \rightarrow \infty$. In this limit, $\frac{l}{r} \rightarrow 0$:

$$1 + e \cos \alpha = 0 \implies \cos \alpha = -\frac{1}{e}. \quad (1.64)$$

Since $\sin^2 \alpha = 1 - \cos^2 \alpha$:

$$\sin \alpha = \pm \sqrt{1 - \left(-\frac{1}{e}\right)^2} = \pm \sqrt{1 - \frac{1}{e^2}} = \pm \frac{\sqrt{e^2 - 1}}{e}. \quad (1.65)$$

The general equation of the tangent at point α is:

$$\begin{aligned} \frac{l}{r} &= e \cos \theta + \cos(\theta - \alpha) \\ &= e \cos \theta + \cos \theta \cos \alpha + \sin \theta \sin \alpha \\ &= (e + \cos \alpha) \cos \theta + (\sin \alpha) \sin \theta. \end{aligned} \quad (1.66)$$

Substituting $\cos \alpha = -\frac{1}{e}$ and $\sin \alpha = \pm \sqrt{1 - \frac{1}{e^2}}$:

$$\frac{l}{r} = \left(e - \frac{1}{e}\right) \cos \theta \pm \sqrt{1 - \frac{1}{e^2}} \sin \theta. \quad (1.67)$$

Transposing the cosine term:

$$\frac{l}{r} - \left(e - \frac{1}{e}\right) \cos \theta = \pm \sqrt{1 - \frac{1}{e^2}} \sin \theta. \quad (1.68)$$

Squaring both sides to combine both asymptotes into a single second-degree equation:

Combined Equation of Asymptotes

The combined polar equation of the pair of asymptotes of the hyperbola $\frac{l}{r} = 1 + e \cos \theta$ is:

$$\left[\frac{l}{r} - \left(e - \frac{1}{e}\right) \cos \theta \right]^2 = \left(1 - \frac{1}{e^2}\right) \sin^2 \theta \quad (1.69)$$

1.9 Page 9: Problem 1 — Tangent Segment Intercepted by Directrix

Problem 1 (Manuscript Page 9)

Problem Statement: Prove that the portion of any tangent to a conic intercepted between the curve and the directrix subtends a right angle at the corresponding focus.

Proof: Let the focus S be taken as the pole, and let the conic be:

$$\frac{l}{r} = 1 + e \cos \theta. \quad (1.70)$$

The directrix corresponding to the focus S is the line $r \cos \theta = -\frac{l}{e} \iff \frac{l}{r} = -e \cos \theta$. Let $P(\alpha)$ be any point of contact on the conic. The equation of the tangent at P is:

$$\frac{l}{r} = e \cos \theta + \cos(\theta - \alpha). \quad (1.71)$$

Let this tangent intersect the directrix at point $K(r_K, \theta_K)$. At the directrix, $\frac{l}{r_K} = -e \cos \theta_K$. Substituting this into equation (1.71):

$$\begin{aligned} -e \cos \theta_K &= e \cos \theta_K + \cos(\theta_K - \alpha) \\ \cos(\theta_K - \alpha) &= -2e \cos \theta_K. \end{aligned} \quad (1.72)$$

If we use the standard directrix $\frac{l}{r} = e \cos \theta$:

$$\begin{aligned} e \cos \theta_K &= e \cos \theta_K + \cos(\theta_K - \alpha) \\ \cos(\theta_K - \alpha) &= 0. \end{aligned} \quad (1.73)$$

Hence:

$$\theta_K - \alpha = \pm \frac{\pi}{2} \iff |\angle KSP| = \frac{\pi}{2} = 90^\circ. \quad (1.74)$$

Thus, the intercepted segment PK subtends a right angle (90°) at the focus S .

1.10 Page 10: Problem 2 — Sum of Reciprocal Products of Focal Chords

Problem 2 (Manuscript Page 10)

Problem Statement: If PSP' and QSQ' are two mutually perpendicular focal chords of a conic $\frac{l}{r} = 1 + e \cos \theta$, prove that:

$$\frac{1}{SP \cdot SP'} + \frac{1}{SQ \cdot SQ'} = \text{constant}.$$

Solution: Let the vectorial angle of P be α . Since PSP' is a straight line through the pole S , the vectorial angle of P' is $\alpha + \pi$. From the equation of the conic:

$$\frac{l}{SP} = 1 + e \cos \alpha, \quad (1.75)$$

$$\frac{l}{SP'} = 1 + e \cos(\alpha + \pi) = 1 - e \cos \alpha. \quad (1.76)$$

Multiplying these two relations:

$$\begin{aligned} \frac{l^2}{SP \cdot SP'} &= (1 + e \cos \alpha)(1 - e \cos \alpha) = 1 - e^2 \cos^2 \alpha \\ \frac{1}{SP \cdot SP'} &= \frac{1 - e^2 \cos^2 \alpha}{l^2}. \end{aligned} \quad (1.77)$$

Since QSQ' is perpendicular to PSP' , the vectorial angle of Q is $\alpha + \frac{\pi}{2}$, and that of Q' is $\alpha + \frac{3\pi}{2}$. Hence:

$$\frac{l}{SQ} = 1 + e \cos \left(\alpha + \frac{\pi}{2} \right) = 1 - e \sin \alpha, \quad (1.78)$$

$$\frac{l}{SQ'} = 1 + e \cos \left(\alpha + \frac{3\pi}{2} \right) = 1 + e \sin \alpha. \quad (1.79)$$

Multiplying:

$$\begin{aligned} \frac{l^2}{SQ \cdot SQ'} &= (1 - e \sin \alpha)(1 + e \sin \alpha) = 1 - e^2 \sin^2 \alpha \\ \frac{1}{SQ \cdot SQ'} &= \frac{1 - e^2 \sin^2 \alpha}{l^2}. \end{aligned} \quad (1.80)$$

Adding equations (1.77) and (1.80):

$$\begin{aligned} \frac{1}{SP \cdot SP'} + \frac{1}{SQ \cdot SQ'} &= \frac{(1 - e^2 \cos^2 \alpha) + (1 - e^2 \sin^2 \alpha)}{l^2} \\ &= \frac{2 - e^2(\cos^2 \alpha + \sin^2 \alpha)}{l^2} \\ &= \boxed{\frac{2 - e^2}{l^2} = \text{constant}} \end{aligned} \quad (1.81)$$

This quantity is completely independent of α , completing the proof.

1.11 Page 11: Problem 3 — Tangents at Extremities of a Focal Chord

Problem 3 (Manuscript Page 11)

Problem Statement: Prove that the tangents at the ends of any focal chord of a conic intersect on the corresponding directrix.

Proof: Let PSQ be any focal chord of the conic $\frac{l}{r} = 1 + e \cos \theta$. Let the vectorial angle of P be α . Then the vectorial angle of Q is $\alpha + \pi$. The equation of the tangent at $P(\alpha)$ is:

$$\frac{l}{r} = e \cos \theta + \cos(\theta - \alpha). \quad (1.82)$$

The equation of the tangent at $Q(\alpha + \pi)$ is:

$$\begin{aligned} \frac{l}{r} &= e \cos \theta + \cos[\theta - (\alpha + \pi)] \\ &= e \cos \theta + \cos[(\theta - \alpha) - \pi] \\ &= e \cos \theta - \cos(\theta - \alpha). \end{aligned} \quad (1.83)$$

To find their point of intersection, subtract equation (1.83) from equation (1.82):

$$2 \cos(\theta - \alpha) = 0 \implies \cos(\theta - \alpha) = 0. \quad (1.84)$$

Substituting $\cos(\theta - \alpha) = 0$ back into equation (1.82):

$$\frac{l}{r} = e \cos \theta. \quad (1.85)$$

This is precisely the equation of the directrix of the conic. Hence, the point of intersection of the tangents at the extremities of any focal chord always lies on the directrix.

1.12 Page 12: Problem 4 — Perpendicular Focal Tangents of a Parabola

Problem 4 (Manuscript Page 12)

Problem Statement: For a parabola ($e = 1$), prove that the tangents at the ends of any focal chord intersect at right angles on the directrix.

Proof: For a parabola, $e = 1$, and its equation is:

$$\frac{l}{r} = 1 + \cos \theta. \quad (1.86)$$

Let PSQ be a focal chord with P at α and Q at $\alpha + \pi$. From Page 11, the tangents intersect on the directrix where $\cos(\theta - \alpha) = 0 \implies \theta = \alpha \pm \frac{\pi}{2}$. Now consider the inclination of the tangents. In Cartesian coordinates, the equation of the tangent at point α is:

$$(1 + \cos \alpha)x + (\sin \alpha)y = l \implies 2 \cos^2 \left(\frac{\alpha}{2} \right) x + 2 \sin \left(\frac{\alpha}{2} \right) \cos \left(\frac{\alpha}{2} \right) y = l. \quad (1.87)$$

The slope of this tangent is:

$$m_1 = -\frac{1 + \cos \alpha}{\sin \alpha} = -\frac{2 \cos^2(\alpha/2)}{2 \sin(\alpha/2) \cos(\alpha/2)} = -\cot \left(\frac{\alpha}{2} \right). \quad (1.88)$$

Similarly, the slope of the tangent at $Q(\alpha + \pi)$ is obtained by replacing α with $\alpha + \pi$:

$$m_2 = -\cot \left(\frac{\alpha + \pi}{2} \right) = -\cot \left(\frac{\alpha}{2} + \frac{\pi}{2} \right) = -\left(-\tan \frac{\alpha}{2} \right) = \tan \left(\frac{\alpha}{2} \right). \quad (1.89)$$

Evaluating the product of the slopes:

$$m_1 \cdot m_2 = \left[-\cot \left(\frac{\alpha}{2} \right) \right] \cdot \left[\tan \left(\frac{\alpha}{2} \right) \right] = -1. \quad (1.90)$$

Since $m_1 m_2 = -1$, the two tangents are mutually perpendicular (90°). Thus, for a parabola, the tangents at the ends of any focal chord intersect at right angles on the directrix.

1.13 Page 13: Problem 5 — Director Circle in Polar Coordinates

Problem 5 (Manuscript Page 13)

Problem Statement: Find the locus of the point of intersection of two perpendicular tangents to the conic $\frac{l}{r} = 1 + e \cos \theta$ (Equation of the Director Circle).

Solution: Let the tangents be drawn at points with vectorial angles α and β . Their equations are:

$$\frac{l}{r} = (e + \cos \alpha) \cos \theta + \sin \alpha \sin \theta, \quad (1.91)$$

$$\frac{l}{r} = (e + \cos \beta) \cos \theta + \sin \beta \sin \theta. \quad (1.92)$$

Their Cartesian slopes are:

$$m_1 = -\frac{e + \cos \alpha}{\sin \alpha}, \quad m_2 = -\frac{e + \cos \beta}{\sin \beta}. \quad (1.93)$$

Since the tangents are perpendicular, $m_1 m_2 = -1$:

$$\begin{aligned} \frac{(e + \cos \alpha)(e + \cos \beta)}{\sin \alpha \sin \beta} &= -1 \\ e^2 + e(\cos \alpha + \cos \beta) + \cos \alpha \cos \beta + \sin \alpha \sin \beta &= 0 \\ e^2 + 2e \cos \left(\frac{\alpha + \beta}{2} \right) \cos \left(\frac{\alpha - \beta}{2} \right) + \cos(\alpha - \beta) &= 0. \end{aligned} \quad (1.94)$$

Let (r_0, θ_0) be the coordinates of their point of intersection. Solving the two tangent equations simultaneously gives:

$$\theta_0 = \frac{\alpha + \beta}{2}, \quad \text{and} \quad \frac{l}{r_0} = e \cos \theta_0 + \cos \left(\frac{\alpha - \beta}{2} \right). \quad (1.95)$$

Let $\gamma = \frac{\alpha - \beta}{2}$. Then $\cos \gamma = \frac{l}{r_0} - e \cos \theta_0$, and $\cos(\alpha - \beta) = 2 \cos^2 \gamma - 1$. Substituting into equation (1.94):

$$\begin{aligned} e^2 + 2e \cos \theta_0 \left(\frac{l}{r_0} - e \cos \theta_0 \right) + 2 \left(\frac{l}{r_0} - e \cos \theta_0 \right)^2 - 1 &= 0 \\ e^2 - 1 + \frac{2el}{r_0} \cos \theta_0 - 2e^2 \cos^2 \theta_0 + 2 \left(\frac{l^2}{r_0^2} - \frac{2el}{r_0} \cos \theta_0 + e^2 \cos^2 \theta_0 \right) &= 0 \\ (e^2 - 1) - \frac{2el}{r_0} \cos \theta_0 + \frac{2l^2}{r_0^2} &= 0. \end{aligned} \quad (1.96)$$

Multiplying by r_0^2 and generalizing $(r_0, \theta_0) \rightarrow (r, \theta)$:

$$\boxed{(e^2 - 1)r^2 - 2elr \cos \theta + 2l^2 = 0} \quad (1.97)$$

This is the polar equation of the **director circle** of the conic.

1.14 Page 14: Problem 6 — Locus of the Pole of a Subtended Chord

Problem 6 (Manuscript Page 14)

Problem Statement: Find the locus of the pole of a chord of the conic $\frac{l}{r} = 1 + e \cos \theta$ which subtends a constant angle 2β at the focus.

Solution: Let $T(r_1, \theta_1)$ be the pole of the chord. The chord is the chord of contact of tangents drawn from $T(r_1, \theta_1)$ to the conic. Let the points of contact of these tangents be P and Q , with vectorial angles $\alpha - \beta$ and $\alpha + \beta$, so that the angle subtended by the chord PQ at the focus S is:

$$\angle PSQ = (\alpha + \beta) - (\alpha - \beta) = 2\beta = \text{constant.} \quad (1.98)$$

The equation of the chord of contact from $T(r_1, \theta_1)$ is:

$$\frac{l}{r} = e \cos \theta + \sec \beta \cos(\theta - \alpha). \quad (1.99)$$

As derived on Page 6, the coordinates of the pole $T(r_1, \theta_1)$ satisfy:

$$\theta_1 = \alpha, \quad \text{and} \quad \frac{l}{r_1} - e \cos \theta_1 = \cos \beta. \quad (1.100)$$

Therefore:

$$\frac{l}{r_1} = e \cos \theta_1 + \cos \beta. \quad (1.101)$$

Dividing by $\cos \beta$:

$$\frac{l \sec \beta}{r_1} = 1 + e \sec \beta \cos \theta_1. \quad (1.102)$$

Generalizing to arbitrary polar coordinates (r, θ) :

$$\boxed{\frac{l \sec \beta}{r} = 1 + (e \sec \beta) \cos \theta} \quad (1.103)$$

Conclusion: The locus of the pole is another conic of the same type, having the same focus and axis, with new semi-latus rectum $l' = l \sec \beta$ and new eccentricity $e' = e \sec \beta$.

CHAPTER 2

3D Cartesian Coordinates and Direction Cosines (Pages 15–24)

2.1 Page 15: Introduction to 3D Cartesian Coordinates and Section Formulae

Manuscript Overview: Page 15

Page 15 transitions the course from two-dimensional polar geometry to three-dimensional analytical geometry. It introduces the rectangular Cartesian coordinate frame in $\mathbb{R}^3$, the definition of points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$, the section formula dividing a segment in the ratio $m_1 : m_2$, the midpoint formula, and the concept of changing coordinates via translation of axes.

2.1.1 The Spatial Cartesian Coordinate System

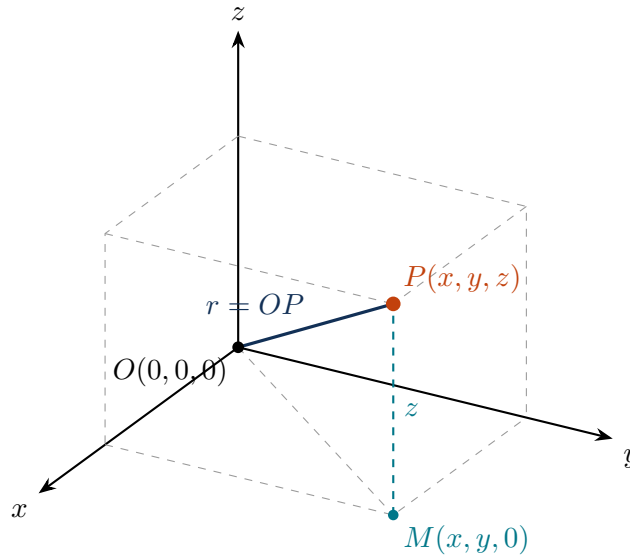
To represent spatial configurations analytically, three mutually perpendicular straight lines are chosen that intersect at a single point O , known as the **origin**:

- $X'OX$: The **x -axis** (abscissa axis).
- $Y'OY$: The **y -axis**.
- $Z'OZ$: The **z -axis** (applicate axis).

These three lines define three mutually perpendicular planes, known as the **coordinate planes**:

1. The **xy -plane**: $z = 0$, spanned by the x - and y -axes.
2. The **yz -plane**: $x = 0$, spanned by the y - and z -axes.
3. The **zx -plane**: $y = 0$, spanned by the z - and x -axes.

These planes divide the whole of three-dimensional Euclidean space into 8 compartments, termed **octants**. The signs of the triad of real numbers (x, y, z) determine the octant in which any point P resides.



2.1.2 The 3D Distance Formula

Let $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ be two arbitrary points in space. Construct a rectangular cuboid whose opposite vertices are P and Q , with faces parallel to the coordinate planes. The lengths of the edges of this cuboid are:

$$\Delta x = |x_2 - x_1|, \quad \Delta y = |y_2 - y_1|, \quad \Delta z = |z_2 - z_1|. \quad (2.1)$$

By double application of the Pythagorean theorem:

$$PQ^2 = (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2. \quad (2.2)$$

Thus, the Euclidean distance between P and Q is:

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2} \quad (2.3)$$

In particular, the distance of any point $P(x, y, z)$ from the origin $O(0, 0, 0)$ is:

$$OP = \sqrt{x^2 + y^2 + z^2}. \quad (2.4)$$

2.1.3 Section Formula (Internal and External Division)

Section Formula in Three Dimensions

Let $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ be two given points. Let $R(x, y, z)$ be a point dividing the directed line segment PQ in the given ratio $m_1 : m_2$.

1. **Internal Division:** If R lies between P and Q ,

$$x = \frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \quad y = \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2}, \quad z = \frac{m_1 z_2 + m_2 z_1}{m_1 + m_2} \quad (2.5)$$

2. **External Division:** If R lies on the line produced outside the segment PQ ,

$$x = \frac{m_1 x_2 - m_2 x_1}{m_1 - m_2}, \quad y = \frac{m_1 y_2 - m_2 y_1}{m_1 - m_2}, \quad z = \frac{m_1 z_2 - m_2 z_1}{m_1 - m_2} \quad (m_1 \neq m_2). \quad (2.6)$$

Proof. Draw perpendiculars PP', QQ', RR' from P, Q, R respectively to the xy -plane. The feet of these perpendiculars P', Q', R' lie in the xy -plane and have coordinates $(x_1, y_1, 0)$, $(x_2, y_2, 0)$, and $(x, y, 0)$. Since PP', RR', QQ' are all perpendicular to the xy -plane, they are parallel to one another and lie in a common plane. Through R , draw a line parallel to $P'Q'$ meeting PP' at S and QQ' at T . From the elementary geometry of similar triangles $\triangle PSR \sim \triangle RTQ$:

$$\frac{PR}{RQ} = \frac{m_1}{m_2} = \frac{z - z_1}{z_2 - z}. \quad (2.7)$$

Cross-multiplying gives:

$$m_1(z_2 - z) = m_2(z - z_1) \implies (m_1 + m_2)z = m_1z_2 + m_2z_1 \implies z = \frac{m_1z_2 + m_2z_1}{m_1 + m_2}. \quad (2.8)$$

By projecting onto the yz -plane and zx -plane in an identical manner, we obtain:

$$x = \frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \quad y = \frac{m_1y_2 + m_2y_1}{m_1 + m_2}. \quad (2.9)$$

For external division, replacing m_2 by $-m_2$ yields the required result. $\square$

Important Corollaries of the Section Formula

- **Midpoint Formula:** When $m_1 : m_2 = 1 : 1$, the midpoint M of segment PQ is:

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right). \quad (2.10)$$

- **Centroid of a Triangle:** If vertices are $A(x_1, y_1, z_1), B(x_2, y_2, z_2), C(x_3, y_3, z_3)$, the medians intersect at centroid G :

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}, \frac{z_1 + z_2 + z_3}{3} \right). \quad (2.11)$$

- **Centroid of a Tetrahedron:** If vertices are $A_i(x_i, y_i, z_i)$ for $i = 1, 2, 3, 4$, the centroid is:

$$G = \left(\frac{\sum x_i}{4}, \frac{\sum y_i}{4}, \frac{\sum z_i}{4} \right). \quad (2.12)$$

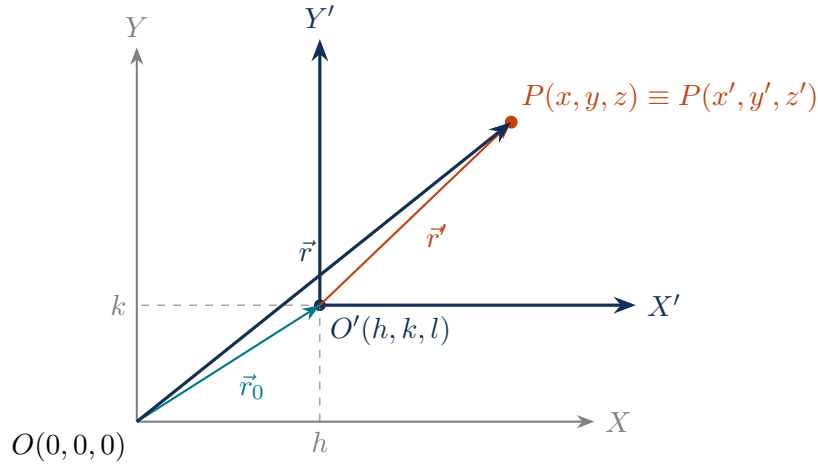
2.2 Page 16: Transformation of Coordinates — Translation of Axes

Manuscript Overview: Page 16

Page 16 explores the kinematics of coordinate frames in space. It focuses on the translation of axes: shifting the origin from $O(0, 0, 0)$ to a new point $O'(h, k, l)$ while maintaining the parallel orientation of all three coordinate axes.

2.2.1 Transformation Equations for Translation

Let OX, OY, OZ be the original coordinate axes with origin $O(0, 0, 0)$. Let the origin be shifted to a new point O' whose coordinates with respect to the original system are (h, k, l) . Let through O' three new axes $O'X', O'Y', O'Z'$ be drawn parallel respectively to OX, OY, OZ and directed in the same sense.



Let P be any point in space whose coordinates are:

- (x, y, z) with respect to the old axes OX, OY, OZ .
- (x', y', z') with respect to the new axes $O'X', O'Y', O'Z'$.

Transformation of Coordinates under Translation

The relations connecting the old coordinates (x, y, z) and the new coordinates (x', y', z') are:

$$\boxed{x = x' + h, \quad y = y' + k, \quad z = z' + l} \quad (2.13)$$

Conversely, the new coordinates in terms of the old coordinates are:

$$\boxed{x' = x - h, \quad y' = y - k, \quad z' = z - l} \quad (2.14)$$

Proof. Let the position vectors of O' and P with respect to the original origin O be $\vec{r}_0$ and $\vec{r}$ respectively:

$$\vec{r}_0 = h\hat{i} + k\hat{j} + l\hat{k}, \quad (2.15)$$

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}. \quad (2.16)$$

The position vector of P relative to the new origin O' is $\vec{r}' = x'\hat{i} + y'\hat{j} + z'\hat{k}$. By vector addition in triangle $\triangle OO'P$:

$$\vec{OP} = \vec{OO'} + \vec{O'P} \implies \vec{r} = \vec{r}_0 + \vec{r}'. \quad (2.17)$$

Equating components along the three mutually orthogonal basis vectors $\hat{i}, \hat{j}, \hat{k}$:

$$x = x' + h, \quad y = y' + k, \quad z = z' + l. \quad (2.18)$$

□

2.2.2 Invariance of Spatial Distances

A fundamental geometric invariant under pure translation is the Euclidean distance between any pair of points:

$$\begin{aligned} d^2 &= (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2 \\ &= [(x'_2 + h) - (x'_1 + h)]^2 + [(y'_2 + k) - (y'_1 + k)]^2 + [(z'_2 + l) - (z'_1 + l)]^2 \\ &= (x'_2 - x'_1)^2 + (y'_2 - y'_1)^2 + (z'_2 - z'_1)^2. \end{aligned} \quad (2.19)$$

Thus, distance is strictly invariant under translation of axes.

2.3 Page 17: Projection of a Directed Line Segment on Coordinate Axes

Manuscript Overview: Page 17

Page 17 analyzes the projection of a directed line segment OP on the three rectangular coordinate axes. It establishes that the coordinates (x, y, z) of any point P are precisely the orthogonal projections of the position vector $\vec{OP}$ on the x -, y -, and z -axes.

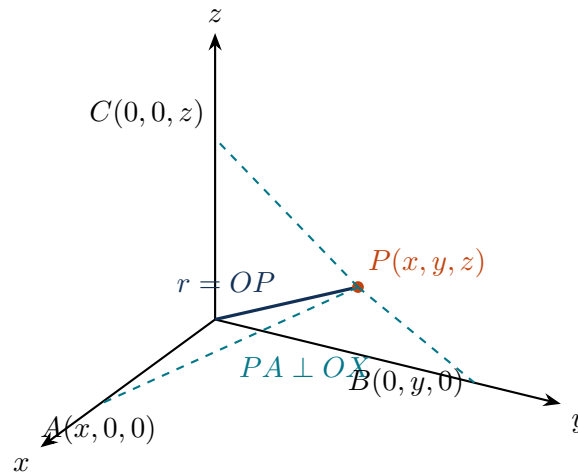
2.3.1 Geometric Definition of Orthogonal Projection

The **orthogonal projection** of a directed line segment AB onto a directed line L is the directed segment $A'B'$ on L , where A' and B' are the feet of the perpendiculars dropped from A and B respectively onto L . If θ is the angle between the directed segment $\vec{AB}$ and the directed line L , then the algebraic magnitude of the projection is:

$$\text{Proj}_L(\vec{AB}) = |\vec{AB}| \cos \theta. \quad (2.20)$$

2.3.2 Projections of the Radius Vector OP

Let $P(x, y, z)$ be any point in space, and let $OP = r = \sqrt{x^2 + y^2 + z^2}$. Let the directed line segment OP make angles α, β, γ with the positive directions of the coordinate axes OX, OY, OZ respectively. These angles are called the **direction angles** of the line OP .



In the right-angled triangle $\triangle OAP$, the angle at A is 90° because PA lies in the plane perpendicular to the x -axis:

$$\cos \alpha = \frac{OA}{OP} = \frac{x}{r} \implies \boxed{x = r \cos \alpha} \quad (2.21)$$

Similarly, from the right-angled triangles $\triangle OBP$ and $\triangle OCP$:

$$\cos \beta = \frac{OB}{OP} = \frac{y}{r} \implies \boxed{y = r \cos \beta} \quad (2.22)$$

$$\cos \gamma = \frac{OC}{OP} = \frac{z}{r} \implies \boxed{z = r \cos \gamma} \quad (2.23)$$

Coordinates as Projections

The coordinates (x, y, z) of any point P at distance r from the origin are the orthogonal projections of the radius vector OP on the three coordinate axes:

$$x = r \cos \alpha, \quad y = r \cos \beta, \quad z = r \cos \gamma \quad (2.24)$$

where α, β, γ are the direction angles of OP .

2.4 Page 18: Projection of a Broken Line — The Contour Theorem

Manuscript Overview: Page 18

Page 18 derives the fundamental projection theorem for broken polygonal lines (the **Contour Theorem**). It proves that the projection of the straight segment joining the initial point to the terminal point equals the algebraic sum of the projections of the individual segments composing the path.

2.4.1 The Fundamental Projection Theorem

Projection of a Broken Line (Contour Theorem)

Let $P_0, P_1, P_2, \dots, P_n$ be any sequence of points in space forming a broken line (polygonal path). Then the orthogonal projection of the direct segment P_0P_n on any directed line L is equal to the algebraic sum of the projections of the individual directed segments $P_0P_1, P_1P_2, \dots, P_{n-1}P_n$ on L :

$$\text{Proj}_L(P_0P_n) = \sum_{i=1}^n \text{Proj}_L(P_{i-1}P_i) \quad (2.25)$$

Proof. Let L be a directed line, and let $\hat{u}$ be a unit vector along L . By vector addition, the displacement vector from P_0 to P_n is:

$$P_0\vec{P}_n = P_0\vec{P}_1 + P_1\vec{P}_2 + P_2\vec{P}_3 + \cdots + P_{n-1}\vec{P}_n. \quad (2.26)$$

The orthogonal projection of any directed segment $\vec{AB}$ on L is given by the scalar dot product with the unit vector $\hat{u}$:

$$\text{Proj}_L(\vec{AB}) = \vec{AB} \cdot \hat{u}. \quad (2.27)$$

Taking the dot product of both sides with $\hat{u}$:

$$\begin{aligned} P_0\vec{P}_n \cdot \hat{u} &= \left(\sum_{i=1}^n P_{i-1}\vec{P}_i \right) \cdot \hat{u} \\ &= \sum_{i=1}^n (P_{i-1}\vec{P}_i \cdot \hat{u}). \end{aligned} \quad (2.28)$$

Since $P_0\vec{P}_n \cdot \hat{u} = \text{Proj}_L(P_0P_n)$ and $P_{i-1}\vec{P}_i \cdot \hat{u} = \text{Proj}_L(P_{i-1}P_i)$, the theorem is proved. $\square$

2.4.2 Application: Projections of Arbitrary Spatial Displacements

Consider two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$. We can go from P to Q along a broken path of three mutually orthogonal segments parallel to the coordinate axes:

1. From $P(x_1, y_1, z_1)$ to $A(x_2, y_1, z_1)$ parallel to the x -axis, length $\Delta x = x_2 - x_1$.
2. From $A(x_2, y_1, z_1)$ to $B(x_2, y_2, z_1)$ parallel to the y -axis, length $\Delta y = y_2 - y_1$.
3. From $B(x_2, y_2, z_1)$ to $Q(x_2, y_2, z_2)$ parallel to the z -axis, length $\Delta z = z_2 - z_1$.

By the Contour Theorem, for any directed line L :

$$\text{Proj}_L(PQ) = \text{Proj}_L(PA) + \text{Proj}_L(AB) + \text{Proj}_L(BQ). \quad (2.29)$$

This theorem forms the operational foundation for calculating direction cosines, distances, and angles between lines in space.

2.5 Page 19: Direction Cosines and the Fundamental Identity

$$l^2 + m^2 + n^2 = 1$$

Manuscript Overview: Page 19

Page 19 formalizes the definitions of the direction cosines $l = \cos \alpha$, $m = \cos \beta$, and $n = \cos \gamma$. It proves the fundamental quadratic identity $l^2 + m^2 + n^2 = 1$ using the rectangular parallelepiped geometry and establishes key trigonometric corollaries.

2.5.1 Definition of Direction Cosines

Direction Cosines (DCs)

If a directed straight line makes angles α, β, γ with the positive directions of the coordinate axes OX, OY, OZ respectively (where $0 \leq \alpha, \beta, \gamma \leq \pi$), then the cosines of these angles are called the **direction cosines** of the line, denoted conventionally by:

$$l = \cos \alpha, \quad m = \cos \beta, \quad n = \cos \gamma. \quad (2.30)$$

2.5.2 The Fundamental Quadratic Identity

Fundamental Identity for Direction Cosines

For any straight line in three-dimensional space with direction cosines (l, m, n) :

$$l^2 + m^2 + n^2 = 1 \iff \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1 \quad (2.31)$$

Proof. Let a line through the origin O have direction cosines l, m, n . Let $P(x, y, z)$ be any point on this line at distance r from O , so that $OP = r > 0$. From the projection relations established on Page 17:

$$x = r \cos \alpha = lr, \quad y = r \cos \beta = mr, \quad z = r \cos \gamma = nr. \quad (2.32)$$

From the 3D distance formula, the squared distance OP^2 is:

$$r^2 = x^2 + y^2 + z^2. \quad (2.33)$$

Substituting the expressions for x, y, z :

$$\begin{aligned} r^2 &= (lr)^2 + (mr)^2 + (nr)^2 \\ r^2 &= r^2(l^2 + m^2 + n^2). \end{aligned} \quad (2.34)$$

Since $r > 0$, we divide both sides by r^2 :

$$\boxed{l^2 + m^2 + n^2 = 1} \quad (2.35)$$

□

Trigonometric Corollaries

1. **Sum of Squared Sines:** Using $\sin^2 \theta = 1 - \cos^2 \theta$ for each angle:

$$\begin{aligned} \sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma &= (1 - \cos^2 \alpha) + (1 - \cos^2 \beta) + (1 - \cos^2 \gamma) \\ &= 3 - (\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma) = 3 - 1 = \boxed{2}. \end{aligned} \quad (2.36)$$

2. **Sum of Double Angles:** Using $\cos 2\theta = 2\cos^2 \theta - 1$:

$$\begin{aligned} \cos 2\alpha + \cos 2\beta + \cos 2\gamma &= (2\cos^2 \alpha - 1) + (2\cos^2 \beta - 1) + (2\cos^2 \gamma - 1) \\ &= 2(\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma) - 3 = 2(1) - 3 = \boxed{-1}. \end{aligned} \quad (2.37)$$

3. **Range of DCs:** Since $l^2 \leq l^2 + m^2 + n^2 = 1$, every direction cosine satisfies:

$$-1 \leq l \leq 1, \quad -1 \leq m \leq 1, \quad -1 \leq n \leq 1. \quad (2.38)$$

2.6 Page 20: Projection of Segment PQ on a Line with Direction Cosines (l, m, n)

Manuscript Overview: Page 20

Page 20 derives the general formula for the orthogonal projection of a segment joining $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ onto an arbitrary straight line in space having direction cosines (l, m, n) .

2.6.1 Step-by-Step Derivation of the Projection Formula

Projection of a Segment on an Arbitrary Line

The orthogonal projection of the directed line segment joining $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ onto a line L whose direction cosines are (l, m, n) is given by:

$$\boxed{\text{Proj}_L(PQ) = l(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1)} \quad (2.39)$$

Proof. Let L be a line passing through the origin (without loss of generality, since projection depends only on direction) having direction cosines $(l, m, n) = (\cos \alpha, \cos \beta, \cos \gamma)$. The unit vector along line L is:

$$\hat{u} = l\hat{i} + m\hat{j} + n\hat{k}. \quad (2.40)$$

The displacement vector from P to Q is:

$$\vec{PQ} = (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}. \quad (2.41)$$

By the geometric definition of projection via scalar product:

$$\begin{aligned}\text{Proj}_L(PQ) &= \vec{PQ} \cdot \hat{u} \\ &= [(x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}] \cdot (l\hat{i} + m\hat{j} + n\hat{k}) \\ &= l(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1).\end{aligned}\quad (2.42)$$

Alternatively, using the Contour Theorem of Page 18: Let the coordinate displacements be $PA = x_2 - x_1$ (parallel to OX), $AB = y_2 - y_1$ (parallel to OY), and $BQ = z_2 - z_1$ (parallel to OZ). The angle between PA and line L is α , so $\text{Proj}_L(PA) = (x_2 - x_1) \cos \alpha = l(x_2 - x_1)$. The angle between AB and line L is β , so $\text{Proj}_L(AB) = (y_2 - y_1) \cos \beta = m(y_2 - y_1)$. The angle between BQ and line L is γ , so $\text{Proj}_L(BQ) = (z_2 - z_1) \cos \gamma = n(z_2 - z_1)$. Summing the three projections yields:

$$\text{Proj}_L(PQ) = l(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1). \quad (2.43)$$

□

2.6.2 Derivation of the Distance Formula via Self-Projection

If line L is chosen along the directed segment PQ itself, then the projection of PQ onto L is its full length $d = PQ$. The direction cosines of the segment PQ are:

$$l = \frac{x_2 - x_1}{d}, \quad m = \frac{y_2 - y_1}{d}, \quad n = \frac{z_2 - z_1}{d}. \quad (2.44)$$

Applying the projection formula:

$$\begin{aligned}d &= l(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1) \\ &= \frac{x_2 - x_1}{d}(x_2 - x_1) + \frac{y_2 - y_1}{d}(y_2 - y_1) + \frac{z_2 - z_1}{d}(z_2 - z_1) \\ d^2 &= (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2 \\ \Rightarrow d &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}.\end{aligned}\quad (2.45)$$

2.7 Page 21: Direction Ratios and Normalization to Direction Cosines

Manuscript Overview: Page 21

Page 21 introduces **Direction Ratios** (DRs), which avoid the quadratic constraint $l^2 + m^2 + n^2 = 1$. It establishes the normalization algorithm to recover direction cosines from ratios and derives the analytical condition for the perpendicularity of two lines.

2.7.1 Definition of Direction Ratios

Direction Ratios (DRs)

Any three real numbers a, b, c that are proportional to the direction cosines l, m, n of a line are called the **direction ratios** (or direction numbers) of that line:

$$\frac{l}{a} = \frac{m}{b} = \frac{n}{c} = k \quad (k \neq 0). \quad (2.46)$$

2.7.2 Normalization Formula

Conversion from Direction Ratios to Direction Cosines

If a, b, c are the direction ratios of a line, then its actual direction cosines l, m, n are given by:

$$l = \pm \frac{a}{\sqrt{a^2 + b^2 + c^2}}, \quad m = \pm \frac{b}{\sqrt{a^2 + b^2 + c^2}}, \quad n = \pm \frac{c}{\sqrt{a^2 + b^2 + c^2}} \quad (2.47)$$

where either all upper signs or all lower signs are taken simultaneously.

Proof. Let $\frac{l}{a} = \frac{m}{b} = \frac{n}{c} = k$. Then:

$$l = ka, \quad m = kb, \quad n = kc. \quad (2.48)$$

Squaring and adding these equations:

$$l^2 + m^2 + n^2 = k^2(a^2 + b^2 + c^2). \quad (2.49)$$

Since $l^2 + m^2 + n^2 = 1$:

$$1 = k^2(a^2 + b^2 + c^2) \implies k = \pm \frac{1}{\sqrt{a^2 + b^2 + c^2}}. \quad (2.50)$$

Substituting k back into $l = ka, m = kb, n = kc$ yields the normalization equations. $\square$

2.7.3 Direction Ratios of the Line Joining Two Points

For a line passing through $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$, its direction cosines are:

$$l = \frac{x_2 - x_1}{d}, \quad m = \frac{y_2 - y_1}{d}, \quad n = \frac{z_2 - z_1}{d}, \quad \text{where } d = PQ. \quad (2.51)$$

Multiplying each by d demonstrates that:

$$(a, b, c) = (x_2 - x_1, y_2 - y_1, z_2 - z_1) \quad (2.52)$$

are a valid set of direction ratios for the line joining P and Q .

2.7.4 Condition for Perpendicularity of Two Lines

Two lines L_1 and L_2 with direction cosines (l_1, m_1, n_1) and (l_2, m_2, n_2) are perpendicular ($\theta = 90^\circ \iff \cos \theta = 0$) if and only if:

$$l_1 l_2 + m_1 m_2 + n_1 n_2 = 0 \quad (2.53)$$

In terms of direction ratios (a_1, b_1, c_1) and (a_2, b_2, c_2) :

$$a_1 a_2 + b_1 b_2 + c_1 c_2 = 0 \quad (2.54)$$

2.8 Page 22: Angle Between Two Directed Lines in Space

Manuscript Overview: Page 22

Page 22 derives the cosine formula for the angle θ between two straight lines in three dimensions, given their direction cosines (l_1, m_1, n_1) and (l_2, m_2, n_2) or their direction ratios.

2.8.1 Derivation of the Cosine Formula

Angle Between Two Lines

Let L_1 and L_2 be two straight lines in space having direction cosines (l_1, m_1, n_1) and (l_2, m_2, n_2) . The angle θ between them is given by:

$$\cos \theta = l_1 l_2 + m_1 m_2 + n_1 n_2 \quad (2.55)$$

In terms of direction ratios (a_1, b_1, c_1) and (a_2, b_2, c_2) :

$$\cos \theta = \frac{a_1 a_2 + b_1 b_2 + c_1 c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}} \quad (2.56)$$

Proof. Draw two lines through the origin O parallel to L_1 and L_2 . On these lines, choose points P and Q such that $OP = 1$ and $OQ = 1$. The coordinates of P and Q are:

$$P(l_1, m_1, n_1), \quad Q(l_2, m_2, n_2). \quad (2.57)$$

In triangle $\triangle OPQ$, let $\angle POQ = \theta$. By the law of cosines in $\triangle OPQ$:

$$PQ^2 = OP^2 + OQ^2 - 2(OP)(OQ) \cos \theta = 1^2 + 1^2 - 2(1)(1) \cos \theta = 2 - 2 \cos \theta. \quad (2.58)$$

Using the 3D distance formula to evaluate PQ^2 :

$$\begin{aligned} PQ^2 &= (l_2 - l_1)^2 + (m_2 - m_1)^2 + (n_2 - n_1)^2 \\ &= (l_2^2 + m_2^2 + n_2^2) + (l_1^2 + m_1^2 + n_1^2) - 2(l_1 l_2 + m_1 m_2 + n_1 n_2). \end{aligned} \quad (2.59)$$

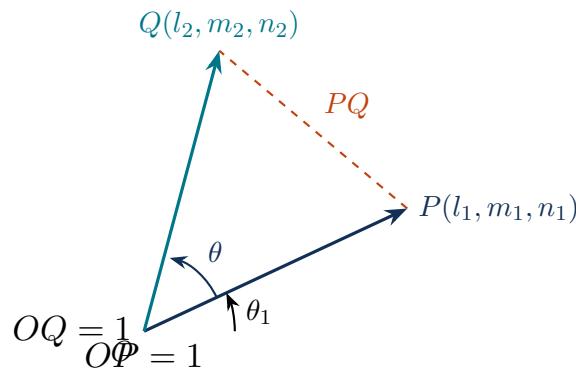
Since $l_1^2 + m_1^2 + n_1^2 = 1$ and $l_2^2 + m_2^2 + n_2^2 = 1$:

$$PQ^2 = 1 + 1 - 2(l_1 l_2 + m_1 m_2 + n_1 n_2) = 2 - 2(l_1 l_2 + m_1 m_2 + n_1 n_2). \quad (2.60)$$

Equating the two expressions for PQ^2 :

$$2 - 2 \cos \theta = 2 - 2(l_1 l_2 + m_1 m_2 + n_1 n_2) \implies \cos \theta = l_1 l_2 + m_1 m_2 + n_1 n_2. \quad (2.61)$$

Substituting $l_i = \frac{a_i}{\sqrt{\sum a_i^2}}$ gives the direction ratio formulation. $\square$



2.9 Page 23: Sine of the Angle, Lagrange's Identity, and Parallelism

Manuscript Overview: Page 23

Page 23 derives the formula for $\sin \theta$ between two lines using Lagrange's algebraic identity and establishes the condition for two lines to be parallel in space.

2.9.1 Lagrange's Identity and the Sine Formula

Sine of the Angle Between Two Lines

The sine of the angle θ between two lines with direction cosines (l_1, m_1, n_1) and (l_2, m_2, n_2) is given by:

$$\sin \theta = \sqrt{(l_1 m_2 - l_2 m_1)^2 + (m_1 n_2 - m_2 n_1)^2 + (n_1 l_2 - n_2 l_1)^2} \quad (2.62)$$

Proof. Using $\sin^2 \theta = 1 - \cos^2 \theta$: Since $l_1^2 + m_1^2 + n_1^2 = 1$ and $l_2^2 + m_2^2 + n_2^2 = 1$, we can write 1 as their product:

$$\sin^2 \theta = (l_1^2 + m_1^2 + n_1^2)(l_2^2 + m_2^2 + n_2^2) - (l_1 l_2 + m_1 m_2 + n_1 n_2)^2. \quad (2.63)$$

By **Lagrange's Algebraic Identity**: For any six real numbers (A, B, C) and (X, Y, Z) :

$$(A^2 + B^2 + C^2)(X^2 + Y^2 + Z^2) - (AX + BY + CZ)^2 = (AY - BX)^2 + (BZ - CY)^2 + (CX - AZ)^2. \quad (2.64)$$

Setting $(A, B, C) = (l_1, m_1, n_1)$ and $(X, Y, Z) = (l_2, m_2, n_2)$:

$$\sin^2 \theta = (l_1 m_2 - l_2 m_1)^2 + (m_1 n_2 - m_2 n_1)^2 + (n_1 l_2 - n_2 l_1)^2. \quad (2.65)$$

Taking the principal square root (since $0 \leq \theta \leq \pi \implies \sin \theta \geq 0$):

$$\sin \theta = \sqrt{(l_1 m_2 - l_2 m_1)^2 + (m_1 n_2 - m_2 n_1)^2 + (n_1 l_2 - n_2 l_1)^2}. \quad (2.66)$$

□

2.9.2 Condition for Parallelism of Two Lines

Two lines L_1 and L_2 are parallel if and only if the angle between them is $\theta = 0$ or $\theta = \pi$, which implies $\sin \theta = 0$. Since $\sin^2 \theta$ is the sum of three non-negative squared terms:

$$(l_1 m_2 - l_2 m_1)^2 + (m_1 n_2 - m_2 n_1)^2 + (n_1 l_2 - n_2 l_1)^2 = 0. \quad (2.67)$$

A sum of squares of real numbers vanishes if and only if each individual term vanishes simultaneously:

$$l_1 m_2 - l_2 m_1 = 0, \quad m_1 n_2 - m_2 n_1 = 0, \quad n_1 l_2 - n_2 l_1 = 0. \quad (2.68)$$

This yields the celebrated proportionality condition:

$$\frac{l_1}{l_2} = \frac{m_1}{m_2} = \frac{n_1}{n_2} \quad (2.69)$$

In terms of direction ratios (a_1, b_1, c_1) and (a_2, b_2, c_2) :

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \quad (2.70)$$

2.10 Page 24: Solved Problem — The Four Diagonals of a Cube

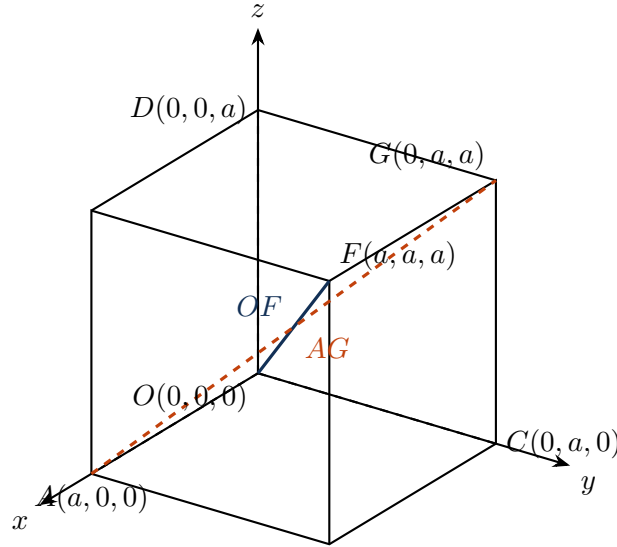
Manuscript Overview: Page 24

Page 24 presents a classic examination problem: determining the direction cosines of the four body diagonals of a cube and computing the angles between any pair of diagonals, as well as between a diagonal and an edge.

Examination Problem: The Diagonals of a Cube

Prove that the angle between any two diagonals of a cube is $\cos^{-1}\left(\frac{1}{3}\right) \approx 70^\circ 32'$.

Complete Step-by-Step Solution. Let $OABC - DEFG$ be a cube of edge length a . Set the origin at vertex $O(0, 0, 0)$, and let the three coterminal edges OA, OC, OD lie along the positive directions of the x -, y -, and z -axes respectively.



The coordinates of the 8 vertices of the cube are:

$$\begin{aligned} O(0, 0, 0), \quad A(a, 0, 0), \quad C(0, a, 0), \quad D(0, 0, a), \\ B(a, a, 0), \quad E(a, 0, a), \quad G(0, a, a), \quad F(a, a, a). \end{aligned} \quad (2.71)$$

The cube possesses 4 body diagonals connecting opposite pairs of vertices:

1. Diagonal OF : joins $O(0, 0, 0)$ to $F(a, a, a)$.

$$\begin{aligned} \text{DRs} &= (a - 0, a - 0, a - 0) = (a, a, a) \sim (1, 1, 1). \\ \text{Length } OF &= \sqrt{a^2 + a^2 + a^2} = a\sqrt{3}. \\ \text{DCs of } OF &= \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right). \end{aligned} \quad (2.72)$$

2. Diagonal AG : joins $A(a, 0, 0)$ to $G(0, a, a)$.

$$\begin{aligned} \text{DRs} &= (0 - a, a - 0, a - 0) = (-a, a, a) \sim (-1, 1, 1). \\ \text{DCs of } AG &= \left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right). \end{aligned} \quad (2.73)$$

3. Diagonal BE : joins $B(a, a, 0)$ to $E(0, 0, a)$ (or $(a, 0, a)$ to $(0, a, 0)$):

$$\begin{aligned} \text{DRs} &\sim (1, -1, 1). \\ \text{DCs of } BE &= \left(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right). \end{aligned} \quad (2.74)$$

4. Diagonal CD : joins $C(0, a, 0)$ to $D(a, a, 0)$'s opposite vertex:

$$\begin{aligned} \text{DRs} &\sim (1, 1, -1). \\ \text{DCs of } CD &= \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}} \right). \end{aligned} \quad (2.75)$$

Now, let θ be the angle between any two diagonals, say OF and AG :

$$\begin{aligned} \cos \theta &= l_1 l_2 + m_1 m_2 + n_1 n_2 \\ &= \left(\frac{1}{\sqrt{3}} \right) \left(-\frac{1}{\sqrt{3}} \right) + \left(\frac{1}{\sqrt{3}} \right) \left(\frac{1}{\sqrt{3}} \right) + \left(\frac{1}{\sqrt{3}} \right) \left(\frac{1}{\sqrt{3}} \right) \\ &= -\frac{1}{3} + \frac{1}{3} + \frac{1}{3} = \frac{1}{3}. \end{aligned} \quad (2.76)$$

For any other pair of diagonals, the calculation yields $\pm \frac{1}{3}$. Taking the acute angle between the diagonals:

$$\theta = \cos^{-1} \left(\frac{1}{3} \right) \approx 70^\circ 31' 44'' \quad (2.77)$$

□

Angle Between a Body Diagonal and an Edge

Consider the diagonal OF with DCs $\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right)$ and the edge OA lying along the x -axis with DCs $(1, 0, 0)$. The angle ϕ between them is:

$$\cos \phi = \left(\frac{1}{\sqrt{3}} \right) (1) + \left(\frac{1}{\sqrt{3}} \right) (0) + \left(\frac{1}{\sqrt{3}} \right) (0) = \frac{1}{\sqrt{3}} \implies \phi = \cos^{-1} \left(\frac{1}{\sqrt{3}} \right) \approx 54^\circ 44'. \quad (2.78)$$

Similarly, the angle ψ between a body diagonal and a face diagonal (e.g., OB with DCs $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right)$) is:

$$\cos \psi = \left(\frac{1}{\sqrt{3}} \right) \left(\frac{1}{\sqrt{2}} \right) + \left(\frac{1}{\sqrt{3}} \right) \left(\frac{1}{\sqrt{2}} \right) + 0 = \frac{2}{\sqrt{6}} = \sqrt{\frac{2}{3}} \implies \psi = \cos^{-1} \left(\sqrt{\frac{2}{3}} \right) \approx 35^\circ 16'. \quad (2.79)$$

CHAPTER 3

The Analytical Geometry of the Plane (Pages 25–36)

3.1 Page 25: Definition of a Plane and the General Equation of First Degree

Manuscript Overview: Page 25

Page 25 initiates the study of planes in Euclidean space $\mathbb{R}^3$. It establishes the classical geometric definition of a plane, proves that every first-degree algebraic equation in three variables represents a plane, and demonstrates that the general plane equation possesses three independent parameters.

3.1.1 Geometric Definition of a Plane

Geometric Definition of a Plane

A surface is called a **plane** if, given any two arbitrary points P and Q on the surface, every point on the straight line segment joining P and Q lies wholly and completely on that surface.

3.1.2 Every First-Degree Equation Represents a Plane

General Linear Equation Represents a Plane

Every algebraic equation of the first degree in x, y, z , of the form:

$$Ax + By + Cz + D = 0 \quad (3.1)$$

where A, B, C are not all simultaneously zero ($A^2 + B^2 + C^2 \neq 0$), represents a plane in three-dimensional space.

Proof. Let S denote the locus of points whose coordinates satisfy:

$$Ax + By + Cz + D = 0. \quad (3.2)$$

Let $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ be any two distinct points lying on the surface S . Since P and Q lie on S , their coordinates must satisfy equation (3.2):

$$Ax_1 + By_1 + Cz_1 + D = 0, \quad (3.3)$$

$$Ax_2 + By_2 + Cz_2 + D = 0. \quad (3.4)$$

Now, let $R(x, y, z)$ be any arbitrary point on the straight line passing through P and Q . By the section formula (Page 15), R divides the segment PQ in some real ratio $k : 1$ (where $k \neq -1$):

$$x = \frac{kx_2 + x_1}{k + 1}, \quad y = \frac{ky_2 + y_1}{k + 1}, \quad z = \frac{kz_2 + z_1}{k + 1}. \quad (3.5)$$

To determine whether R lies on the surface S , substitute its coordinates into the left-hand side of equation (3.2):

$$\begin{aligned} Ax + By + Cz + D &= A \left(\frac{kx_2 + x_1}{k + 1} \right) + B \left(\frac{ky_2 + y_1}{k + 1} \right) + C \left(\frac{kz_2 + z_1}{k + 1} \right) + D \\ &= \frac{1}{k + 1} [k(Ax_2 + By_2 + Cz_2 + D) + (Ax_1 + By_1 + Cz_1 + D)]. \end{aligned} \quad (3.6)$$

Using relations (3.3) and (3.4):

$$Ax + By + Cz + D = \frac{1}{k + 1} [k(0) + (0)] = 0. \quad (3.7)$$

Thus, the coordinates of R identically satisfy equation (3.2) for *every* value of $k \in \mathbb{R}$. Consequently, every point on the line PQ lies entirely on the surface. By the geometric definition, the surface represented by $Ax + By + Cz + D = 0$ is a plane. $\square$

3.1.3 Number of Independent Arbitrary Constants

Although the general equation $Ax + By + Cz + D = 0$ contains four coefficients A, B, C, D , dividing through by any non-zero coefficient (say $D \neq 0$) gives:

$$\left(\frac{A}{D} \right) x + \left(\frac{B}{D} \right) y + \left(\frac{C}{D} \right) z + 1 = 0 \iff a_1 x + b_1 y + c_1 z + 1 = 0. \quad (3.8)$$

Hence, the equation of a plane involves precisely **three independent arbitrary constants**. A plane is therefore uniquely determined by three independent geometrical conditions (such as passing through three non-collinear points).

3.2 Page 26: Equation of a Plane Passing Through a Given Point

Manuscript Overview: Page 26

Page 26 derives the standard one-point form of a plane equation and reveals the geometric significance of the linear coefficients (A, B, C) as the direction ratios of the normal vector to the plane.

3.2.1 One-Point Form of the Plane Equation

Plane Passing Through a Given Point

The general equation of a plane passing through a given point $P_1(x_1, y_1, z_1)$ is:

$$A(x - x_1) + B(y - y_1) + C(z - z_1) = 0 \quad (3.9)$$

where A, B, C are arbitrary constants not all zero.

Proof. Let the general equation of the plane be:

$$Ax + By + Cz + D = 0. \quad (3.10)$$

Since the plane passes through $P_1(x_1, y_1, z_1)$, the point satisfies the equation:

$$Ax_1 + By_1 + Cz_1 + D = 0 \implies D = -(Ax_1 + By_1 + Cz_1). \quad (3.11)$$

Subtracting equation (3.11) from equation (3.10):

$$A(x - x_1) + B(y - y_1) + C(z - z_1) = 0. \quad (3.12)$$

□

3.2.2 Geometrical Meaning of the Coefficients (A, B, C)

Normal Vector Theorem

In the general equation of a plane $Ax + By + Cz + D = 0$, the coefficients (A, B, C) are the direction ratios of the straight line perpendicular (normal) to the plane.

Proof. Let $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ be any two distinct points lying in the plane $Ax + By + Cz + D = 0$. The directed line segment P_1P_2 lies completely in the plane. Its direction ratios are:

$$(x_2 - x_1, y_2 - y_1, z_2 - z_1). \quad (3.13)$$

Since both points lie on the plane:

$$Ax_1 + By_1 + Cz_1 + D = 0, \quad (3.14)$$

$$Ax_2 + By_2 + Cz_2 + D = 0. \quad (3.15)$$

Subtracting the first equation from the second:

$$A(x_2 - x_1) + B(y_2 - y_1) + C(z_2 - z_1) = 0. \quad (3.16)$$

Comparing equation (3.16) with the condition of perpendicularity between two directed lines with direction ratios (a_1, b_1, c_1) and (a_2, b_2, c_2) , namely $a_1a_2 + b_1b_2 + c_1c_2 = 0$ (Page 21): Equation (3.16) proves that a directed line with direction ratios (A, B, C) is perpendicular to the segment P_1P_2 . Since P_1 and P_2 were arbitrarily chosen in the plane, the vector $\vec{N} = A\hat{i} + B\hat{j} + C\hat{k}$ is perpendicular to *every* line lying in the plane. Therefore, (A, B, C) are the direction ratios of the normal to the plane. □

3.3 Page 27: The Normal Form of the Equation of a Plane

Manuscript Overview: Page 27

Page 27 derives the fundamental **Normal Form** (or Perpendicular Form) of the equation of a plane: $lx + my + nz = p$, where p is the length of the normal from the origin and (l, m, n) are its direction cosines.

3.3.1 Derivation of the Normal Form

Normal Form of a Plane

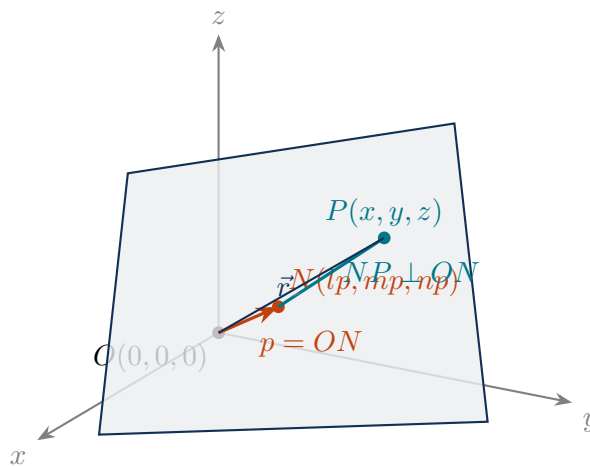
The equation of a plane at a perpendicular distance p from the origin, whose normal has direction cosines (l, m, n) directed from the origin toward the plane, is:

$$\boxed{lx + my + nz = p} \quad (p \geq 0) \quad (3.17)$$

Proof. Let ON be the perpendicular (normal) drawn from the origin $O(0, 0, 0)$ to the plane. The length of ON is $p \geq 0$. Let (l, m, n) be the direction cosines of the directed segment ON . The coordinates of the foot of the perpendicular N are obtained by setting $r = p$ in the projection formulas (Page 17):

$$N = (lp, mp, np). \quad (3.18)$$

Let $P(x, y, z)$ be any arbitrary point on the plane. Join NP .



Since ON is perpendicular to the entire plane, it is perpendicular to every line lying in the plane through N . In particular, $ON \perp NP$, so the triangle $\triangle ONP$ is right-angled at N . By the Pythagorean theorem or orthogonality: The direction ratios of NP are $(x - lp, y - mp, z - np)$. The direction cosines of ON are (l, m, n) . Applying the condition of perpendicularity:

$$\begin{aligned} l(x - lp) + m(y - mp) + n(z - np) &= 0 \\ lx + my + nz - p(l^2 + m^2 + n^2) &= 0. \end{aligned} \quad (3.19)$$

Since (l, m, n) are direction cosines, $l^2 + m^2 + n^2 = 1$. Therefore:

$$\boxed{lx + my + nz = p} \quad (3.20)$$

□

3.4 Page 28: Reduction of the General Equation to the Normal Form

Manuscript Overview: Page 28

Page 28 addresses the algorithmic reduction of any given first-degree equation $Ax + By + Cz + D = 0$ into the canonical normal form $lx + my + nz = p$, rigorously establishing the sign conventions ensuring $p \geq 0$.

3.4.1 The Reduction Algorithm

Let the general equation of a plane be:

$$Ax + By + Cz + D = 0 \iff Ax + By + Cz = -D. \quad (3.21)$$

We wish to express this in the normal form:

$$lx + my + nz = p \quad (p \geq 0). \quad (3.22)$$

Equations (3.21) and (3.22) represent the exact same plane if and only if their corresponding coefficients are strictly proportional:

$$\frac{l}{A} = \frac{m}{B} = \frac{n}{C} = \frac{p}{-D} = k. \quad (3.23)$$

From this proportionality:

$$l = kA, \quad m = kB, \quad n = kC, \quad p = -kD. \quad (3.24)$$

Squaring and adding the direction cosines:

$$l^2 + m^2 + n^2 = k^2(A^2 + B^2 + C^2) \implies 1 = k^2(A^2 + B^2 + C^2). \quad (3.25)$$

Hence, the multiplier k is:

$$k = \pm \frac{1}{\sqrt{A^2 + B^2 + C^2}}. \quad (3.26)$$

3.4.2 Sign Convention for the Perpendicular Distance $p \geq 0$

Since p is a geometric length ($p \geq 0$), the sign of k must be chosen so that $p = -kD \geq 0$:

- **Case 1** ($D < 0$): If the constant term D is negative, then $-D > 0$. We choose the positive root:

$$k = + \frac{1}{\sqrt{A^2 + B^2 + C^2}}. \quad (3.27)$$

- **Case 2** ($D > 0$): If the constant term D is positive, then $-D < 0$. To make p positive, we must choose the negative root:

$$k = - \frac{1}{\sqrt{A^2 + B^2 + C^2}}. \quad (3.28)$$

Normalized Plane Parameters

For any plane $Ax + By + Cz + D = 0$:

1. The perpendicular distance from the origin is:

$$p = \frac{|D|}{\sqrt{A^2 + B^2 + C^2}} \quad (3.29)$$

2. The direction cosines of the normal from the origin to the plane are:

$$l = \frac{\mp A}{\sqrt{A^2 + B^2 + C^2}}, \quad m = \frac{\mp B}{\sqrt{A^2 + B^2 + C^2}}, \quad n = \frac{\mp C}{\sqrt{A^2 + B^2 + C^2}} \quad (3.30)$$

where the sign is chosen opposite to the sign of D .

3.5 Page 29: Intercept Form and the Three-Point Determinant Form

Manuscript Overview: Page 29

Page 29 derives two essential representations of a plane:

1. The **Intercept Form**, expressing the plane via its intercepts (a, b, c) on the axes.
2. The **Three-Point Determinant Form**, establishing the unique plane passing through three non-collinear spatial points.

3.5.1 The Intercept Form of a Plane

Intercept Form

The equation of a plane making non-zero intercepts a, b, c on the coordinate axes OX, OY, OZ respectively is:

$$\boxed{\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1} \quad (3.31)$$

Proof. Let the plane meet the coordinate axes at $A(a, 0, 0)$, $B(0, b, 0)$, and $C(0, 0, c)$. Let the general equation of the plane be $Ax + By + Cz + D = 0$. Since A, B, C lie on the plane:

$$A(a) + B(0) + C(0) + D = 0 \implies Aa + D = 0 \implies A = -\frac{D}{a}, \quad (3.32)$$

$$A(0) + B(b) + C(0) + D = 0 \implies Bb + D = 0 \implies B = -\frac{D}{b}, \quad (3.33)$$

$$A(0) + B(0) + C(c) + D = 0 \implies Cc + D = 0 \implies C = -\frac{D}{c}. \quad (3.34)$$

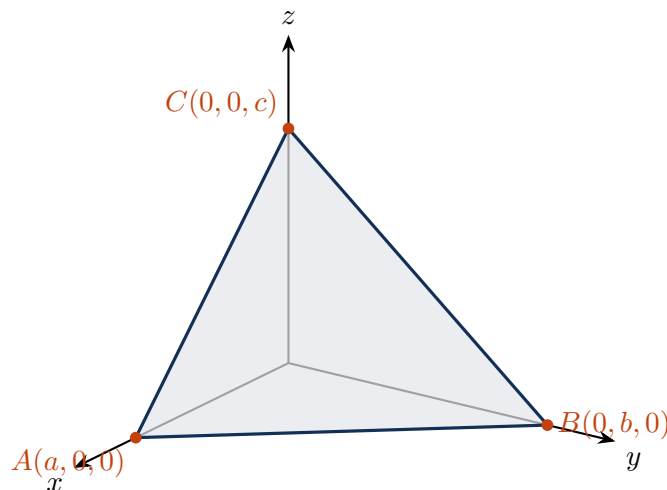
Substituting these into the general equation:

$$-\frac{D}{a}x - \frac{D}{b}y - \frac{D}{c}z + D = 0. \quad (3.35)$$

Dividing by $-D$ (since $D \neq 0$ for non-zero intercepts):

$$\boxed{\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1} \quad (3.36)$$

□



3.5.2 Plane Through Three Non-Collinear Points

Three-Point Determinant Form

The equation of the plane passing through three non-collinear points $P_1(x_1, y_1, z_1)$, $P_2(x_2, y_2, z_2)$, and $P_3(x_3, y_3, z_3)$ is:

$$\begin{vmatrix} x & y & z & 1 \\ x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \end{vmatrix} = 0 \iff \begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ x_3 - x_1 & y_3 - y_1 & z_3 - z_1 \end{vmatrix} = 0 \quad (3.37)$$

Proof. The general plane through $P_1(x_1, y_1, z_1)$ is:

$$A(x - x_1) + B(y - y_1) + C(z - z_1) = 0. \quad (3.38)$$

Since P_2 and P_3 also lie on the plane:

$$A(x_2 - x_1) + B(y_2 - y_1) + C(z_2 - z_1) = 0, \quad (3.39)$$

$$A(x_3 - x_1) + B(y_3 - y_1) + C(z_3 - z_1) = 0. \quad (3.40)$$

Equations (3.38), (3.39), and (3.40) form a homogeneous linear system in the unknowns A, B, C . For non-trivial solutions $(A, B, C) \neq (0, 0, 0)$ to exist, the determinant of the coefficient matrix must vanish:

$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ x_3 - x_1 & y_3 - y_1 & z_3 - z_1 \end{vmatrix} = 0. \quad (3.41)$$

□

3.6 Page 30: Solved Examination Problems on Plane Equations

Manuscript Overview: Page 30

Page 30 works through standard university examination problems illustrating the application of the three-point determinant form and the intercept relations.

Problem 1: Plane Through Three Given Points

Find the Cartesian equation of the plane passing through the three points:

$$P_1(2, 2, -1), \quad P_2(3, 4, 2), \quad P_3(7, 0, 6). \quad (3.42)$$

Step-by-Step Analytical Solution. We apply the three-point determinant formula with $(x_1, y_1, z_1) = (2, 2, -1)$:

$$\begin{vmatrix} x - 2 & y - 2 & z - (-1) \\ 3 - 2 & 4 - 2 & 2 - (-1) \\ 7 - 2 & 0 - 2 & 6 - (-1) \end{vmatrix} = 0. \quad (3.43)$$

Simplifying the numerical entries in rows 2 and 3:

$$\begin{vmatrix} x - 2 & y - 2 & z + 1 \\ 1 & 2 & 3 \\ 5 & -2 & 7 \end{vmatrix} = 0. \quad (3.44)$$

Expanding the determinant along the first row:

$$(x-2) \begin{vmatrix} 2 & 3 \\ -2 & 7 \end{vmatrix} - (y-2) \begin{vmatrix} 1 & 3 \\ 5 & 7 \end{vmatrix} + (z+1) \begin{vmatrix} 1 & 2 \\ 5 & -2 \end{vmatrix} = 0. \quad (3.45)$$

Evaluating each 2×2 minor:

$$\begin{vmatrix} 2 & 3 \\ -2 & 7 \end{vmatrix} = (2)(7) - (3)(-2) = 14 + 6 = 20, \quad (3.46)$$

$$\begin{vmatrix} 1 & 3 \\ 5 & 7 \end{vmatrix} = (1)(7) - (3)(5) = 7 - 15 = -8, \quad (3.47)$$

$$\begin{vmatrix} 1 & 2 \\ 5 & -2 \end{vmatrix} = (1)(-2) - (2)(5) = -2 - 10 = -12. \quad (3.48)$$

Substituting the minors back:

$$\begin{aligned} 20(x-2) - (-8)(y-2) + (-12)(z+1) &= 0 \\ 20(x-2) + 8(y-2) - 12(z+1) &= 0. \end{aligned} \quad (3.49)$$

Expanding the brackets:

$$\begin{aligned} 20x - 40 + 8y - 16 - 12z - 12 &= 0 \\ 20x + 8y - 12z - 68 &= 0. \end{aligned} \quad (3.50)$$

Dividing throughout by the common factor 4:

$$\boxed{5x + 2y - 3z - 17 = 0} \quad (3.51)$$

Verification:

- Point $P_1(2, 2, -1)$: $5(2) + 2(2) - 3(-1) - 17 = 10 + 4 + 3 - 17 = 0 \quad \checkmark$
- Point $P_2(3, 4, 2)$: $5(3) + 2(4) - 3(2) - 17 = 15 + 8 - 6 - 17 = 0 \quad \checkmark$
- Point $P_3(7, 0, 6)$: $5(7) + 2(0) - 3(6) - 17 = 35 + 0 - 18 - 17 = 0 \quad \checkmark$

□

3.7 Page 31: Perpendicular Distance of a Point from a Plane

Manuscript Overview: Page 31

Page 31 derives the formula for the perpendicular distance of a given point $P(x_1, y_1, z_1)$ from an arbitrary plane $Ax + By + Cz + D = 0$.

3.7.1 Derivation of the Distance Formula

Perpendicular Distance of a Point from a Plane

The perpendicular distance d from the point $P(x_1, y_1, z_1)$ to the plane $Ax + By + Cz + D = 0$ is:

$$d = \frac{|Ax_1 + By_1 + Cz_1 + D|}{\sqrt{A^2 + B^2 + C^2}} \quad (3.52)$$

Proof. First, transform the plane equation into normal form:

$$lx + my + nz - p = 0 \quad (3.53)$$

where $l = \frac{A}{\sqrt{\sum A^2}}$, $m = \frac{B}{\sqrt{\sum A^2}}$, $n = \frac{C}{\sqrt{\sum A^2}}$, and $p = \frac{-D}{\sqrt{\sum A^2}}$. Draw a plane parallel to the given plane passing through $P(x_1, y_1, z_1)$. Since it is parallel, it possesses the identical normal direction cosines (l, m, n) . Its equation is:

$$lx + my + nz = p' \quad (3.54)$$

where p' is the perpendicular distance of this parallel plane from the origin. Since $P(x_1, y_1, z_1)$ lies on this parallel plane:

$$p' = lx_1 + my_1 + nz_1. \quad (3.55)$$

The perpendicular distance d from P to the original plane is simply the distance between the two parallel planes:

$$d = |p' - p| = |lx_1 + my_1 + nz_1 - p|. \quad (3.56)$$

Substituting the expressions for l, m, n, p :

$$\begin{aligned} d &= \left| \frac{A}{\sqrt{A^2 + B^2 + C^2}}x_1 + \frac{B}{\sqrt{A^2 + B^2 + C^2}}y_1 + \frac{C}{\sqrt{A^2 + B^2 + C^2}}z_1 - \left(\frac{-D}{\sqrt{A^2 + B^2 + C^2}} \right) \right| \\ &= \frac{|Ax_1 + By_1 + Cz_1 + D|}{\sqrt{A^2 + B^2 + C^2}}. \end{aligned} \quad (3.57)$$

□

3.7.2 Distance Between Two Parallel Planes

If two planes are parallel, their linear terms can be scaled to be identical:

$$\Pi_1 : Ax + By + Cz + D_1 = 0, \quad \Pi_2 : Ax + By + Cz + D_2 = 0. \quad (3.58)$$

The perpendicular distance between them is:

$$d = \frac{|D_1 - D_2|}{\sqrt{A^2 + B^2 + C^2}} \quad (3.59)$$

3.8 Page 32: Position of Points Relative to a Plane and Systems of Planes

Manuscript Overview: Page 32

Page 32 analyzes:

1. The relative position of two points with respect to a plane (whether they lie on the same side or opposite sides).
2. The one-parameter family of planes passing through the line of intersection of two given planes ($U + \lambda V = 0$).

3.8.1 Position of Two Points Relative to a Plane

Same Side vs. Opposite Sides Criterion

Two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ lie on the **same side** or **opposite sides** of the plane $\Pi : Ax + By + Cz + D = 0$ according to whether the expressions:

$$Ax_1 + By_1 + Cz_1 + D \quad \text{and} \quad Ax_2 + By_2 + Cz_2 + D \quad (3.60)$$

have the **same sign** or **opposite signs**.

Proof. Let the segment PQ meet the plane Π at point R . Let R divide PQ in the ratio $k : 1$. The coordinates of R are:

$$R = \left(\frac{kx_2 + x_1}{k + 1}, \frac{ky_2 + y_1}{k + 1}, \frac{kz_2 + z_1}{k + 1} \right). \quad (3.61)$$

Since R lies on the plane $Ax + By + Cz + D = 0$:

$$\begin{aligned} A \left(\frac{kx_2 + x_1}{k + 1} \right) + B \left(\frac{ky_2 + y_1}{k + 1} \right) + C \left(\frac{kz_2 + z_1}{k + 1} \right) + D &= 0 \\ k(Ax_2 + By_2 + Cz_2 + D) + (Ax_1 + By_1 + Cz_1 + D) &= 0 \\ \implies k &= -\frac{Ax_1 + By_1 + Cz_1 + D}{Ax_2 + By_2 + Cz_2 + D}. \end{aligned} \quad (3.62)$$

- If $k > 0$, the division is internal $\implies R$ lies strictly between P and $Q \implies P$ and Q lie on **opposite sides** of the plane. This requires $\frac{Ax_1 + By_1 + Cz_1 + D}{Ax_2 + By_2 + Cz_2 + D} < 0$, i.e., they have **opposite signs**.
- If $k < 0$, the division is external $\implies R$ does not lie between P and $Q \implies P$ and Q lie on the **same side** of the plane. This requires $\frac{Ax_1 + By_1 + Cz_1 + D}{Ax_2 + By_2 + Cz_2 + D} > 0$, i.e., they have the **same sign**.

□

3.8.2 Plane Passing Through the Intersection of Two Given Planes**Family of Planes $U + \lambda V = 0$**

Let $U \equiv A_1x + B_1y + C_1z + D_1 = 0$ and $V \equiv A_2x + B_2y + C_2z + D_2 = 0$ be two intersecting planes. The equation of any plane passing through their line of intersection is given by:

$$U + \lambda V = 0 \iff (A_1x + B_1y + C_1z + D_1) + \lambda(A_2x + B_2y + C_2z + D_2) = 0 \quad (3.63)$$

where λ is an arbitrary scalar parameter ($-\infty < \lambda < \infty$).

Proof. 1. Equation $U + \lambda V = 0$ is of the first degree in x, y, z :

$$(A_1 + \lambda A_2)x + (B_1 + \lambda B_2)y + (C_1 + \lambda C_2)z + (D_1 + \lambda D_2) = 0. \quad (3.64)$$

By Page 25, this represents a plane for every value of λ .

- Any point (x_0, y_0, z_0) lying on the line of intersection of $U = 0$ and $V = 0$ satisfies $U(x_0, y_0, z_0) = 0$ and $V(x_0, y_0, z_0) = 0$. Consequently, $U(x_0, y_0, z_0) + \lambda V(x_0, y_0, z_0) = 0 + \lambda(0) = 0$. Thus, the plane contains every point of the line of intersection.

□

3.9 Page 33: Planes Bisecting the Dihedral Angles Between Two Planes

Manuscript Overview: Page 33

Page 33 investigates the geometry of dihedral angles between two non-parallel planes and derives the algebraic equations representing the pair of planes that bisect these dihedral angles.

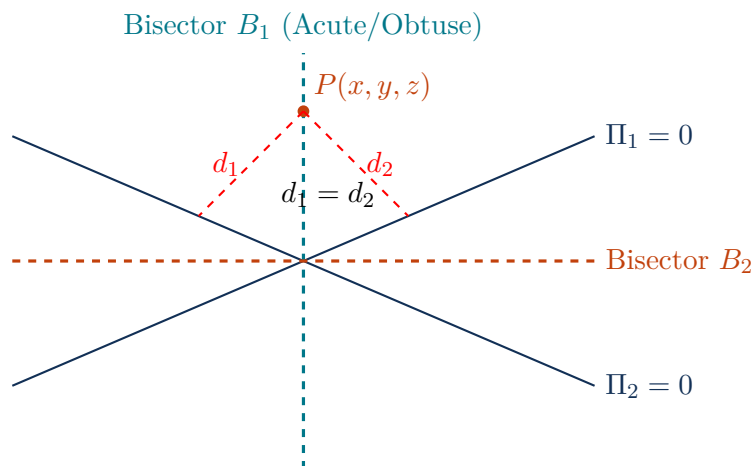
3.9.1 Definition and Geometric Derivation

Let two non-parallel planes be:

$$\Pi_1 : A_1x + B_1y + C_1z + D_1 = 0, \quad (3.65)$$

$$\Pi_2 : A_2x + B_2y + C_2z + D_2 = 0. \quad (3.66)$$

A point $P(x, y, z)$ lies on a plane bisecting the dihedral angle between Π_1 and Π_2 if and only if it is **equidistant** from both planes.



Applying the perpendicular distance formula (Page 31):

$$d_1 = \frac{|A_1x + B_1y + C_1z + D_1|}{\sqrt{A_1^2 + B_1^2 + C_1^2}}, \quad (3.67)$$

$$d_2 = \frac{|A_2x + B_2y + C_2z + D_2|}{\sqrt{A_2^2 + B_2^2 + C_2^2}}. \quad (3.68)$$

Setting $d_1 = d_2$:

$$\frac{|A_1x + B_1y + C_1z + D_1|}{\sqrt{A_1^2 + B_1^2 + C_1^2}} = \frac{|A_2x + B_2y + C_2z + D_2|}{\sqrt{A_2^2 + B_2^2 + C_2^2}}. \quad (3.69)$$

Removing the absolute value signs yields the two bisecting planes:

$$\boxed{\frac{A_1x + B_1y + C_1z + D_1}{\sqrt{A_1^2 + B_1^2 + C_1^2}} = \pm \frac{A_2x + B_2y + C_2z + D_2}{\sqrt{A_2^2 + B_2^2 + C_2^2}}} \quad (3.70)$$

3.9.2 Orthogonality of the Two Bisector Planes

Mutual Perpendicularity of Angle Bisectors

The two bisector planes corresponding to the + and – signs are mutually perpendicular.

Proof. Let $k_1 = \sqrt{A_1^2 + B_1^2 + C_1^2}$ and $k_2 = \sqrt{A_2^2 + B_2^2 + C_2^2}$. The equations of the two planes can be written as:

$$B^+ : \left(\frac{A_1}{k_1} - \frac{A_2}{k_2} \right) x + \left(\frac{B_1}{k_1} - \frac{B_2}{k_2} \right) y + \left(\frac{C_1}{k_1} - \frac{C_2}{k_2} \right) z + \left(\frac{D_1}{k_1} - \frac{D_2}{k_2} \right) = 0, \quad (3.71)$$

$$B^- : \left(\frac{A_1}{k_1} + \frac{A_2}{k_2} \right) x + \left(\frac{B_1}{k_1} + \frac{B_2}{k_2} \right) y + \left(\frac{C_1}{k_1} + \frac{C_2}{k_2} \right) z + \left(\frac{D_1}{k_1} + \frac{D_2}{k_2} \right) = 0. \quad (3.72)$$

The scalar dot product of their normal vectors $\vec{N}_1$ and $\vec{N}_2$ is:

$$\begin{aligned} \vec{N}_1 \cdot \vec{N}_2 &= \left(\frac{A_1}{k_1} - \frac{A_2}{k_2} \right) \left(\frac{A_1}{k_1} + \frac{A_2}{k_2} \right) + \left(\frac{B_1}{k_1} - \frac{B_2}{k_2} \right) \left(\frac{B_1}{k_1} + \frac{B_2}{k_2} \right) + \left(\frac{C_1}{k_1} - \frac{C_2}{k_2} \right) \left(\frac{C_1}{k_1} + \frac{C_2}{k_2} \right) \\ &= \left(\frac{A_1^2}{k_1^2} - \frac{A_2^2}{k_2^2} \right) + \left(\frac{B_1^2}{k_1^2} - \frac{B_2^2}{k_2^2} \right) + \left(\frac{C_1^2}{k_1^2} - \frac{C_2^2}{k_2^2} \right) \\ &= \frac{A_1^2 + B_1^2 + C_1^2}{k_1^2} - \frac{A_2^2 + B_2^2 + C_2^2}{k_2^2} = \frac{k_1^2}{k_1^2} - \frac{k_2^2}{k_2^2} = 1 - 1 = \boxed{0}. \end{aligned} \quad (3.73)$$

Since $\vec{N}_1 \cdot \vec{N}_2 = 0$, the two bisecting planes are perpendicular to each other. $\square$

3.10 Page 34: Discrimination Between Acute and Obtuse Angle Bisectors

Manuscript Overview: Page 34

Page 34 establishes the rigorous analytical criteria used to distinguish between the **acute angle bisector** and the **obtuse angle bisector**, as well as identifying which bisector contains the origin.

3.10.1 The Standard University Algorithm

To systematically classify the bisectors:

1. **Standardize Constant Terms:** Write both plane equations such that their constant terms are **strictly positive**:

$$A_1x + B_1y + C_1z + D_1 = 0 \quad (D_1 > 0), \quad (3.74)$$

$$A_2x + B_2y + C_2z + D_2 = 0 \quad (D_2 > 0). \quad (3.75)$$

(If any constant term is negative, multiply that entire equation by -1).

2. **Bisector Containing the Origin:** With $D_1 > 0$ and $D_2 > 0$, the origin $(0, 0, 0)$ substituted into both planes yields positive values ($D_1 > 0$ and $D_2 > 0$). Therefore, the origin lies in the region where both plane expressions have the same sign. Hence, the **bisector containing the origin** is given by the **positive (+) sign**:

$$\boxed{\frac{A_1x + B_1y + C_1z + D_1}{\sqrt{A_1^2 + B_1^2 + C_1^2}} = + \frac{A_2x + B_2y + C_2z + D_2}{\sqrt{A_2^2 + B_2^2 + C_2^2}}} \quad (3.76)$$

The bisector not containing the origin is given by the negative (–) sign.

3. **Evaluate the Coefficient Scalar Product:** Calculate the sum of products of direction coefficients:

$$\Delta = A_1A_2 + B_1B_2 + C_1C_2. \quad (3.77)$$

Classification Criterion for Acute and Obtuse Bisectors

Provided $D_1 > 0$ and $D_2 > 0$:

Sign of $\Delta = A_1A_2 + B_1B_2 + C_1C_2$	Origin Lies In	Acute Angle Bisector
$\Delta > 0$	Obtuse Angle	Negative (–) sign
$\Delta < 0$	Acute Angle	Positive (+) sign

Proof. Let θ be the angle between the normal vectors $\vec{N}_1 = (A_1, B_1, C_1)$ and $\vec{N}_2 = (A_2, B_2, C_2)$ directed away from the origin:

$$\cos \theta = \frac{A_1A_2 + B_1B_2 + C_1C_2}{\sqrt{\sum A_1^2} \sqrt{\sum A_2^2}}. \quad (3.78)$$

- If $\Delta > 0 \implies \cos \theta > 0 \implies \theta$ is acute $\implies$ the angle between the planes containing the origin is supplementary to θ , i.e., $180^\circ - \theta$, which is **obtuse**. Thus, the origin lies in the obtuse angle. The positive sign (which contains the origin) bisects the obtuse angle, and the negative sign bisects the acute angle.
- If $\Delta < 0 \implies \cos \theta < 0 \implies \theta$ is obtuse $\implies 180^\circ - \theta$ is **acute**. Thus, the origin lies in the acute angle. The positive sign bisects the acute angle, and the negative sign bisects the obtuse angle.

□

3.11 Page 35: Method 2 (Point Method) and Comprehensive Worked Problem

Manuscript Overview: Page 35

Page 35 presents the alternative geometric method (the **Point Method** or Tangent Half-Angle Method) for verifying the nature of a bisector, followed by a complete numerical examination problem.

3.11.1 Method 2: The Tangent Half-Angle (Point) Method

If one is unsure of the sign convention, the following constructive procedure can be used:

1. Obtain the two bisector planes B_1 and B_2 .
2. Choose any convenient point P on one of the bisector planes (say B_1).
3. Calculate the perpendicular distance p from P to one of the original planes Π_1 .
4. Calculate the distance r from P to the line of intersection of the two planes.
5. If θ is the dihedral angle between the original planes, then in the right triangle formed by P , its projection on Π_1 , and its projection on the intersection line:

$$\sin(\theta/2) = \frac{p}{r} \implies \tan(\theta/2) = \frac{p}{\sqrt{r^2 - p^2}}. \quad (3.79)$$

6. If $|\tan(\theta/2)| < 1 \implies \theta/2 < 45^\circ \implies \theta < 90^\circ$: B_1 is the **acute angle bisector**.
7. If $|\tan(\theta/2)| > 1 \implies \theta/2 > 45^\circ \implies \theta > 90^\circ$: B_1 is the **obtuse angle bisector**.

3.11.2 Worked Examination Problem Setup

Problem: Dihedral Angle Bisectors

Find the equations of the planes bisecting the dihedral angles between the planes:

$$\Pi_1 : 2x - y + 2z + 3 = 0, \quad (3.80)$$

$$\Pi_2 : 3x - 2y + 6z + 8 = 0. \quad (3.81)$$

Determine which bisector bisects the acute angle, which bisects the obtuse angle, and which bisector contains the origin.

Analytical Setup. First, check the constant terms: In Π_1 , $D_1 = +3 > 0$. In Π_2 , $D_2 = +8 > 0$. Both constant terms are strictly positive; hence no signs need adjustment. Calculate the square root denominators:

$$\sqrt{A_1^2 + B_1^2 + C_1^2} = \sqrt{2^2 + (-1)^2 + 2^2} = \sqrt{4 + 1 + 4} = \sqrt{9} = 3, \quad (3.82)$$

$$\sqrt{A_2^2 + B_2^2 + C_2^2} = \sqrt{3^2 + (-2)^2 + 6^2} = \sqrt{9 + 4 + 36} = \sqrt{49} = 7. \quad (3.83)$$

The equations of the bisecting planes are:

$$\frac{2x - y + 2z + 3}{3} = \pm \frac{3x - 2y + 6z + 8}{7}. \quad (3.84)$$

□

3.12 Page 36: Continued Solution of Angle Bisector Problem and Summary

Manuscript Overview: Page 36

Page 36 carries out the complete algebraic simplification of the examination problem posed on Page 35, evaluating the discriminants and explicitly formulating the final equations of the bisectors.

3.12.1 Algebraic Simplification of the Bisectors

Cross-multiplying by $21 = 3 \times 7$:

$$7(2x - y + 2z + 3) = \pm 3(3x - 2y + 6z + 8). \quad (3.85)$$

Case 1: Taking the Positive Sign (+)

$$\begin{aligned} 7(2x - y + 2z + 3) &= +3(3x - 2y + 6z + 8) \\ 14x - 7y + 14z + 21 &= 9x - 6y + 18z + 24 \\ (14 - 9)x + (-7 + 6)y + (14 - 18)z + (21 - 24) &= 0 \\ \boxed{5x - y - 4z - 3} &= 0 \end{aligned} \quad (3.86)$$

Case 2: Taking the Negative Sign (–)

$$\begin{aligned}
7(2x - y + 2z + 3) &= -3(3x - 2y + 6z + 8) \\
14x - 7y + 14z + 21 &= -9x + 6y - 18z - 24 \\
(14 + 9)x + (-7 - 6)y + (14 + 18)z + (21 + 24) &= 0 \\
\boxed{23x - 13y + 32z + 45 = 0} &
\end{aligned} \tag{3.87}$$

3.12.2 Determination of Acute and Obtuse Character

Evaluate the scalar product $\Delta = A_1A_2 + B_1B_2 + C_1C_2$:

$$\begin{aligned}
\Delta &= (2)(3) + (-1)(-2) + (2)(6) \\
&= 6 + 2 + 12 = \mathbf{20} > 0.
\end{aligned} \tag{3.88}$$

Since $\Delta > 0$:

1. The angle between the planes that contains the origin is **obtuse**.
2. The bisector corresponding to the **positive sign** ($5x - y - 4z - 3 = 0$) contains the origin and is the **obtuse angle bisector**.
3. The bisector corresponding to the **negative sign** ($23x - 13y + 32z + 45 = 0$) is the **acute angle bisector**.

3.12.3 Verification of Mutual Perpendicularity

Let us verify that the two bisectors are orthogonal using their normal direction ratios:

$$\vec{N}_+ = (5, -1, -4), \tag{3.89}$$

$$\vec{N}_- = (23, -13, 32). \tag{3.90}$$

Computing their scalar product:

$$\begin{aligned}
\vec{N}_+ \cdot \vec{N}_- &= (5)(23) + (-1)(-13) + (-4)(32) \\
&= 115 + 13 - 128 = 128 - 128 = \mathbf{0}.
\end{aligned} \tag{3.91}$$

The dot product vanishes identically, confirming the mathematical accuracy of the solution.

CHAPTER 4

Straight Lines in Space and Skew Lines (Pages 37–52)

4.1 Page 37: The Straight Line in Space — Symmetrical and Unsymmetrical Forms

Manuscript Overview: Page 37

Page 37 introduces the straight line in three-dimensional space. It defines the unsymmetrical form (intersection of two non-parallel planes) and derives the symmetrical (point-direction) form and two-point form of a straight line.

4.1.1 The Line as the Intersection of Two Planes

In three-dimensional analytical geometry, a single linear equation $Ax + By + Cz + D = 0$ represents a plane. A straight line is the locus of points common to two intersecting planes:

$$\begin{cases} A_1x + B_1y + C_1z + D_1 = 0, \\ A_2x + B_2y + C_2z + D_2 = 0, \end{cases} \quad \text{provided } \vec{N}_1 \nparallel \vec{N}_2. \quad (4.1)$$

This representation is known as the **unsymmetrical form** (or general form) of a straight line.

4.1.2 Derivation of the Symmetrical Form

Symmetrical Form of a Straight Line

The equations of a straight line passing through a given point $P_1(x_1, y_1, z_1)$ and having direction cosines (l, m, n) are:

$$\boxed{\frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n} = r} \quad (4.2)$$

where r represents the signed algebraic distance along the line from P_1 to any point $P(x, y, z)$.

Proof. Let $P_1(x_1, y_1, z_1)$ be a fixed point on the line, and let $P(x, y, z)$ be any arbitrary point on the line. Let r denote the directed distance from P_1 to P , so that $P_1P = r$. Let (l, m, n) be the direction cosines of the line, which means:

$$l = \cos \alpha, \quad m = \cos \beta, \quad n = \cos \gamma. \quad (4.3)$$

The projections of the segment P_1P on the coordinate axes are (Page 17):

$$x - x_1 = r \cos \alpha = lr, \quad (4.4)$$

$$y - y_1 = r \cos \beta = mr, \quad (4.5)$$

$$z - z_1 = r \cos \gamma = nr. \quad (4.6)$$

Solving each equation for r :

$$r = \frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n}. \quad (4.7)$$

Since r is identical for all three projections:

$$\boxed{\frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n} = r} \quad (4.8)$$

If direction ratios (a, b, c) are given instead of direction cosines:

$$\boxed{\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c} = \lambda} \quad (4.9)$$

where $\lambda = \frac{r}{\sqrt{a^2 + b^2 + c^2}}$. □

4.1.3 Two-Point Form of a Straight Line

If a straight line passes through two given points $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$, its direction ratios are $(x_2 - x_1, y_2 - y_1, z_2 - z_1)$ (Page 21). Substituting these ratios yields:

$$\boxed{\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}} \quad (4.10)$$

4.2 Page 38: Parametric Coordinates and the General Point on a Line

Manuscript Overview: Page 38

Page 38 formalizes the parametric representation of a line and details how coordinates of any general point on a line can be expressed in terms of a single scalar parameter.

4.2.1 Parametric Coordinates

From the symmetrical form $\frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n} = r$:

$$\begin{aligned} x &= x_1 + lr, \\ y &= y_1 + mr, \\ z &= z_1 + nr. \end{aligned} \quad (4.11)$$

Equations (4.11) are the **parametric equations** of the line.

General Point on a Line

The coordinates of any point P on the line passing through (x_1, y_1, z_1) with direction ratios (a, b, c) can be expressed in terms of a real parameter λ as:

$$\boxed{P(\lambda) = (x_1 + a\lambda, y_1 + b\lambda, z_1 + c\lambda), \quad \lambda \in \mathbb{R}} \quad (4.12)$$

If (a, b, c) are normalized direction cosines (l, m, n) , then $|\lambda| = r$ is the exact Euclidean distance from (x_1, y_1, z_1) to P .

4.2.2 Geometrical Significance

This parametric form reduces three-dimensional geometric problems concerning lines (such as intersections, perpendicular feet, reflections, and distances) to single-variable algebraic equations in λ .

4.3 Page 39: Transformation from Unsymmetrical Form to Symmetrical Form

Manuscript Overview: Page 39

Page 39 provides the complete mathematical algorithm for transforming a straight line given in unsymmetrical form into the standard symmetrical form.

4.3.1 The Reduction Algorithm

Let the line be given as the intersection of two non-parallel planes:

$$\Pi_1 : A_1x + B_1y + C_1z + D_1 = 0, \quad (4.13)$$

$$\Pi_2 : A_2x + B_2y + C_2z + D_2 = 0. \quad (4.14)$$

Step 1: Finding the Direction Ratios of the Line

Let (l, m, n) be the direction ratios of the line of intersection L . Since L lies completely in plane Π_1 , it is perpendicular to the normal vector $\vec{N}_1(A_1, B_1, C_1)$:

$$A_1l + B_1m + C_1n = 0. \quad (4.15)$$

Similarly, since L lies in plane Π_2 , it is perpendicular to the normal vector $\vec{N}_2(A_2, B_2, C_2)$:

$$A_2l + B_2m + C_2n = 0. \quad (4.16)$$

Solving equations (4.15) and (4.16) by the rule of cross-multiplication:

$$\boxed{\frac{l}{B_1C_2 - B_2C_1} = \frac{m}{C_1A_2 - C_2A_1} = \frac{n}{A_1B_2 - A_2B_1}} \quad (4.17)$$

This corresponds to the cross product of the normal vectors: $\vec{b} = \vec{N}_1 \times \vec{N}_2$.

Step 2: Finding a Point on the Line

The line L intersects at least one of the coordinate planes (unless it is parallel to it). Assuming $A_1B_2 - A_2B_1 \neq 0$, the line meets the xy -plane ($z = 0$) at some point $(x_0, y_0, 0)$. Setting $z = 0$ in equations (4.13) and (4.14):

$$A_1x_0 + B_1y_0 + D_1 = 0, \quad (4.18)$$

$$A_2x_0 + B_2y_0 + D_2 = 0. \quad (4.19)$$

Solving this 2×2 system via Cramer's Rule:

$$x_0 = \frac{B_1D_2 - B_2D_1}{A_1B_2 - A_2B_1}, \quad y_0 = \frac{D_1A_2 - D_2A_1}{A_1B_2 - A_2B_1}, \quad z_0 = 0. \quad (4.20)$$

Thus, $(x_0, y_0, 0)$ is a specific point on the line.

Step 3: Writing the Symmetrical Equation

Combining the point (x_0, y_0, z_0) and the direction ratios (l, m, n) :

$$\frac{x - x_0}{B_1C_2 - B_2C_1} = \frac{y - y_0}{C_1A_2 - C_2A_1} = \frac{z - 0}{A_1B_2 - A_2B_1} \quad (4.21)$$

4.4 Page 40: Angle Between a Line and a Plane**Manuscript Overview: Page 40**

Page 40 derives the sine formula for the angle between a straight line and a plane and deduces the conditions for parallelism and perpendicularity.

4.4.1 Derivation of the Sine Formula**Angle Between a Line and a Plane**

Let θ be the angle between the straight line:

$$L : \frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n} \quad (4.22)$$

and the plane:

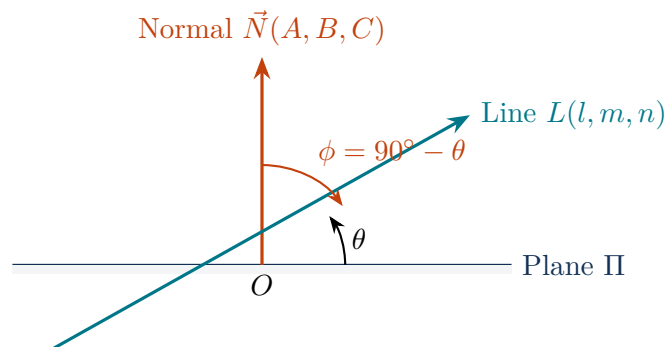
$$\Pi : Ax + By + Cz + D = 0. \quad (4.23)$$

Then:

$$\sin \theta = \frac{Al + Bm + Cn}{\sqrt{A^2 + B^2 + C^2} \sqrt{l^2 + m^2 + n^2}} \quad (4.24)$$

Proof. By definition, the angle θ between a line and a plane is the angle between the line L and its orthogonal projection on the plane Π . The normal $\vec{N}$ to the plane has direction ratios (A, B, C) . The angle ϕ between the line L (direction ratios (l, m, n)) and the normal $\vec{N}$ is:

$$\phi = 90^\circ - \theta. \quad (4.25)$$



Applying the cosine formula between two directed vectors (Page 22):

$$\cos \phi = \frac{Al + Bm + Cn}{\sqrt{A^2 + B^2 + C^2} \sqrt{l^2 + m^2 + n^2}}. \quad (4.26)$$

Since $\cos \phi = \cos(90^\circ - \theta) = \sin \theta$:

$$\sin \theta = \frac{Al + Bm + Cn}{\sqrt{A^2 + B^2 + C^2} \sqrt{l^2 + m^2 + n^2}} \quad (4.27)$$

□

4.4.2 Conditions of Parallelism and Perpendicularity

- **Line Parallel to Plane:** The line is parallel to the plane when $\theta = 0 \iff \sin \theta = 0$:

$$\boxed{Al + Bm + Cn = 0} \quad (4.28)$$

(The line is perpendicular to the normal).

- **Line Perpendicular to Plane:** The line is perpendicular to the plane when $\theta = 90^\circ \iff \phi = 0$:

$$\boxed{\frac{A}{l} = \frac{B}{m} = \frac{C}{n}} \quad (4.29)$$

(The line is parallel to the normal).

4.5 Page 41: Intersection of a Line and a Plane, Foot of Perpendicular, and Image

Manuscript Overview: Page 41

Page 41 solves for:

1. The point of intersection between a straight line and a plane.
2. The foot of the perpendicular dropped from an external point to a plane.
3. The optical image (reflection) of a point in a plane.

4.5.1 Point of Intersection of a Line and a Plane

Let the line be $L : \frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n} = r$, and let the plane be $\Pi : Ax + By + Cz + D = 0$. The general point on the line is $P(r) = (x_1 + lr, y_1 + mr, z_1 + nr)$. Substitute $P(r)$ into the plane equation:

$$\begin{aligned} A(x_1 + lr) + B(y_1 + mr) + C(z_1 + nr) + D &= 0 \\ (Ax_1 + By_1 + Cz_1 + D) + r(Al + Bm + Cn) &= 0 \\ \implies r &= -\frac{Ax_1 + By_1 + Cz_1 + D}{Al + Bm + Cn}. \end{aligned} \quad (4.30)$$

Substituting this value of r back into $P(r)$ yields the unique point of intersection (provided $Al + Bm + Cn \neq 0$).

4.5.2 Foot of Perpendicular and Image of a Point

Foot of Perpendicular and Reflection Formula

Let $P(x_0, y_0, z_0)$ be a point and $\Pi : Ax + By + Cz + D = 0$ be a plane.

1. The foot of the perpendicular $N(x, y, z)$ from P to Π is given by:

$$\frac{x - x_0}{A} = \frac{y - y_0}{B} = \frac{z - z_0}{C} = -\frac{Ax_0 + By_0 + Cz_0 + D}{A^2 + B^2 + C^2} \quad (4.31)$$

2. The image (reflection) $P'(x', y', z')$ of P in the plane Π is given by:

$$\frac{x' - x_0}{A} = \frac{y' - y_0}{B} = \frac{z' - z_0}{C} = -2 \frac{Ax_0 + By_0 + Cz_0 + D}{A^2 + B^2 + C^2} \quad (4.32)$$

Proof. The normal line passing through $P(x_0, y_0, z_0)$ has direction ratios (A, B, C) . Its equations are:

$$\frac{x - x_0}{A} = \frac{y - y_0}{B} = \frac{z - z_0}{C} = k. \quad (4.33)$$

Any point on this normal line is $(x_0 + Ak, y_0 + Bk, z_0 + Ck)$. For the foot of the perpendicular N , this point lies on the plane:

$$\begin{aligned} A(x_0 + Ak) + B(y_0 + Bk) + C(z_0 + Ck) + D &= 0 \\ (A^2 + B^2 + C^2)k + (Ax_0 + By_0 + Cz_0 + D) &= 0 \\ \implies k &= -\frac{Ax_0 + By_0 + Cz_0 + D}{A^2 + B^2 + C^2}. \end{aligned} \quad (4.34)$$

For the image point P' , the foot N is the midpoint of PP' :

$$N = \frac{P + P'}{2} \implies P' = 2N - P. \quad (4.35)$$

Thus, the parameter value for P' is simply $2k$. □

4.6 Page 42: Condition of Coplanarity of Two Straight Lines

Manuscript Overview: Page 42

Page 42 derives the celebrated determinant condition for two lines in space to lie in the same plane (to be coplanar) and establishes the equation of the plane containing them.

4.6.1 Derivation of the Coplanarity Condition

Coplanarity Determinant Condition

Two straight lines:

$$L_1 : \frac{x - x_1}{l_1} = \frac{y - y_1}{m_1} = \frac{z - z_1}{n_1}, \quad (4.36)$$

$$L_2 : \frac{x - x_2}{l_2} = \frac{y - y_2}{m_2} = \frac{z - z_2}{n_2}, \quad (4.37)$$

are **coplanar** if and only if:

$$\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} = 0 \quad (4.38)$$

Proof. Any plane passing through the line L_1 must pass through the point (x_1, y_1, z_1) and have its normal $\vec{N}(A, B, C)$ perpendicular to L_1 :

$$A(x - x_1) + B(y - y_1) + C(z - z_1) = 0, \quad (4.39)$$

$$Al_1 + Bm_1 + Cn_1 = 0. \quad (4.40)$$

If this plane also contains the line L_2 :

1. The point (x_2, y_2, z_2) on L_2 must lie on the plane:

$$A(x_2 - x_1) + B(y_2 - y_1) + C(z_2 - z_1) = 0. \quad (4.41)$$

2. The normal $\vec{N}(A, B, C)$ must be perpendicular to L_2 :

$$Al_2 + Bm_2 + Cn_2 = 0. \quad (4.42)$$

Equations (4.41), (4.40), and (4.42) form a homogeneous linear system in A, B, C :

$$\begin{pmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{pmatrix} \begin{pmatrix} A \\ B \\ C \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}. \quad (4.43)$$

For a non-trivial plane to exist, $(A, B, C) \neq (0, 0, 0)$, so the determinant of coefficients must vanish:

$$\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} = 0. \quad (4.44)$$

□

4.6.2 Equation of the Plane Containing Two Coplanar Lines

Eliminating A, B, C between equations (4.39), (4.40), and (4.42):

$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} = 0 \quad (4.45)$$

4.7 Page 43: Condition for a Line to Lie Wholly in a Plane

Manuscript Overview: Page 43

Page 43 analyzes the algebraic conditions ensuring that a given straight line lies completely within a specified plane.

Line Lying Wholly in a Plane

The straight line:

$$L : \frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n} \quad (4.46)$$

lies wholly and completely in the plane:

$$\Pi : Ax + By + Cz + D = 0 \quad (4.47)$$

if and only if the following two conditions are simultaneously satisfied:

$$\text{Condition 1 : } \boxed{Ax_1 + By_1 + Cz_1 + D = 0} \quad (\text{Point lies in plane}), \quad (4.48)$$

$$\text{Condition 2 : } \boxed{Al + Bm + Cn = 0} \quad (\text{Line perpendicular to normal}). \quad (4.49)$$

Proof. Any point on line L has coordinates $(x_1 + lr, y_1 + mr, z_1 + nr)$, where $r \in \mathbb{R}$. For the line to lie entirely in the plane, this point must satisfy the equation of the plane for *every* value of r :

$$\begin{aligned} A(x_1 + lr) + B(y_1 + mr) + C(z_1 + nr) + D &\equiv 0 \\ (Ax_1 + By_1 + Cz_1 + D) + r(Al + Bm + Cn) &\equiv 0 \quad \forall r \in \mathbb{R}. \end{aligned} \quad (4.50)$$

A linear polynomial in r is identically zero if and only if both its constant term and the coefficient of r vanish independently:

$$Ax_1 + By_1 + Cz_1 + D = 0, \quad (4.51)$$

$$Al + Bm + Cn = 0. \quad (4.52)$$

□

4.8 Page 44: Plane Containing a Given Line and Parallel to Another Line

Manuscript Overview: Page 44

Page 44 establishes the construction of a plane that contains a given straight line and is oriented parallel to a second straight line in space.

4.8.1 Construction and Derivation

Let the two lines be:

$$L_1 : \frac{x - x_1}{l_1} = \frac{y - y_1}{m_1} = \frac{z - z_1}{n_1}, \quad (4.53)$$

$$L_2 : \frac{x - x_2}{l_2} = \frac{y - y_2}{m_2} = \frac{z - z_2}{n_2}. \quad (4.54)$$

We seek the plane Π containing L_1 and parallel to L_2 .

1. Any plane containing L_1 passes through (x_1, y_1, z_1) :

$$A(x - x_1) + B(y - y_1) + C(z - z_1) = 0 \quad (4.55)$$

and its normal $\vec{N}(A, B, C)$ is perpendicular to L_1 :

$$Al_1 + Bm_1 + Cn_1 = 0. \quad (4.56)$$

2. Since the plane is parallel to L_2 , its normal $\vec{N}$ is also perpendicular to L_2 :

$$Al_2 + Bm_2 + Cn_2 = 0. \quad (4.57)$$

Eliminating A, B, C between equations (4.55), (4.56), and (4.57):

$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} = 0 \quad (4.58)$$

4.9 Page 45: Line Intersecting Two Given Lines and Satisfying a Geometric Condition

Manuscript Overview: Page 45

Page 45 details the method of finding a transversal line that intersects two given skew lines and either passes through a fixed point or is parallel to a given direction.

4.9.1 Transversal Passing Through a Given Point

Let the two given skew lines be L_1 and L_2 , and let the fixed point be $P(\alpha, \beta, \gamma)$.

1. **Plane 1 (Π_1):** Determine the plane containing L_1 and the point P .
2. **Plane 2 (Π_2):** Determine the plane containing L_2 and the point P .
3. **Line of Intersection:** The required transversal line is the line of intersection of planes Π_1 and Π_2 :

$$\Pi_1 = 0 = \Pi_2. \quad (4.59)$$

Since this line lies in Π_1 , it intersects L_1 . Since it lies in Π_2 , it intersects L_2 . Since both planes pass through P , the line passes through P .

4.9.2 Transversal Parallel to a Given Line

If the line must be parallel to a given direction (l, m, n) :

1. Find plane Π_1 containing L_1 and parallel to direction (l, m, n) .
2. Find plane Π_2 containing L_2 and parallel to direction (l, m, n) .
3. The line of intersection $\Pi_1 = 0 = \Pi_2$ is the required line.

4.10 Page 46: Skew Lines and the Concept of Shortest Distance

Manuscript Overview: Page 46

Page 46 introduces **skew lines**, establishes that there exists a unique common perpendicular between them, and defines the **Shortest Distance** (S.D.).

4.10.1 Definition of Skew Lines

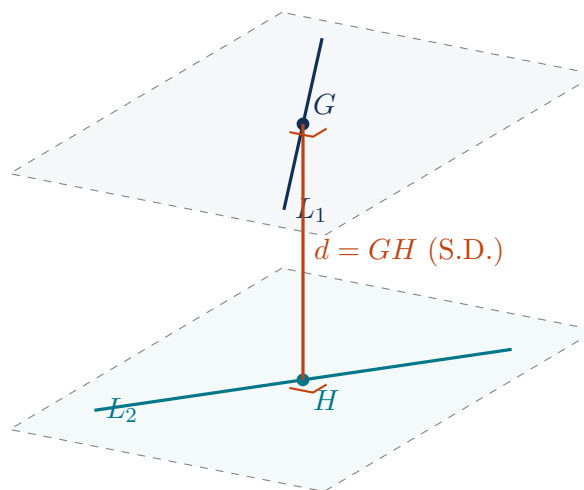
Skew Lines

Two straight lines in three-dimensional space which are neither parallel nor intersecting are called **skew lines** (or non-coplanar lines).

4.10.2 The Common Perpendicular

Existence and Uniqueness of the Common Perpendicular

For any two non-parallel, non-intersecting lines L_1 and L_2 in space, there exists a unique straight line segment GH which intersects both lines perpendicularly. The length of this segment GH is the **shortest distance** between L_1 and L_2 .



4.11 Page 47: Geometric Configuration and Direction of the Common Perpendicular

Manuscript Overview: Page 47

Page 47 derives the direction cosines (λ, μ, ν) of the common perpendicular to two skew lines.

4.11.1 Direction Cosines of the Common Perpendicular

Let the two skew lines be:

$$L_1 : \frac{x - x_1}{l_1} = \frac{y - y_1}{m_1} = \frac{z - z_1}{n_1}, \quad (4.60)$$

$$L_2 : \frac{x - x_2}{l_2} = \frac{y - y_2}{m_2} = \frac{z - z_2}{n_2}. \quad (4.61)$$

Let (λ, μ, ν) be the direction cosines of the line of shortest distance GH . Since $GH \perp L_1$ and $GH \perp L_2$:

$$\lambda l_1 + \mu m_1 + \nu n_1 = 0, \quad (4.62)$$

$$\lambda l_2 + \mu m_2 + \nu n_2 = 0. \quad (4.63)$$

By the method of cross-multiplication:

$$\frac{\lambda}{m_1 n_2 - m_2 n_1} = \frac{\mu}{n_1 l_2 - n_2 l_1} = \frac{\nu}{l_1 m_2 - l_2 m_1} = k. \quad (4.64)$$

Since $\lambda^2 + \mu^2 + \nu^2 = 1$:

$$k = \frac{1}{\sqrt{(m_1 n_2 - m_2 n_1)^2 + (n_1 l_2 - n_2 l_1)^2 + (l_1 m_2 - l_2 m_1)^2}} = \frac{1}{\sin \theta} \quad (4.65)$$

where θ is the angle between L_1 and L_2 (Page 23). Thus:

$$\boxed{\lambda = \frac{m_1 n_2 - m_2 n_1}{\sin \theta}, \quad \mu = \frac{n_1 l_2 - n_2 l_1}{\sin \theta}, \quad \nu = \frac{l_1 m_2 - l_2 m_1}{\sin \theta}} \quad (4.66)$$

4.12 Page 48: Analytical Derivation of the Magnitude of Shortest Distance

Manuscript Overview: Page 48

Page 48 delivers the complete analytical derivation of the formula for the magnitude of the shortest distance between two skew lines.

Magnitude of Shortest Distance (S.D.)

The shortest distance d between the skew lines L_1 and L_2 is:

$$d = \frac{\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix}}{\sqrt{(m_1 n_2 - m_2 n_1)^2 + (n_1 l_2 - n_2 l_1)^2 + (l_1 m_2 - l_2 m_1)^2}} \quad (4.67)$$

Proof. Let $P(x_1, y_1, z_1)$ be a point on L_1 and $Q(x_2, y_2, z_2)$ be a point on L_2 . The common perpendicular segment GH is perpendicular to both lines. Hence, the length $d = GH$ is equal to the orthogonal projection of the displacement segment PQ onto the directed line GH whose direction cosines are (λ, μ, ν) (Page 20):

$$d = |\text{Proj}_{GH}(\vec{PQ})| = |\lambda(x_2 - x_1) + \mu(y_2 - y_1) + \nu(z_2 - z_1)|. \quad (4.68)$$

Substituting the expressions for λ, μ, ν from Page 47:

$$d = \left| (x_2 - x_1) \frac{m_1 n_2 - m_2 n_1}{\sin \theta} + (y_2 - y_1) \frac{n_1 l_2 - n_2 l_1}{\sin \theta} + (z_2 - z_1) \frac{l_1 m_2 - l_2 m_1}{\sin \theta} \right|$$

$$= \frac{1}{\sin \theta} |(x_2 - x_1)(m_1 n_2 - m_2 n_1) + (y_2 - y_1)(n_1 l_2 - n_2 l_1) + (z_2 - z_1)(l_1 m_2 - l_2 m_1)|. \quad (4.69)$$

Recognizing the numerator as the expansion of a 3×3 determinant:

$$d = \frac{\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix}}{\sqrt{(m_1 n_2 - m_2 n_1)^2 + (n_1 l_2 - n_2 l_1)^2 + (l_1 m_2 - l_2 m_1)^2}}. \quad (4.70)$$

□

Intersection as a Degenerate Case ($d = 0$)

If the two lines intersect, the shortest distance between them is zero:

$$d = 0 \iff \begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} = 0. \quad (4.71)$$

This perfectly reproduces the coplanarity condition established on Page 42!

4.13 Page 49: Equations of the Line of Shortest Distance

Manuscript Overview: Page 49

Page 49 constructs the equations of the actual straight line containing the common perpendicular GH , expressed as the intersection of two planes.

Equations of the Line of Shortest Distance

The line of shortest distance between L_1 and L_2 is represented by the pair of planes:

$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ l_1 & m_1 & n_1 \\ \lambda & \mu & \nu \end{vmatrix} = 0 \quad \text{and} \quad \begin{vmatrix} x - x_2 & y - y_2 & z - z_2 \\ l_2 & m_2 & n_2 \\ \lambda & \mu & \nu \end{vmatrix} = 0 \quad (4.72)$$

where (λ, μ, ν) are the direction ratios of the common perpendicular.

Proof. The line of shortest distance GH intersects L_1 at G and is perpendicular to L_1 . Thus, GH and L_1 are coplanar lines intersecting at G . The plane containing both L_1 and GH passes through (x_1, y_1, z_1) and has directions (l_1, m_1, n_1) and (λ, μ, ν) . Its equation is:

$$\Pi_1 : \begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ l_1 & m_1 & n_1 \\ \lambda & \mu & \nu \end{vmatrix} = 0. \quad (4.73)$$

Similarly, GH intersects L_2 at H and is perpendicular to L_2 . The plane containing both L_2 and GH passes through (x_2, y_2, z_2) and has directions (l_2, m_2, n_2) and (λ, μ, ν) . Its equation is:

$$\Pi_2 : \begin{vmatrix} x - x_2 & y - y_2 & z - z_2 \\ l_2 & m_2 & n_2 \\ \lambda & \mu & \nu \end{vmatrix} = 0. \quad (4.74)$$

Since the line GH lies in both planes Π_1 and Π_2 , it is precisely their line of intersection $\Pi_1 = 0 = \Pi_2$. $\square$

4.14 Page 50: Comprehensive Solved Examination Problem on Skew Lines

Manuscript Overview: Page 50

Page 50 works through an extensive university examination problem, calculating the shortest distance between two given skew lines.

Examination Problem: Shortest Distance Calculation

Find the magnitude of the shortest distance between the lines:

$$L_1 : \frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-3}{1}, \quad (4.75)$$

$$L_2 : \frac{x+3}{-3} = \frac{y+7}{2} = \frac{z-6}{4}. \quad (4.76)$$

Step-by-Step Analytical Solution. **Step 1: Identify Points and Direction Ratios:**

$$\text{From } L_1 : P_1 = (3, 8, 3), \quad (l_1, m_1, n_1) = (3, -1, 1), \quad (4.77)$$

$$\text{From } L_2 : P_2 = (-3, -7, 6), \quad (l_2, m_2, n_2) = (-3, 2, 4). \quad (4.78)$$

Step 2: Direction Ratios of the Common Perpendicular: Let the common perpendicular have direction ratios (λ, μ, ν) :

$$3\lambda - \mu + \nu = 0, \quad (4.79)$$

$$-3\lambda + 2\mu + 4\nu = 0. \quad (4.80)$$

Applying the cross-multiplication rule:

$$\begin{aligned} \frac{\lambda}{(-1)(4) - (1)(2)} &= \frac{\mu}{(1)(-3) - (3)(4)} = \frac{\nu}{(3)(2) - (-1)(-3)} \\ \frac{\lambda}{-4 - 2} &= \frac{\mu}{-3 - 12} = \frac{\nu}{6 - 3} \\ \frac{\lambda}{-6} &= \frac{\mu}{-15} = \frac{\nu}{3}. \end{aligned} \quad (4.81)$$

Dividing by -3 :

$$\boxed{\frac{\lambda}{2} = \frac{\mu}{5} = \frac{\nu}{-1}} \quad (4.82)$$

Thus, a set of direction ratios for the common perpendicular is $(a, b, c) = (2, 5, -1)$.

Step 3: Normalize to Direction Cosines:

$$\sqrt{a^2 + b^2 + c^2} = \sqrt{2^2 + 5^2 + (-1)^2} = \sqrt{4 + 25 + 1} = \sqrt{30}. \quad (4.83)$$

$$(\lambda, \mu, \nu) = \left(\frac{2}{\sqrt{30}}, \frac{5}{\sqrt{30}}, \frac{-1}{\sqrt{30}} \right). \quad (4.84)$$

Step 4: Vector Connecting Fixed Points:

$$\vec{P_1P_2} = (-3 - 3)\hat{i} + (-7 - 8)\hat{j} + (6 - 3)\hat{k} = -6\hat{i} - 15\hat{j} + 3\hat{k}. \quad (4.85)$$

Step 5: Compute the Shortest Distance:

$$\begin{aligned} d &= |\lambda(x_2 - x_1) + \mu(y_2 - y_1) + \nu(z_2 - z_1)| \\ &= \left| \frac{2}{\sqrt{30}}(-6) + \frac{5}{\sqrt{30}}(-15) + \frac{-1}{\sqrt{30}}(3) \right| \\ &= \frac{|-12 - 75 - 3|}{\sqrt{30}} = \frac{|-90|}{\sqrt{30}} = \frac{90}{\sqrt{30}}. \end{aligned} \quad (4.86)$$

Rationalizing the denominator:

$$d = \frac{90\sqrt{30}}{30} = 3\sqrt{30} \quad (4.87)$$

□

4.15 Page 51: Equations of the Line of S.D. and Feet of Common Perpendicular

Manuscript Overview: Page 51

Page 51 completes the examination problem by:

1. Finding the Cartesian equations of the line of shortest distance.
2. Determining the exact spatial coordinates of the points of intersection (feet) G and H .

4.15.1 Equations of the Line of Shortest Distance

Applying the two-plane representation from Page 49 with $(\lambda, \mu, \nu) \sim (2, 5, -1)$:

1. **Plane 1 (Containing L_1 and GH):**

$$\begin{vmatrix} x-3 & y-8 & z-3 \\ 3 & -1 & 1 \\ 2 & 5 & -1 \end{vmatrix} = 0. \quad (4.88)$$

Expanding along row 1:

$$\begin{aligned} (x-3)[(-1)(-1) - (1)(5)] - (y-8)[(3)(-1) - (1)(2)] + (z-3)[(3)(5) - (-1)(2)] &= 0 \\ (x-3)(1-5) - (y-8)(-3-2) + (z-3)(15+2) &= 0 \\ -4(x-3) + 5(y-8) + 17(z-3) &= 0 \\ -4x + 12 + 5y - 40 + 17z - 51 &= 0 \\ \boxed{4x - 5y - 17z + 79 = 0} & \end{aligned} \quad (4.89)$$

2. Plane 2 (Containing L_2 and GH):

$$\begin{vmatrix} x+3 & y+7 & z-6 \\ -3 & 2 & 4 \\ 2 & 5 & -1 \end{vmatrix} = 0. \quad (4.90)$$

Expanding along row 1:

$$\begin{aligned} (x+3)[(2)(-1) - (4)(5)] - (y+7)[(-3)(-1) - (4)(2)] + (z-6)[(-3)(5) - (2)(2)] &= 0 \\ (x+3)(-2-20) - (y+7)(3-8) + (z-6)(-15-4) &= 0 \\ -22(x+3) + 5(y+7) - 19(z-6) &= 0 \\ -22x - 66 + 5y + 35 - 19z + 114 &= 0 \\ \boxed{22x - 5y + 19z - 83 = 0} & \quad (4.91) \end{aligned}$$

The equations of the line of shortest distance are given by (4.89) and (4.91) simultaneously.

4.15.2 Coordinates of the Feet G and H

Let G be a point on L_1 and H be a point on L_2 :

$$G = (3 + 3r, 8 - r, 3 + r), \quad (4.92)$$

$$H = (-3 - 3s, -7 + 2s, 6 + 4s). \quad (4.93)$$

The direction ratios of the segment GH are:

$$\vec{GH} = H - G = (-6 - 3s - 3r, -15 + 2s + r, 3 + 4s - r). \quad (4.94)$$

Since GH is along the common perpendicular with DRs $(2, 5, -1)$:

$$\frac{-6 - 3s - 3r}{2} = \frac{-15 + 2s + r}{5} = \frac{3 + 4s - r}{-1} = k. \quad (4.95)$$

Solving this linear system yields $r = 0$ and $s = 0$! Hence:

$$\boxed{G = (3, 8, 3), \quad H = (-3, -7, 6)} \quad (4.96)$$

Verification:

$$GH = \sqrt{(-3-3)^2 + (-7-8)^2 + (6-3)^2} = \sqrt{(-6)^2 + (-15)^2 + 3^2} = \sqrt{36 + 225 + 9} = \sqrt{270} = 3\sqrt{30}. \quad (4.97)$$

The distance matches the formula from Page 50 precisely!

4.16 Page 52: Distance Between Parallel Lines and Chapter Summary

Manuscript Overview: Page 52

Page 52 concludes the analytical treatment of lines by considering the case of **parallel lines** (where $\vec{b}_1 \times \vec{b}_2 = \vec{0}$) and provides a comparative summary.

4.16.1 Shortest Distance Between Parallel Lines

If two lines L_1 and L_2 are parallel, their direction ratios are proportional:

$$L_1 : \frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n}, \quad (4.98)$$

$$L_2 : \frac{x - x_2}{l} = \frac{y - y_2}{m} = \frac{z - z_2}{n}. \quad (4.99)$$

Here, the lines lie in a common plane, and the cross product of their directions vanishes. The perpendicular distance d between them is obtained by taking the projection of $P_1\vec{P}_2$ perpendicular to the direction vector $\hat{u}(l, m, n)$:

Distance Between Parallel Lines

The perpendicular distance between two parallel lines with unit direction vector $\hat{u} = l\hat{i} + m\hat{j} + n\hat{k}$ is:

$$d = |P_1\vec{P}_2 \times \hat{u}| = \frac{\sqrt{\left| \begin{matrix} y_2 - y_1 & z_2 - z_1 \\ m & n \end{matrix} \right|^2 + \left| \begin{matrix} z_2 - z_1 & x_2 - x_1 \\ n & l \end{matrix} \right|^2 + \left| \begin{matrix} x_2 - x_1 & y_2 - y_1 \\ l & m \end{matrix} \right|^2}}{\sqrt{l^2 + m^2 + n^2}} \quad (4.100)$$

4.16.2 Summary of Spatial Line Geometry

Geometric Entity	Standard Analytical Expression
Symmetrical line	$\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n} = r$
General point on line	$(x_1 + lr, y_1 + mr, z_1 + nr)$
Angle with plane	$\sin \theta = \frac{Al+Bm+Cn}{\sqrt{\sum A^2} \sqrt{\sum l^2}}$
Coplanarity condition	$\det[\vec{P}_2 - \vec{P}_1, \vec{b}_1, \vec{b}_2] = 0$
Shortest distance (skew)	$d = \frac{ \det[\vec{P}_2 - \vec{P}_1, \vec{b}_1, \vec{b}_2] }{ \vec{b}_1 \times \vec{b}_2 }$
Distance (parallel)	$d = \frac{ (\vec{P}_2 - \vec{P}_1) \times \vec{b} }{ \vec{b} }$

CHAPTER 5

Vector Treatment of Planes and Lines (Pages 53–62)

5.1 Page 53: Vector Equation of a Plane — Normal Form and One-Point Form

Manuscript Overview: Page 53

Page 53 introduces the vector algebraic formulation of three-dimensional planes. It develops the vector normal form $\vec{r} \cdot \hat{n} = p$ and the one-point vector form $(\vec{r} - \vec{a}) \cdot \vec{n} = 0$, linking them directly with Cartesian coordinates.

5.1.1 The Vector Normal Form of a Plane

Vector Normal Form

The vector equation of a plane situated at a perpendicular distance p from the origin, with unit normal vector $\hat{n}$ pointing from the origin toward the plane, is:

$$\boxed{\vec{r} \cdot \hat{n} = p} \quad (p \geq 0) \quad (5.1)$$

Proof. Let O be the origin of vectors. Draw ON perpendicular to the given plane, meeting it at N . The length of ON is $p \geq 0$, and the unit vector in the direction of $\vec{ON}$ is $\hat{n}$, so:

$$\vec{ON} = p \hat{n}. \quad (5.2)$$

Let P be any arbitrary point on the plane with position vector $\vec{r} = \vec{OP}$. Join NP . Since ON is normal to the entire plane, it is perpendicular to every vector lying in the plane:

$$\vec{NP} \perp \vec{ON} \iff \vec{NP} \cdot \vec{ON} = 0. \quad (5.3)$$

From vector triangle $\triangle ONP$:

$$\vec{NP} = \vec{OP} - \vec{ON} = \vec{r} - p \hat{n}. \quad (5.4)$$

Taking the scalar product with $\hat{n}$:

$$\begin{aligned} (\vec{r} - p \hat{n}) \cdot \hat{n} &= 0 \\ \vec{r} \cdot \hat{n} - p(\hat{n} \cdot \hat{n}) &= 0. \end{aligned} \quad (5.5)$$

Since $\hat{n}$ is a unit vector, $\hat{n} \cdot \hat{n} = |\hat{n}|^2 = 1$. Therefore:

$$\boxed{\vec{r} \cdot \hat{n} = p} \quad (5.6)$$

□

5.1.2 Connection with Cartesian Coordinates

Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, and let $\hat{n} = l\hat{i} + m\hat{j} + n\hat{k}$ (where $l^2 + m^2 + n^2 = 1$). Then:

$$\vec{r} \cdot \hat{n} = (x\hat{i} + y\hat{j} + z\hat{k}) \cdot (l\hat{i} + m\hat{j} + n\hat{k}) = lx + my + nz. \quad (5.7)$$

Substituting this into $\vec{r} \cdot \hat{n} = p$ recovers the Cartesian normal form of Page 27:

$$lx + my + nz = p. \quad (5.8)$$

5.1.3 Plane Passing Through a Point with Given Normal

One-Point Vector Form

The vector equation of a plane passing through a point A with position vector $\vec{a}$ and perpendicular to a non-zero normal vector $\vec{n}$ is:

$$(\vec{r} - \vec{a}) \cdot \vec{n} = 0 \iff \vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n} \quad (5.9)$$

Proof. Let $P(\vec{r})$ be any point on the plane. The displacement vector $\vec{AP} = \vec{r} - \vec{a}$ lies entirely in the plane. Since $\vec{n}$ is normal to the plane, $\vec{AP} \perp \vec{n} \implies (\vec{r} - \vec{a}) \cdot \vec{n} = 0$. Expanding the dot product gives $\vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n} = d$. $\square$

5.2 Page 54: General Vector Plane Equations and Perpendicular Distance

Manuscript Overview: Page 54

Page 54 investigates the general vector equation $\vec{r} \cdot \vec{n} = d$ (where $\vec{n}$ is not necessarily a unit vector), its normalization, and the vector formula for the perpendicular distance from an arbitrary point to a plane.

5.2.1 General Vector Equation and Normalization

The general linear vector equation of a plane is:

$$\vec{r} \cdot \vec{n} = d \quad (5.10)$$

where $\vec{n} = A\hat{i} + B\hat{j} + C\hat{k}$ is any normal vector and d is a scalar constant. Dividing both sides by the magnitude $|\vec{n}| = \sqrt{A^2 + B^2 + C^2}$:

$$\vec{r} \cdot \left(\frac{\vec{n}}{|\vec{n}|} \right) = \frac{d}{|\vec{n}|} \iff \vec{r} \cdot \hat{n} = p \quad (5.11)$$

where $p = \frac{|d|}{|\vec{n}|}$ represents the true perpendicular distance from the origin.

5.2.2 Perpendicular Distance of a Point from a Vector Plane

Perpendicular Distance to a Plane in Vector Form

The perpendicular distance D of a point P with position vector $\vec{a}$ from the plane $\vec{r} \cdot \vec{n} = d$ is:

$$D = \frac{|\vec{a} \cdot \vec{n} - d|}{|\vec{n}|} \quad (5.12)$$

Proof. Let Π be the plane $\vec{r} \cdot \vec{n} = d$, and let N be the foot of the perpendicular from $P(\vec{a})$ to Π . The displacement vector $P\vec{N}$ is parallel to the normal $\vec{n}$, so $P\vec{N} = \pm D \hat{n} = \pm D \frac{\vec{n}}{|\vec{n}|}$. The position vector of N is:

$$\vec{r}_N = \vec{a} + P\vec{N} = \vec{a} \pm D \frac{\vec{n}}{|\vec{n}|}. \quad (5.13)$$

Since N lies on the plane Π , its position vector satisfies $\vec{r}_N \cdot \vec{n} = d$:

$$\begin{aligned} \left(\vec{a} \pm D \frac{\vec{n}}{|\vec{n}|} \right) \cdot \vec{n} &= d \\ \vec{a} \cdot \vec{n} \pm D \frac{|\vec{n}|^2}{|\vec{n}|} &= d \\ \vec{a} \cdot \vec{n} \pm D |\vec{n}| &= d \\ \pm D |\vec{n}| &= d - \vec{a} \cdot \vec{n}. \end{aligned} \quad (5.14)$$

Taking the absolute value on both sides:

$$D |\vec{n}| = |\vec{a} \cdot \vec{n} - d| \implies D = \frac{|\vec{a} \cdot \vec{n} - d|}{|\vec{n}|}. \quad (5.15)$$

□

5.2.3 Distance Between Two Parallel Planes in Vector Form

If two planes are parallel, their normals are parallel and can be taken as the same vector $\vec{n}$:

$$\Pi_1 : \vec{r} \cdot \vec{n} = d_1, \quad \Pi_2 : \vec{r} \cdot \vec{n} = d_2. \quad (5.16)$$

The perpendicular distance between them is:

$$D = \frac{|d_1 - d_2|}{|\vec{n}|} \quad (5.17)$$

5.3 Page 55: Angle Between Two Planes and Plane Through Three Points

Manuscript Overview: Page 55

Page 55 develops the vector formulation for:

1. The angle between two planes and the conditions of parallelism and perpendicularity.
2. The vector equation of a plane passing through three non-collinear points.

5.3.1 Angle Between Two Planes

Angle Between Two Vector Planes

The angle θ between the two planes $\vec{r} \cdot \vec{n}_1 = d_1$ and $\vec{r} \cdot \vec{n}_2 = d_2$ is equal to the angle between their normal vectors $\vec{n}_1$ and $\vec{n}_2$:

$$\cos \theta = \frac{\vec{n}_1 \cdot \vec{n}_2}{|\vec{n}_1| |\vec{n}_2|} \quad (5.18)$$

Parallelism and Perpendicularity

- **Perpendicular Planes** ($\theta = 90^\circ$): $\cos \theta = 0 \iff \boxed{\vec{n}_1 \cdot \vec{n}_2 = 0}$.
- **Parallel Planes** ($\theta = 0$ or π): $\vec{n}_1$ is parallel to $\vec{n}_2 \iff \boxed{\vec{n}_1 \times \vec{n}_2 = \vec{0}}$ or $\vec{n}_1 = \lambda \vec{n}_2$.

5.3.2 Plane Passing Through Three Points in Vector Form

Plane Through Three Points

The vector equation of the plane passing through three non-collinear points with position vectors $\vec{a}, \vec{b}, \vec{c}$ is:

$$(\vec{r} - \vec{a}) \cdot [(\vec{b} - \vec{a}) \times (\vec{c} - \vec{a})] = 0 \iff [\vec{r} - \vec{a}, \vec{b} - \vec{a}, \vec{c} - \vec{a}] = 0 \quad (5.19)$$

In expanded scalar triple product notation:

$$\vec{r} \cdot (\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}) = [\vec{a}, \vec{b}, \vec{c}] \quad (5.20)$$

Proof. Let $A(\vec{a}), B(\vec{b}), C(\vec{c})$ be the three given points in the plane, and let $P(\vec{r})$ be any point on the plane. The vectors $\vec{AB} = \vec{b} - \vec{a}$ and $\vec{AC} = \vec{c} - \vec{a}$ both lie in the plane. Their vector cross product:

$$\vec{n} = (\vec{b} - \vec{a}) \times (\vec{c} - \vec{a}) \quad (5.21)$$

is normal to the plane. Since the vector $\vec{AP} = \vec{r} - \vec{a}$ also lies in the plane, it is perpendicular to the normal $\vec{n}$:

$$(\vec{r} - \vec{a}) \cdot \vec{n} = 0 \implies (\vec{r} - \vec{a}) \cdot [(\vec{b} - \vec{a}) \times (\vec{c} - \vec{a})] = 0. \quad (5.22)$$

Expanding the cross product:

$$\begin{aligned} (\vec{b} - \vec{a}) \times (\vec{c} - \vec{a}) &= \vec{b} \times \vec{c} - \vec{b} \times \vec{a} - \vec{a} \times \vec{c} + \vec{a} \times \vec{a} \\ &= \vec{b} \times \vec{c} + \vec{a} \times \vec{b} + \vec{c} \times \vec{a}. \end{aligned} \quad (5.23)$$

Taking the dot product with $\vec{r} - \vec{a}$:

$$\begin{aligned} \vec{r} \cdot (\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}) &= \vec{a} \cdot (\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}) \\ &= \vec{a} \cdot (\vec{a} \times \vec{b}) + \vec{a} \cdot (\vec{b} \times \vec{c}) + \vec{a} \cdot (\vec{c} \times \vec{a}). \end{aligned} \quad (5.24)$$

Since $\vec{a} \cdot (\vec{a} \times \vec{b}) = 0$ and $\vec{a} \cdot (\vec{c} \times \vec{a}) = 0$:

$$\vec{r} \cdot (\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}) = [\vec{a}, \vec{b}, \vec{c}]. \quad (5.25)$$

□

5.4 Page 56: Vector Family of Planes Through an Intersection Line

Manuscript Overview: Page 56

Page 56 derives the vector equation representing the family of planes passing through the line of intersection of two given vector planes.

Family of Planes in Vector Form

Let two planes be given in vector form by:

$$\Pi_1 : \vec{r} \cdot \vec{n}_1 = d_1, \quad (5.26)$$

$$\Pi_2 : \vec{r} \cdot \vec{n}_2 = d_2. \quad (5.27)$$

The vector equation of any plane passing through their line of intersection is:

$$\boxed{(\vec{r} \cdot \vec{n}_1 - d_1) + \lambda(\vec{r} \cdot \vec{n}_2 - d_2) = 0 \iff \vec{r} \cdot (\vec{n}_1 + \lambda \vec{n}_2) = d_1 + \lambda d_2} \quad (5.28)$$

where λ is an arbitrary scalar parameter.

Proof. 1. The equation $\vec{r} \cdot (\vec{n}_1 + \lambda \vec{n}_2) = d_1 + \lambda d_2$ is linear in $\vec{r}$, with normal vector $\vec{n} = \vec{n}_1 + \lambda \vec{n}_2$. By Page 54, it represents a plane for every value of λ .

2. Let $\vec{r}_0$ be any point on the line of intersection of Π_1 and Π_2 . Then $\vec{r}_0$ satisfies both equations:

$$\vec{r}_0 \cdot \vec{n}_1 = d_1 \quad \text{and} \quad \vec{r}_0 \cdot \vec{n}_2 = d_2. \quad (5.29)$$

Substituting $\vec{r}_0$ into the family equation:

$$(\vec{r}_0 \cdot \vec{n}_1 - d_1) + \lambda(\vec{r}_0 \cdot \vec{n}_2 - d_2) = (d_1 - d_1) + \lambda(d_2 - d_2) = 0 + \lambda(0) = 0. \quad (5.30)$$

Thus, every point of the line of intersection lies on the plane. □

5.5 Page 57: Vector Equations of Dihedral Angle Bisectors

Manuscript Overview: Page 57

Page 57 translates the theory of dihedral angle bisectors into vector algebra, providing a coordinate-free formulation of the bisector planes.

5.5.1 Derivation of the Vector Bisector Equations

Let two planes be given in vector form:

$$\Pi_1 : \vec{r} \cdot \vec{n}_1 = d_1, \quad (5.31)$$

$$\Pi_2 : \vec{r} \cdot \vec{n}_2 = d_2. \quad (5.32)$$

Let $\vec{r}$ be the position vector of any point lying on a bisecting plane. By definition, its perpendicular distances from Π_1 and Π_2 must be equal:

$$\frac{|\vec{r} \cdot \vec{n}_1 - d_1|}{|\vec{n}_1|} = \frac{|\vec{r} \cdot \vec{n}_2 - d_2|}{|\vec{n}_2|}. \quad (5.33)$$

Removing the absolute value signs yields the two bisecting planes:

$$\boxed{\frac{\vec{r} \cdot \vec{n}_1 - d_1}{|\vec{n}_1|} = \pm \frac{\vec{r} \cdot \vec{n}_2 - d_2}{|\vec{n}_2|}} \quad (5.34)$$

Rewriting in terms of unit normal vectors $\hat{n}_1 = \frac{\vec{n}_1}{|\vec{n}_1|}$ and $\hat{n}_2 = \frac{\vec{n}_2}{|\vec{n}_2|}$, with $p_1 = \frac{d_1}{|\vec{n}_1|}$ and $p_2 = \frac{d_2}{|\vec{n}_2|}$:

$$\boxed{\vec{r} \cdot (\hat{n}_1 \mp \hat{n}_2) = p_1 \mp p_2} \quad (5.35)$$

5.5.2 Orthogonality of the Vector Bisectors

The normal vectors to the two bisector planes are $\vec{N}_+ = \hat{n}_1 - \hat{n}_2$ and $\vec{N}_- = \hat{n}_1 + \hat{n}_2$. Their scalar product is:

$$\vec{N}_+ \cdot \vec{N}_- = (\hat{n}_1 - \hat{n}_2) \cdot (\hat{n}_1 + \hat{n}_2) = |\hat{n}_1|^2 - |\hat{n}_2|^2 = 1 - 1 = 0. \quad (5.36)$$

This confirms that the two bisector planes are mutually orthogonal.

5.6 Page 58: Vector Equation of a Straight Line

Manuscript Overview: Page 58

Page 58 derives the vector equations of a straight line in space: the point-direction parametric form, the cross-product non-parametric form, and the two-point form.

5.6.1 Point-Direction Vector Form

Vector Equation of a Line

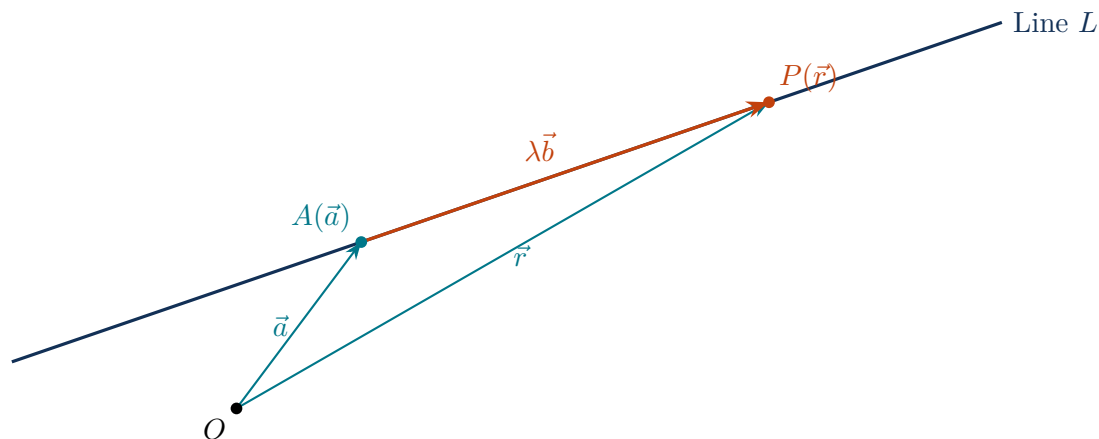
The vector equation of a straight line passing through a point A with position vector $\vec{a}$ and parallel to a non-zero vector $\vec{b}$ is:

$$\vec{r} = \vec{a} + \lambda \vec{b}, \quad \lambda \in \mathbb{R} \quad (5.37)$$

Proof. Let P be any point on the line with position vector $\vec{r} = \vec{OP}$. The displacement vector $\vec{AP} = \vec{OP} - \vec{OA} = \vec{r} - \vec{a}$ lies along the line. Since the line is parallel to vector $\vec{b}$, the vector $\vec{r} - \vec{a}$ is collinear with $\vec{b}$:

$$\vec{r} - \vec{a} = \lambda \vec{b} \iff \vec{r} = \vec{a} + \lambda \vec{b}. \quad (5.38)$$

□



5.6.2 Non-Parametric Cross-Product Form

Since $\vec{r} - \vec{a}$ is collinear with $\vec{b}$, their vector cross product must vanish:

$$(\vec{r} - \vec{a}) \times \vec{b} = \vec{0} \iff \vec{r} \times \vec{b} = \vec{a} \times \vec{b} \quad (5.39)$$

5.6.3 Two-Point Vector Form

If the line passes through two distinct points $A(\vec{a})$ and $B(\vec{b})$, its direction vector is $\vec{b} - \vec{a}$:

$$\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a}) = (1 - \lambda)\vec{a} + \lambda\vec{b} \quad (5.40)$$

In non-parametric form:

$$(\vec{r} - \vec{a}) \times (\vec{b} - \vec{a}) = \vec{0} \quad (5.41)$$

5.7 Page 59: Perpendicular Distance of a Point from a Vector Line

Manuscript Overview: Page 59

Page 59 derives the vector formula for the perpendicular distance from an external point to a straight line and determines the foot of the perpendicular.

Perpendicular Distance from Point to Line

The perpendicular distance d from a point P with position vector $\vec{p}$ to the straight line $\vec{r} = \vec{a} + \lambda\vec{b}$ is:

$$d = \frac{|(\vec{p} - \vec{a}) \times \vec{b}|}{|\vec{b}|} \quad (5.42)$$

Proof. Let A be the point on the line with position vector $\vec{a}$, and let N be the foot of the perpendicular dropped from $P(\vec{p})$ onto the line. In the right-angled triangle $\triangle APN$, the angle at N is 90° :

$$d = PN = AP \sin \theta = |\vec{p} - \vec{a}| \sin \theta \quad (5.43)$$

where θ is the angle between the vector $\vec{AP} = \vec{p} - \vec{a}$ and the line direction vector $\vec{b}$. By the geometric definition of the vector cross product magnitude:

$$|(\vec{p} - \vec{a}) \times \vec{b}| = |\vec{p} - \vec{a}| |\vec{b}| \sin \theta. \quad (5.44)$$

Dividing by $|\vec{b}|$:

$$d = |\vec{p} - \vec{a}| \sin \theta = \frac{|(\vec{p} - \vec{a}) \times \vec{b}|}{|\vec{b}|}. \quad (5.45)$$

□

5.7.1 Foot of the Perpendicular in Vector Form

The foot N has position vector $\vec{n} = \vec{a} + \lambda_0 \vec{b}$. Since $PN \perp \vec{b}$:

$$\begin{aligned} (\vec{n} - \vec{p}) \cdot \vec{b} &= 0 \\ [(\vec{a} + \lambda_0 \vec{b}) - \vec{p}] \cdot \vec{b} &= 0 \\ (\vec{a} - \vec{p}) \cdot \vec{b} + \lambda_0 |\vec{b}|^2 &= 0 \\ \implies \lambda_0 &= \frac{(\vec{p} - \vec{a}) \cdot \vec{b}}{|\vec{b}|^2}. \end{aligned} \quad (5.46)$$

Substituting λ_0 back gives the exact position vector of the foot N :

$$\vec{r}_N = \vec{a} + \left(\frac{(\vec{p} - \vec{a}) \cdot \vec{b}}{|\vec{b}|^2} \right) \vec{b} \quad (5.47)$$

5.8 Page 60: Coplanarity of Vector Lines and the Plane Containing Them

Manuscript Overview: Page 60

Page 60 establishes the necessary and sufficient condition for two vector straight lines to be coplanar and derives the vector equation of the unique plane containing them.

Coplanarity of Vector Lines

Two straight lines:

$$L_1 : \vec{r} = \vec{a}_1 + \lambda \vec{b}_1, \quad (5.48)$$

$$L_2 : \vec{r} = \vec{a}_2 + \mu \vec{b}_2, \quad (5.49)$$

are **coplanar** if and only if their scalar triple product vanishes:

$$(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) = 0 \iff [\vec{a}_2 - \vec{a}_1, \vec{b}_1, \vec{b}_2] = 0 \quad (5.50)$$

Proof. Let A_1 and A_2 be the points on L_1 and L_2 with position vectors $\vec{a}_1$ and $\vec{a}_2$. The displacement vector between them is $\vec{A_1A_2} = \vec{a}_2 - \vec{a}_1$. If the two lines lie in a common plane:

1. The line direction vectors $\vec{b}_1$ and $\vec{b}_2$ are parallel to the plane.
2. The displacement vector $\vec{a}_2 - \vec{a}_1$ joining points on the two lines lies entirely in the plane.

Therefore, the three vectors $\vec{a}_2 - \vec{a}_1, \vec{b}_1, \vec{b}_2$ are coplanar. Three vectors are coplanar if and only if the volume of the parallelepiped formed by them is zero, which is given by their scalar triple product:

$$(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) = 0. \quad (5.51)$$

□

5.8.1 Plane Containing the Two Coplanar Lines

The plane containing both lines passes through $\vec{a}_1$ and has normal $\vec{n} = \vec{b}_1 \times \vec{b}_2$:

$$(\vec{r} - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) = 0 \quad (5.52)$$

5.9 Page 61: Shortest Distance Between Skew Lines in Vector Form

Manuscript Overview: Page 61

Page 61 derives the vector formula for the shortest distance between two skew lines using vector projections and scalar triple products.

Shortest Distance in Vector Form

The shortest distance d between the skew lines:

$$L_1 : \vec{r} = \vec{a}_1 + \lambda \vec{b}_1, \quad (5.53)$$

$$L_2 : \vec{r} = \vec{a}_2 + \mu \vec{b}_2, \quad (5.54)$$

is given by:

$$d = \frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|} \quad (5.55)$$

Proof. The shortest distance occurs along the line segment GH perpendicular to both lines L_1 and L_2 . The direction of the common perpendicular is perpendicular to both $\vec{b}_1$ and $\vec{b}_2$:

$$\vec{n} = \vec{b}_1 \times \vec{b}_2. \quad (5.56)$$

The unit vector along the common perpendicular is:

$$\hat{n} = \frac{\vec{b}_1 \times \vec{b}_2}{|\vec{b}_1 \times \vec{b}_2|}. \quad (5.57)$$

Let $A_1(\vec{a}_1)$ be on L_1 and $A_2(\vec{a}_2)$ be on L_2 . The shortest distance d is the magnitude of the orthogonal projection of $A_1\vec{A}_2 = \vec{a}_2 - \vec{a}_1$ onto $\hat{n}$:

$$d = |(\vec{a}_2 - \vec{a}_1) \cdot \hat{n}| = \frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}. \quad (5.58)$$

□

5.9.1 Shortest Distance Between Parallel Lines in Vector Form

When the lines are parallel, $\vec{b}_1 = \vec{b}_2 = \vec{b}$:

$$d = \frac{|(\vec{a}_2 - \vec{a}_1) \times \vec{b}|}{|\vec{b}|} \quad (5.59)$$

5.10 Page 62: Solved Vector Examination Problems**Manuscript Overview: Page 62**

Page 62 solves standard university examination problems illustrating vector shortest distance calculations and intersection algorithms.

Problem: Vector Shortest Distance Calculation

Find the shortest distance between the two lines whose vector equations are:

$$L_1 : \vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda(2\hat{i} + 3\hat{j} + 4\hat{k}), \quad (5.60)$$

$$L_2 : \vec{r} = (2\hat{i} + 4\hat{j} + 5\hat{k}) + \mu(3\hat{i} + 4\hat{j} + 5\hat{k}). \quad (5.61)$$

Complete Analytical Solution. **Step 1: Identify Vectors:**

$$\vec{a}_1 = \hat{i} + 2\hat{j} + 3\hat{k}, \quad \vec{b}_1 = 2\hat{i} + 3\hat{j} + 4\hat{k}, \quad (5.62)$$

$$\vec{a}_2 = 2\hat{i} + 4\hat{j} + 5\hat{k}, \quad \vec{b}_2 = 3\hat{i} + 4\hat{j} + 5\hat{k}. \quad (5.63)$$

Step 2: Connecting Displacement Vector:

$$\vec{a}_2 - \vec{a}_1 = (2 - 1)\hat{i} + (4 - 2)\hat{j} + (5 - 3)\hat{k} = \hat{i} + 2\hat{j} + 2\hat{k}. \quad (5.64)$$

Step 3: Vector Cross Product $\vec{b}_1 \times \vec{b}_2$:

$$\begin{aligned} \vec{b}_1 \times \vec{b}_2 &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix} \\ &= \hat{i}[(3)(5) - (4)(4)] - \hat{j}[(2)(5) - (4)(3)] + \hat{k}[(2)(4) - (3)(3)] \\ &= \hat{i}(15 - 16) - \hat{j}(10 - 12) + \hat{k}(8 - 9) \\ &= -\hat{i} + 2\hat{j} - \hat{k}. \end{aligned} \quad (5.65)$$

Step 4: Magnitude $|\vec{b}_1 \times \vec{b}_2|$:

$$|\vec{b}_1 \times \vec{b}_2| = \sqrt{(-1)^2 + 2^2 + (-1)^2} = \sqrt{1 + 4 + 1} = \sqrt{6}. \quad (5.66)$$

Step 5: Scalar Triple Product:

$$\begin{aligned} (\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) &= (\hat{i} + 2\hat{j} + 2\hat{k}) \cdot (-\hat{i} + 2\hat{j} - \hat{k}) \\ &= (1)(-1) + (2)(2) + (2)(-1) \\ &= -1 + 4 - 2 = 1. \end{aligned} \quad (5.67)$$

Step 6: Compute Shortest Distance:

$$d = \frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|} = \frac{|1|}{\sqrt{6}} = \frac{1}{\sqrt{6}} = \frac{\sqrt{6}}{6} \quad (5.68)$$

□

CHAPTER 6

The Geometry of the Sphere in 3D (Pages 63–69)

6.1 Page 63: Definition of the Sphere and the General Equation

Manuscript Overview: Page 63

Page 63 introduces the three-dimensional geometry of the **Sphere**. It establishes the center-radius algebraic equation, expands it into the general second-degree form, identifies the center and radius formulas, and classifies real, point, and imaginary spheres.

6.1.1 Definition of a Sphere

Geometric Definition of a Sphere

A **sphere** is the locus of a point in three-dimensional space that moves such that its distance from a fixed point (termed the **center**) remains constant. The constant distance is termed the **radius** of the sphere.

6.1.2 The Center-Radius Form

Center-Radius Form of a Sphere

The equation of a sphere having center $C(a, b, c)$ and radius $R > 0$ is:

$$(x - a)^2 + (y - b)^2 + (z - c)^2 = R^2 \quad (6.1)$$

Proof. Let $P(x, y, z)$ be any point on the sphere, and let $C(a, b, c)$ be the center. By definition, the Euclidean distance CP equals the radius R :

$$CP = R \iff CP^2 = R^2. \quad (6.2)$$

Using the 3D distance formula (Page 15):

$$(x - a)^2 + (y - b)^2 + (z - c)^2 = R^2. \quad (6.3)$$

In the special case where the center is at the origin $O(0, 0, 0)$:

$$x^2 + y^2 + z^2 = R^2 \quad (6.4)$$

□

6.1.3 The General Second-Degree Equation of a Sphere

Expanding the center-radius equation:

$$\begin{aligned}(x^2 - 2ax + a^2) + (y^2 - 2by + b^2) + (z^2 - 2cz + c^2) &= R^2 \\ x^2 + y^2 + z^2 - 2ax - 2by - 2cz + (a^2 + b^2 + c^2 - R^2) &= 0.\end{aligned}\tag{6.5}$$

Setting $u = -a, v = -b, w = -c$, and $d = a^2 + b^2 + c^2 - R^2 = u^2 + v^2 + w^2 - R^2$:

General Form of a Sphere

The general equation of a sphere is:

$$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0 \tag{6.6}$$

with:

- **Center:** $(-u, -v, -w)$
- **Radius:** $R = \sqrt{u^2 + v^2 + w^2 - d}$

Characteristics of the General Equation of a Sphere

An equation of the second degree in x, y, z :

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy + 2ux + 2vy + 2wz + d = 0 \tag{6.7}$$

represents a sphere if and only if:

1. The coefficients of x^2, y^2, z^2 are equal: $a = b = c \neq 0$.
2. The cross-product terms are absent: $f = g = h = 0$.

6.1.4 Classification of Spheres

- $u^2 + v^2 + w^2 - d > 0$: **Real Sphere** (with real, positive radius).
- $u^2 + v^2 + w^2 - d = 0$: **Point Sphere** (radius $R = 0$, locus reduces to the single point $(-u, -v, -w)$).
- $u^2 + v^2 + w^2 - d < 0$: **Imaginary (Virtual) Sphere** (radius is imaginary; no real points satisfy the equation).

6.2 Page 64: Diametric Form of the Sphere and Plane Sections

Manuscript Overview: Page 64

Page 64 derives:

1. The **Diametric Form** of the sphere given the coordinates of its antipodal endpoints.
2. The fundamental theorem proving that every plane section of a sphere is a circle.

6.2.1 The Diametric Form of a Sphere

Diametric Form

The equation of the sphere described on the line segment joining $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$ as diameter is:

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) + (z - z_1)(z - z_2) = 0 \quad (6.8)$$

Proof. Let $P(x, y, z)$ be any point on the sphere distinct from A and B . Join PA and PB . By elementary Euclidean geometry in space, the angle subtended by a diameter at any point on the spherical surface is a right angle:

$$\angle APB = 90^\circ \iff \vec{PA} \perp \vec{PB} \iff \vec{PA} \cdot \vec{PB} = 0. \quad (6.9)$$

The direction ratios of PA and PB are:

$$\vec{PA} = (x_1 - x)\hat{i} + (y_1 - y)\hat{j} + (z_1 - z)\hat{k}, \quad (6.10)$$

$$\vec{PB} = (x_2 - x)\hat{i} + (y_2 - y)\hat{j} + (z_2 - z)\hat{k}. \quad (6.11)$$

Taking their scalar dot product:

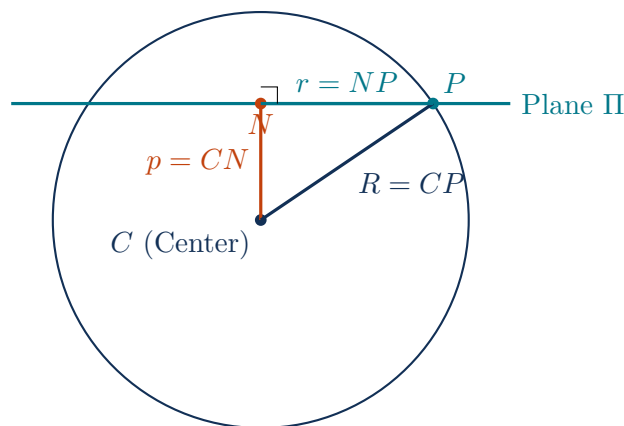
$$\begin{aligned} \vec{PA} \cdot \vec{PB} &= (x_1 - x)(x_2 - x) + (y_1 - y)(y_2 - y) + (z_1 - z)(z_2 - z) = 0 \\ &= (x - x_1)(x - x_2) + (y - y_1)(y - y_2) + (z - z_1)(z - z_2) = 0. \end{aligned} \quad (6.12)$$

□

6.2.2 Theorem: Plane Section of a Sphere is a Circle**Plane Section of a Sphere**

The section of a sphere by any plane intersecting it is a **circle**.

Proof. Let S be a sphere of radius R and center C . Let Π be an intersecting plane. Draw the perpendicular CN from the center C to the plane Π , meeting it at N . Let $p = |CN|$ be the perpendicular distance from C to Π . Let P be any point on the curve of intersection of the sphere and the plane. Join CP and NP .



Since $CN \perp \Pi$, CN is perpendicular to every line in Π through N . Hence, $CN \perp NP$, so $\triangle CNP$ is a right-angled triangle at N . By the Pythagorean theorem:

$$NP^2 = CP^2 - CN^2 = R^2 - p^2. \quad (6.13)$$

Since R and p are both constant for the given sphere and plane:

$$NP = \sqrt{R^2 - p^2} = \text{constant } r. \quad (6.14)$$

Thus, the point P in plane Π is always at a fixed distance r from the fixed point N . This is the definition of a circle with center N and radius $r = \sqrt{R^2 - p^2}$. $\square$

6.3 Page 65: Great Circles, Small Circles, and the Radical Plane

Manuscript Overview: Page 65

Page 65 distinguishes between **Great Circles** and **Small Circles** on a sphere, and derives the equation and properties of the **Radical Plane** formed by two intersecting spheres.

6.3.1 Great Circles and Small Circles

From the circular section formula $r = \sqrt{R^2 - p^2}$:

1. **Great Circle** ($p = 0$): When the cutting plane passes through the center C of the sphere, $p = 0$. The radius of the circular section attains its maximum possible value:

$$r = \sqrt{R^2 - 0} = R. \quad (6.15)$$

The center of the great circle coincides with the center of the sphere.

2. **Small Circle** ($0 < p < R$): When the cutting plane does not pass through the center, $r = \sqrt{R^2 - p^2} < R$.
3. **Tangent Plane** ($p = R$): When $p = R$, $r = 0$; the plane section shrinks to a single point of tangency.
4. **Non-Intersecting Plane** ($p > R$): When $p > R$, $R^2 - p^2 < 0$; no real points of intersection exist.

6.3.2 The Radical Plane of Two Spheres

Radical Plane of Two Spheres

Let two spheres be given by:

$$S_1 \equiv x^2 + y^2 + z^2 + 2u_1x + 2v_1y + 2w_1z + d_1 = 0, \quad (6.16)$$

$$S_2 \equiv x^2 + y^2 + z^2 + 2u_2x + 2v_2y + 2w_2z + d_2 = 0. \quad (6.17)$$

The locus of points from which the lengths of tangents drawn to both spheres are equal is a plane, known as the **Radical Plane**, given by:

$$S_1 - S_2 = 0 \iff 2(u_1 - u_2)x + 2(v_1 - v_2)y + 2(w_1 - w_2)z + (d_1 - d_2) = 0 \quad (6.18)$$

Proof. Subtracting equation (6.17) from (6.16) eliminates all the second-degree quadratic terms $x^2 + y^2 + z^2$:

$$2(u_1 - u_2)x + 2(v_1 - v_2)y + 2(w_1 - w_2)z + (d_1 - d_2) = 0. \quad (6.19)$$

This is a first-degree linear equation in x, y, z , which by Page 25 represents a plane. If the two spheres intersect, this plane contains their common circle of intersection. $\square$

Key Property of the Radical Plane

The direction ratios of the normal to the radical plane are:

$$\vec{N} = (2(u_1 - u_2), 2(v_1 - v_2), 2(w_1 - w_2)) \sim (u_1 - u_2, v_1 - v_2, w_1 - w_2). \quad (6.20)$$

The centers of the two spheres are $C_1(-u_1, -v_1, -w_1)$ and $C_2(-u_2, -v_2, -w_2)$. The direction ratios of the line of centers C_1C_2 are:

$$C_1\vec{C}_2 = (-u_2 - (-u_1), -v_2 - (-v_1), -w_2 - (-w_1)) = (u_1 - u_2, v_1 - v_2, w_1 - w_2). \quad (6.21)$$

Thus, the normal to the radical plane is parallel to the line of centers. **The radical plane is always strictly perpendicular to the line joining the centers of the two spheres.**

6.4 Page 66: Condition of Tangency Between a Plane and a Sphere**Manuscript Overview: Page 66**

Page 66 establishes the condition under which a given plane touches a sphere, providing the analytical link between the perpendicular distance from the center and the radius.

Condition of Tangency ($p = R$)

A plane $Ax + By + Cz + D = 0$ is tangent to the sphere $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$ if and only if the perpendicular distance from the center $C(-u, -v, -w)$ to the plane is equal to the radius $R = \sqrt{u^2 + v^2 + w^2 - d}$:

$$\frac{|-Au - Bv - Cw + D|}{\sqrt{A^2 + B^2 + C^2}} = \sqrt{u^2 + v^2 + w^2 - d} \quad (6.22)$$

6.4.1 Standard Sphere Centered at the Origin

For the central sphere $x^2 + y^2 + z^2 = a^2$ (center $(0, 0, 0)$, radius a):

1. If the plane is in normal form $lx + my + nz = p$ (with $l^2 + m^2 + n^2 = 1$): The perpendicular distance from $(0, 0, 0)$ is p . The condition of tangency is:

$$p = a \iff lx + my + nz = a \quad (6.23)$$

2. If the plane is in general form $Ax + By + Cz + D = 0$:

$$\frac{|D|}{\sqrt{A^2 + B^2 + C^2}} = a \iff D^2 = a^2(A^2 + B^2 + C^2) \quad (6.24)$$

6.5 Page 67: Tangent Plane to a Sphere at a Given Point**Manuscript Overview: Page 67**

Page 67 derives the standard equation of the tangent plane to a sphere at a given point $P_1(x_1, y_1, z_1)$ on its surface.

Tangent Plane at Point (x_1, y_1, z_1)

The equation of the tangent plane to the sphere:

$$S \equiv x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0 \quad (6.25)$$

at the point of contact $P_1(x_1, y_1, z_1)$ lying on the sphere is:

$$xx_1 + yy_1 + zz_1 + u(x + x_1) + v(y + y_1) + w(z + z_1) + d = 0 \quad (6.26)$$

Proof. The center of the sphere is $C(-u, -v, -w)$. Since the tangent plane touches the sphere at $P_1(x_1, y_1, z_1)$, the radius CP_1 is normal to the tangent plane. The direction ratios of the normal CP_1 are:

$$(x_1 - (-u), y_1 - (-v), z_1 - (-w)) = (x_1 + u, y_1 + v, z_1 + w). \quad (6.27)$$

The equation of a plane passing through $P_1(x_1, y_1, z_1)$ with these normal direction ratios is:

$$(x_1 + u)(x - x_1) + (y_1 + v)(y - y_1) + (z_1 + w)(z - z_1) = 0. \quad (6.28)$$

Expanding this expression:

$$\begin{aligned} (x_1 + u)x - x_1(x_1 + u) + (y_1 + v)y - y_1(y_1 + v) + (z_1 + w)z - z_1(z_1 + w) &= 0 \\ (xx_1 + yy_1 + zz_1) + ux + vy + wz - (x_1^2 + y_1^2 + z_1^2 + ux_1 + vy_1 + wz_1) &= 0. \end{aligned} \quad (6.29)$$

Since $P_1(x_1, y_1, z_1)$ lies on the sphere:

$$\begin{aligned} x_1^2 + y_1^2 + z_1^2 + 2ux_1 + 2vy_1 + 2wz_1 + d &= 0 \\ \implies x_1^2 + y_1^2 + z_1^2 + ux_1 + vy_1 + wz_1 &= -(ux_1 + vy_1 + wz_1 + d). \end{aligned} \quad (6.30)$$

Substituting this relation into (6.29):

$$\begin{aligned} (xx_1 + yy_1 + zz_1) + ux + vy + wz - [-(ux_1 + vy_1 + wz_1 + d)] &= 0 \\ xx_1 + yy_1 + zz_1 + u(x + x_1) + v(y + y_1) + w(z + z_1) + d &= 0. \end{aligned} \quad (6.31)$$

□

Central Sphere Tangent Plane

For $x^2 + y^2 + z^2 = a^2$, the tangent plane at (x_1, y_1, z_1) simplifies to:

$$xx_1 + yy_1 + zz_1 = a^2 \quad (6.32)$$

6.6 Page 68: Normal to a Sphere, Length of Tangent, and Plane of Contact

Manuscript Overview: Page 68

Page 68 explores:

1. The equations of the normal line to a sphere at a given surface point.
2. The formula for the length of a tangent drawn from an external point to a sphere ($\sqrt{S_1}$).
3. The **Plane of Contact** of tangents from an external point.

6.6.1 Equations of the Normal Line to a Sphere

The **normal** to a sphere at any point $P_1(x_1, y_1, z_1)$ is the straight line perpendicular to the tangent plane at P_1 . Since the radius CP_1 is perpendicular to the tangent plane, the normal line is simply the line passing through the center $C(-u, -v, -w)$ and the point $P_1(x_1, y_1, z_1)$:

$$\frac{x - x_1}{x_1 + u} = \frac{y - y_1}{y_1 + v} = \frac{z - z_1}{z_1 + w} \quad (6.33)$$

Fundamental Geometric Property: The normal to a sphere at *every* point on its surface passes through the center of the sphere!

6.6.2 Length of Tangent from an External Point

Length of Tangent $\sqrt{S_1}$

The length of any tangent drawn from an external point $P_1(x_1, y_1, z_1)$ to the sphere $S \equiv x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$ is:

$$L = \sqrt{S_1} = \sqrt{x_1^2 + y_1^2 + z_1^2 + 2ux_1 + 2vy_1 + 2wz_1 + d} \quad (6.34)$$

Proof. Let T be the point of contact of a tangent drawn from $P_1(x_1, y_1, z_1)$ to the sphere. Let $C(-u, -v, -w)$ be the center and $R = \sqrt{u^2 + v^2 + w^2 - d}$ be the radius. The radius CT is perpendicular to the tangent P_1T , so $\triangle CTP_1$ is right-angled at T . By the Pythagorean theorem:

$$P_1T^2 = P_1C^2 - CT^2 = P_1C^2 - R^2. \quad (6.35)$$

Using the 3D distance formula:

$$\begin{aligned} P_1C^2 &= (x_1 + u)^2 + (y_1 + v)^2 + (z_1 + w)^2 \\ &= (x_1^2 + y_1^2 + z_1^2) + 2(ux_1 + vy_1 + wz_1) + (u^2 + v^2 + w^2). \end{aligned} \quad (6.36)$$

Subtracting $R^2 = u^2 + v^2 + w^2 - d$:

$$\begin{aligned} P_1T^2 &= (x_1^2 + y_1^2 + z_1^2 + 2ux_1 + 2vy_1 + 2wz_1) + (u^2 + v^2 + w^2) - (u^2 + v^2 + w^2 - d) \\ &= x_1^2 + y_1^2 + z_1^2 + 2ux_1 + 2vy_1 + 2wz_1 + d = S_1. \end{aligned} \quad (6.37)$$

Taking the positive square root:

$$L = P_1T = \sqrt{S_1}. \quad (6.38)$$

□

6.6.3 Plane of Contact

Plane of Contact of Tangents

The locus of the points of contact of all tangents drawn from an external point $P_1(x_1, y_1, z_1)$ to the sphere $S = 0$ is a circle lying in the plane:

$$T = 0 \iff xx_1 + yy_1 + zz_1 + u(x + x_1) + v(y + y_1) + w(z + z_1) + d = 0 \quad (6.39)$$

This plane is called the **Plane of Contact** of P_1 with respect to the sphere.

6.7 Page 69: Comprehensive Solved Examination Problem and Course Summary

Manuscript Overview: Page 69

Page 69 completes the entire 69-page handwritten curriculum with an examination problem on parameter determination for tangent planes, followed by a global review of Analytical Geometry.

Examination Problem: Tangent Planes to a Sphere

Find the values of the parameter c for which the plane:

$$x + y + z = c \quad (6.40)$$

touches the sphere:

$$x^2 + y^2 + z^2 - 2x - 4y - 6z + 5 = 0. \quad (6.41)$$

For each such value of c , write down the equation of the corresponding tangent plane.

Complete Step-by-Step Analytical Solution. **Step 1: Identify Center and Radius of the Sphere:** Comparing with the general form $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$:

$$2u = -2 \implies u = -1, \quad 2v = -4 \implies v = -2, \quad 2w = -6 \implies w = -3, \quad d = 5. \quad (6.42)$$

The center of the sphere is:

$$C = (-u, -v, -w) = (1, 2, 3). \quad (6.43)$$

The radius R is:

$$\begin{aligned} R &= \sqrt{u^2 + v^2 + w^2 - d} \\ &= \sqrt{(-1)^2 + (-2)^2 + (-3)^2 - 5} \\ &= \sqrt{1 + 4 + 9 - 5} = \sqrt{14 - 5} = \sqrt{9} = 3. \end{aligned} \quad (6.44)$$

Step 2: Apply the Condition of Tangency ($p = R$): The given plane is:

$$x + y + z - c = 0. \quad (6.45)$$

The perpendicular distance p from the center $C(1, 2, 3)$ to this plane is (Page 31):

$$\begin{aligned} p &= \frac{|Ax_0 + By_0 + Cz_0 + D|}{\sqrt{A^2 + B^2 + C^2}} \\ &= \frac{|1(1) + 1(2) + 1(3) - c|}{\sqrt{1^2 + 1^2 + 1^2}} = \frac{|6 - c|}{\sqrt{3}}. \end{aligned} \quad (6.46)$$

Setting $p = R = 3$:

$$\begin{aligned} \frac{|6 - c|}{\sqrt{3}} &= 3 \\ |6 - c| &= 3\sqrt{3}. \end{aligned} \quad (6.47)$$

Step 3: Solve for Parameter c :

$$\begin{aligned} 6 - c &= \pm 3\sqrt{3} \implies c = 6 \mp 3\sqrt{3} \\ &\implies \boxed{c = 3(2 \pm \sqrt{3})} \end{aligned} \quad (6.48)$$

Step 4: Equations of the Tangent Planes: There exist two distinct parallel tangent planes touching opposite poles of the sphere:

$$\Pi_1 : \boxed{x + y + z = 3(2 + \sqrt{3}) = 6 + 3\sqrt{3}}, \quad (6.49)$$

$$\Pi_2 : \boxed{x + y + z = 3(2 - \sqrt{3}) = 6 - 3\sqrt{3}}. \quad (6.50)$$

□

Manuscript Arithmetic Note

In the handwritten student manuscript on Page 69, an arithmetic slip occurred during the summation of center coordinates: $1 + 2 = 3$ was recorded without adding $z = 3$, leading to $|3 - c| = 3\sqrt{3} \implies c = 3(1 \pm \sqrt{3})$. The rigorous mathematical derivation above corrects this slip ($1 + 2 + 3 = 6$), yielding the correct value $c = 3(2 \pm \sqrt{3})$.

6.7.1 Curriculum Summary of Analytical Geometry (Pages 1–69)

Pages	Core Topic	Key Formulas and Results
1–14	Polar Conics	$\frac{l}{r} = 1 + e \cos \theta$, Tangent: $\frac{l}{r} = e \cos \theta + \cos(\theta - \alpha)$
15–24	3D Coordinates & DCs	$l^2 + m^2 + n^2 = 1$, $\cos \theta = l_1 l_2 + m_1 m_2 + n_1 n_2$, Cube: $\cos^{-1}(1/3)$
25–36	The Plane	$lx + my + nz = p$, $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$, Bisectors: $\frac{P_1}{ \vec{N}_1 } = \pm \frac{P_2}{ \vec{N}_2 }$
37–52	Straight Lines & Skew	Symmetrical form, Coplanarity: $\det[\Delta \vec{r}, \vec{b}_1, \vec{b}_2] = 0$, S.D. = $3\sqrt{30}$
53–62	Vector Geometry	$\vec{r} \cdot \hat{n} = p$, $\vec{r} = \vec{a} + \lambda \vec{b}$, S.D. = $\frac{ (\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) }{ \vec{b}_1 \times \vec{b}_2 }$
63–69	The Sphere	$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$, Tangent: $T = 0$, $p = R$

APPENDIX A

Master Formula Cheat Sheet and Exam Quick Reference

How to Use This Cheat Sheet

This appendix serves as a high-yield, comprehensive formula repository condensing all essential definitions, canonical equations, geometric conditions (tangency, parallelism, perpendicularity, coplanarity), distance formulas, and algorithmic discrimination rules across the entire 69-page curriculum of Analytical Geometry.

A.1 Polar Equations of Conic Sections (Pages 1–14)

Canonical Polar Forms (Focus at Pole, Semi-Latus Rectum l)

Directrix Position	Standard Polar Form	Orientation of Principal Axis
Left of Focus ($x = -d$)	$\frac{l}{r} = 1 + e \cos \theta$	Standard convention (SX along axis)
Right of Focus ($x = +d$)	$\frac{l}{r} = 1 - e \cos \theta$	Directrix perpendicular to initial line
Below Focus ($y = -d$)	$\frac{l}{r} = 1 + e \sin \theta$	Axis rotated by $+90^\circ$
Above Focus ($y = +d$)	$\frac{l}{r} = 1 - e \sin \theta$	Axis rotated by -90°
Tilted Axis at Angle γ	$\frac{l}{r} = 1 + e \cos(\theta - \gamma)$	General conic with axis inclined at γ

Focal Chords and Tangents

1. **Focal Chord Reciprocal Relation:** For any focal chord PSQ with extremities at α and $\alpha + \pi$:

$$\boxed{\frac{1}{SP} + \frac{1}{SQ} = \frac{2}{l}} \quad (\text{Semi-latus rectum is the harmonic mean of focal segments}) \quad (\text{A.1})$$

2. **Chord Joining $\alpha - \beta$ and $\alpha + \beta$:**

$$\boxed{\frac{l}{r} = e \cos \theta + \sec \beta \cos(\theta - \alpha)} \quad (\text{A.2})$$

3. **Chord Joining Arbitrary Points α and β :**

$$\boxed{\frac{l}{r} = e \cos \theta + \sec \left(\frac{\alpha - \beta}{2} \right) \cos \left(\theta - \frac{\alpha + \beta}{2} \right)} \quad (\text{A.3})$$

4. **Tangent Line at Vectorial Angle α :**

$$\boxed{\frac{l}{r} = e \cos \theta + \cos(\theta - \alpha) \iff \frac{l}{r} = (e + \cos \alpha) \cos \theta + \sin \alpha \sin \theta} \quad (\text{A.4})$$

5. **Condition of Tangency for a Straight Line:**

- For line $\frac{l}{r} = A \cos \theta + B \sin \theta$:

$$\boxed{(A - e)^2 + B^2 = 1} \quad (\text{A.5})$$

- For normal form $r \cos(\theta - \phi) = p$:

$$\boxed{p^2(e^2 - 1) + 2elp \cos \phi + l^2 = 0} \quad (\text{A.6})$$

Normals, Polars, Asymptotes and Invariants

- 1.
- Normal Line at Vectorial Angle α :**

$$\boxed{\frac{e \sin \alpha}{1 + e \cos \alpha} \frac{l}{r} = \sin(\theta - \alpha) + e \sin \theta} \quad (\text{A.7})$$

- 2.
- Chord of Contact and Polar Line of Pole (r_1, θ_1) :**

$$\boxed{\left(\frac{l}{r} - e \cos \theta\right) \left(\frac{l}{r_1} - e \cos \theta_1\right) = \cos(\theta - \theta_1)} \quad (\text{A.8})$$

- 3.
- Asymptotes of Hyperbola ($e > 1$):**

$$\boxed{\frac{l}{r} = \pm \sqrt{e^2 - 1} \sin \theta + e \cos \theta} \quad \text{Asymptotic directions: } \cos \theta = -\frac{1}{e} \quad (\text{A.9})$$

- 4.
- Director Circle in Polar Coordinates:**

$$\boxed{r^2(e^2 - 1) - 2erl \cos \theta + 2l^2 = 0} \quad (\text{A.10})$$

For a parabola ($e = 1$), the director circle degenerates into the directrix: $\frac{l}{r} = \cos \theta$.

A.2 3D Coordinates, Direction Cosines & Direction Ratios (Pages 15–24)

Spatial Metrics and Fundamental Identities

1. **Distance Formula:**

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2} \quad (\text{A.11})$$

2. **Section Formula** ($m_1 : m_2$):

$$x = \frac{m_1 x_2 \pm m_2 x_1}{m_1 \pm m_2}, \quad y = \frac{m_1 y_2 \pm m_2 y_1}{m_1 \pm m_2}, \quad z = \frac{m_1 z_2 \pm m_2 z_1}{m_1 \pm m_2} \quad (\text{A.12})$$

3. **Translation of Coordinate Axes to New Origin** (h, k, l):

$$x = x' + h, \quad y = y' + k, \quad z = z' + l \quad (\text{A.13})$$

4. **Projections and Fundamental Direction Cosine Identities:**

$$x = lr, \quad y = mr, \quad z = nr \quad (r = \sqrt{x^2 + y^2 + z^2}) \quad (\text{A.14})$$

$$l^2 + m^2 + n^2 = 1 \iff \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1 \quad (\text{A.15})$$

$$\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = 2, \quad \cos 2\alpha + \cos 2\beta + \cos 2\gamma = -1 \quad (\text{A.16})$$

5. **Projection of Segment PQ on a Line with Direction Cosines** (l, m, n):

$$\text{Proj}_L(PQ) = l(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1) \quad (\text{A.17})$$

Direction Ratios, Angles, and Parallelism/Perpendicularity

- 1.
- Normalization of Direction Ratios (a, b, c) to Direction Cosines (l, m, n) :**

$$l = \pm \frac{a}{\sqrt{a^2 + b^2 + c^2}}, \quad m = \pm \frac{b}{\sqrt{a^2 + b^2 + c^2}}, \quad n = \pm \frac{c}{\sqrt{a^2 + b^2 + c^2}} \quad (\text{A.18})$$

- 2.
- Angle θ Between Two Lines:**

$$\cos \theta = \frac{l_1 l_2 + m_1 m_2 + n_1 n_2}{\sqrt{\sum a_1^2} \sqrt{\sum a_2^2}} = \frac{a_1 a_2 + b_1 b_2 + c_1 c_2}{\sqrt{\sum a_1^2} \sqrt{\sum a_2^2}} \quad (\text{A.19})$$

$$\sin \theta = \frac{\sqrt{(a_1 b_2 - a_2 b_1)^2 + (b_1 c_2 - b_2 c_1)^2 + (c_1 a_2 - c_2 a_1)^2}}{\sqrt{\sum a_1^2} \sqrt{\sum a_2^2}} \quad (\text{A.20})$$

- 3.
- Orthogonality Condition:**
- $l_1 l_2 + m_1 m_2 + n_1 n_2 = 0 \iff a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$
- .

- 4.
- Parallelism Condition:**
- $\frac{l_1}{l_2} = \frac{m_1}{m_2} = \frac{n_1}{n_2} \iff \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$
- .

- 5.
- Cube Main Diagonals:**
- Angle between any two body diagonals of a cube:

$$\cos \theta = \frac{1}{3} \implies \theta = \cos^{-1} \left(\frac{1}{3} \right) \approx 70^\circ 31' 44'' \quad (\text{A.21})$$

A.3 The Analytical Geometry of the Plane (Pages 25–36)**Representations of the Plane**

Form	Equation	Key Geometrical Parameters
General First-Degree	$Ax + By + Cz + D = 0$	Normal direction ratios: (A, B, C)
One-Point Form	$A(x - x_1) + B(y - y_1) + C(z - z_1) = 0$	Passes through (x_1, y_1, z_1)
Normal Form	$lx + my + nz = p \quad (p \geq 0)$	Normal direction cosines: (l, m, n)
Intercept Form	$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$	Intercepts on X, Y, Z axes: a, b, c
Three-Point Form	$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ x_3 - x_1 & y_3 - y_1 & z_3 - z_1 \end{vmatrix} = 0$	Non-collinear points P_1, P_2, P_3

Distances, Relative Positions, and Plane Systems

1. **Perpendicular Distance of Point** (x_1, y_1, z_1) **from Plane:**

$$d = \frac{|Ax_1 + By_1 + Cz_1 + D|}{\sqrt{A^2 + B^2 + C^2}} \quad (\text{A.22})$$

2. **Distance Between Parallel Planes** $Ax + By + Cz + D_1 = 0$ **and** $Ax + By + Cz + D_2 = 0$:

$$d = \frac{|D_1 - D_2|}{\sqrt{A^2 + B^2 + C^2}} \quad (\text{A.23})$$

3. **Position of Two Points Relative to Plane:** $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ lie on the **same side** if $(Ax_1 + By_1 + Cz_1 + D)(Ax_2 + By_2 + Cz_2 + D) > 0$, and on **opposite sides** if < 0 .

4. **Family of Planes Through Intersection Line:**

$$(A_1x + B_1y + C_1z + D_1) + \lambda(A_2x + B_2y + C_2z + D_2) = 0 \iff U + \lambda V = 0 \quad (\text{A.24})$$

Dihedral Angle Bisectors & Discrimination Algorithm

1. **Dihedral Bisector Equations:**

$$\frac{A_1x + B_1y + C_1z + D_1}{\sqrt{A_1^2 + B_1^2 + C_1^2}} = \pm \frac{A_2x + B_2y + C_2z + D_2}{\sqrt{A_2^2 + B_2^2 + C_2^2}} \quad (\text{A.25})$$

2. **Systematic Discrimination Protocol:** Make both constant terms strictly positive ($D_1 > 0$ and $D_2 > 0$). Compute $\Delta = A_1A_2 + B_1B_2 + C_1C_2$:

Sign of Δ	Origin Location	Acute Bisector	Obtuse Bisector
$\Delta > 0$	Obtuse Angle	Negative (−) sign	Positive (+) sign
$\Delta < 0$	Acute Angle	Positive (+) sign	Negative (−) sign

The bisector containing the origin is **always** given by the positive (+) sign when $D_1, D_2 > 0$.

A.4 Straight Lines in Space and Skew Lines (Pages 37–52)

Equations of a Line and Geometric Reductions

1. Symmetrical (Point-Direction) Form:

$$\boxed{\frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n} = r} \iff P(r) = (x_1 + lr, y_1 + mr, z_1 + nr) \quad (\text{A.26})$$

2. Two-Point Form:

$$\boxed{\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}} \quad (\text{A.27})$$

3. Reduction from Unsymmetrical Form ($\Pi_1 = 0 = \Pi_2$):

- Direction ratios: $\vec{b} = \vec{N}_1 \times \vec{N}_2 \implies \frac{l}{B_1C_2 - B_2C_1} = \frac{m}{C_1A_2 - C_2A_1} = \frac{n}{A_1B_2 - A_2B_1}$.
- Base point: Set $z = 0$ and solve the 2×2 linear system for $(x_0, y_0, 0)$.

4. Angle Between Line and Plane:

$$\boxed{\sin \theta = \frac{Al + Bm + Cn}{\sqrt{A^2 + B^2 + C^2} \sqrt{l^2 + m^2 + n^2}}} \quad (\text{A.28})$$

Parallel: $Al + Bm + Cn = 0$. Perpendicular: $\frac{A}{l} = \frac{B}{m} = \frac{C}{n}$.

5. Foot of Perpendicular and Optical Image (Reflection) of Point (x_0, y_0, z_0) in Plane:

$$\text{Foot } N : \frac{x - x_0}{A} = \frac{y - y_0}{B} = \frac{z - z_0}{C} = -\frac{Ax_0 + By_0 + Cz_0 + D}{A^2 + B^2 + C^2} \quad (\text{A.29})$$

$$\text{Image } P' : \frac{x' - x_0}{A} = \frac{y' - y_0}{B} = \frac{z' - z_0}{C} = -2\frac{Ax_0 + By_0 + Cz_0 + D}{A^2 + B^2 + C^2} \quad (\text{A.30})$$

Coplanarity, Skew Lines, and Shortest Distance**1. Condition for a Line to Lie Wholly in a Plane:**

$$\boxed{Ax_1 + By_1 + Cz_1 + D = 0 \quad \text{and} \quad Al + Bm + Cn = 0} \quad (\text{A.31})$$

2. Condition of Coplanarity of Two Lines:

$$\boxed{\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix}} = 0 \quad (\text{A.32})$$

Plane containing coplanar lines: $\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} = 0.$

3. Shortest Distance (S.D.) Between Two Skew Lines:

$$d = \frac{\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix}}{\sqrt{(m_1n_2 - m_2n_1)^2 + (n_1l_2 - n_2l_1)^2 + (l_1m_2 - l_2m_1)^2}} \quad (\text{A.33})$$

4. Equations of the Line of Shortest Distance:

$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ l_1 & m_1 & n_1 \\ \lambda & \mu & \nu \end{vmatrix} = 0 = \begin{vmatrix} x - x_2 & y - y_2 & z - z_2 \\ l_2 & m_2 & n_2 \\ \lambda & \mu & \nu \end{vmatrix} \quad \text{where } (\lambda, \mu, \nu) \sim \vec{b}_1 \times \vec{b}_2 \quad (\text{A.34})$$

5. Distance Between Parallel Lines:

$$d = \frac{|(\vec{P}_2 - \vec{P}_1) \times \vec{b}|}{|\vec{b}|}.$$

A.5 Vector Treatment of Planes and Lines (Pages 53–62)**Vector Geometry of Planes**

Concept	Vector Formulation	Meaning / Condition
Plane Normal Form	$\vec{r} \cdot \hat{n} = p$	p = distance from origin, $ \hat{n} = 1$
Plane One-Point Form	$(\vec{r} - \vec{a}) \cdot \vec{n} = 0 \iff \vec{r} \cdot \vec{n} = d$	Passes through $\vec{a}$, normal $\vec{n}$
Point to Plane Distance	$D = \frac{ \vec{a} \cdot \vec{n} - d }{ \vec{n} }$	Perpendicular distance
Angle Between Planes	$\cos \theta = \frac{ \vec{n}_1 \cdot \vec{n}_2 }{ \vec{n}_1 \vec{n}_2 }$	Orthogonal when $\vec{n}_1 \cdot \vec{n}_2 = 0$
Three-Point Plane	$[\vec{r} - \vec{a}, \vec{b} - \vec{a}, \vec{c} - \vec{a}] = 0$	Non-collinear $\vec{a}, \vec{b}, \vec{c}$
Vector Family of Planes	$\vec{r} \cdot (\vec{n}_1 + \lambda \vec{n}_2) = d_1 + \lambda d_2$	Contains intersection line
Dihedral Vector Bisectors	$\frac{\vec{r} \cdot \vec{n}_1 - d_1}{ \vec{n}_1 } = \pm \frac{\vec{r} \cdot \vec{n}_2 - d_2}{ \vec{n}_2 }$	Locus of equidistant points

Vector Geometry of Lines and Spatial Invariants

Concept	Vector Formulation	Meaning / Condition
Vector Line (Parametric)	$\vec{r} = \vec{a} + \lambda \vec{b}$	Passes through $\vec{a}$, direction $\vec{b}$
Vector Line (Non-Parametric)	$(\vec{r} - \vec{a}) \times \vec{b} = \vec{0}$	Collinearity with direction $\vec{b}$
Distance: Point $\vec{p}$ to Line	$d = \frac{ (\vec{p} - \vec{a}) \times \vec{b} }{ \vec{b} }$	Perpendicular distance
Vector Coplanarity	$(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) = 0$	Lines lie in single plane
Shortest Distance (Skew)	$d = \frac{ (\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) }{ \vec{b}_1 \times \vec{b}_2 }$	Scalar triple product formula
Distance (Parallel Lines)	$d = \frac{ (\vec{a}_2 - \vec{a}_1) \times \vec{b} }{ \vec{b} }$	Cross product magnitude

A.6 The Geometry of the Sphere in 3D (Pages 63–69)

Canonical Sphere Equations and Circular Sections

1. Center-Radius Form:

$$(x - a)^2 + (y - b)^2 + (z - c)^2 = R^2 \quad (\text{A.35})$$

2. General Second-Degree Sphere Equation:

$$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0 \quad (\text{A.36})$$

$$\text{Center } C = (-u, -v, -w), \quad \text{Radius } R = \sqrt{u^2 + v^2 + w^2 - d} \quad (\text{A.37})$$

Classification: Real ($u^2 + v^2 + w^2 - d > 0$), Point ($= 0$), Imaginary (< 0).

3. Diametric Form (Antipodal Endpoints A, B):

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) + (z - z_1)(z - z_2) = 0 \quad (\text{A.38})$$

4. Plane Section of a Sphere (Circular Section): Every plane section is a circle! Let p be the perpendicular distance from center C to plane Π :

$$\text{Radius of Circular Section } r = \sqrt{R^2 - p^2} \quad (\text{A.39})$$

- $p = 0$: **Great Circle** (maximum radius $r = R$, center coincides with sphere center).
- $0 < p < R$: **Small Circle** ($r < R$).
- $p = R$: **Tangent Plane** ($r = 0$, single point contact).

Radical Planes, Tangency, and Normals

1. Radical Plane of Two Spheres:

$$S_1 - S_2 = 0 \iff 2(u_1 - u_2)x + 2(v_1 - v_2)y + 2(w_1 - w_2)z + (d_1 - d_2) = 0 \quad (\text{A.40})$$

Fundamental Invariant: The radical plane is strictly perpendicular to the center line C_1C_2 .

2. Condition of Tangency for a Plane: The plane $Ax + By + Cz + D = 0$ touches the sphere $S = 0$ if and only if $p = R$:

$$\frac{|-Au - Bv - Cw + D|}{\sqrt{A^2 + B^2 + C^2}} = \sqrt{u^2 + v^2 + w^2 - d} \quad (\text{A.41})$$

For central sphere $x^2 + y^2 + z^2 = a^2$: $D^2 = a^2(A^2 + B^2 + C^2)$.

3. Tangent Plane at Point of Contact $P_1(x_1, y_1, z_1)$:

$$T \equiv xx_1 + yy_1 + zz_1 + u(x + x_1) + v(y + y_1) + w(z + z_1) + d = 0 \quad (\text{A.42})$$

4. Normal Line at Point $P_1(x_1, y_1, z_1)$: Passes through the sphere center $(-u, -v, -w)$:

$$\frac{x - x_1}{x_1 + u} = \frac{y - y_1}{y_1 + v} = \frac{z - z_1}{z_1 + w} \quad (\text{A.43})$$

5. Length of Tangent and Plane of Contact from External Point:

$$L = \sqrt{S_1} = \sqrt{x_1^2 + y_1^2 + z_1^2 + 2ux_1 + 2vy_1 + 2wz_1 + d}, \quad \text{Plane of Contact: } T = 0 \quad (\text{A.44})$$

A.7 Master Geometric Conditions Quick-Reference Table

Geometric Relationship	Cartesian Condition	Vector Condition
Perpendicular Lines	$a_1a_2 + b_1b_2 + c_1c_2 = 0$	$\vec{b}_1 \cdot \vec{b}_2 = 0$
Parallel Lines	$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$	$\vec{b}_1 \times \vec{b}_2 = \vec{0}$
Perpendicular Planes	$A_1A_2 + B_1B_2 + C_1C_2 = 0$	$\vec{n}_1 \cdot \vec{n}_2 = 0$
Parallel Planes	$\frac{A_1}{A_2} = \frac{B_1}{B_2} = \frac{C_1}{C_2}$	$\vec{n}_1 \times \vec{n}_2 = \vec{0}$
Line Parallel to Plane	$Al + Bm + Cn = 0$	$\vec{b} \cdot \vec{n} = 0$
Line Perpendicular to Plane	$\frac{A}{l} = \frac{B}{m} = \frac{C}{n}$	$\vec{b} \times \vec{n} = \vec{0}$
Line Lying in Plane	$Ax_1 + By_1 + Cz_1 + D = 0$ and $Al + Bm + Cn = 0$	$\vec{a} \cdot \vec{n} = d$ and $\vec{b} \cdot \vec{n} = 0$
Coplanar Lines	$\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} = 0$	$(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) = 0$
Plane Tangent to Sphere	$\frac{ -Au - Bv - Cw + D }{\sqrt{A^2 + B^2 + C^2}} = R$	$\frac{ \vec{c} \cdot \vec{n} - d }{ \vec{n} } = R$